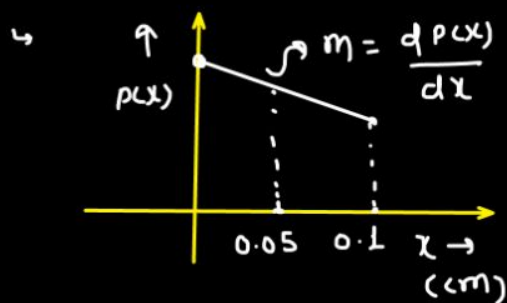


EDC PYQ

Q.

A non-uniformly doped p-type semiconductor is maintained at a temperature of 400 K. The hole concentration in the semiconductor decreases linearly from $x = 0$ to $x = 0.1$ cm. At 400 K, the measured hole diffusion current density is 20 A/cm^2 . If the hole concentration at $x = 0$ is $3 \times 10^{18} \text{ cm}^{-3}$, determine the hole concentration at $x = 0.05$ cm. Given that, μ_p at room's Temp. is $550 \text{ cm}^2/\text{V.s}$.



$$p(0) = 3 \times 10^{18} \text{ cm}^{-3}$$

$$p(0.05 \text{ cm}) = ?$$

$$\int_{p(0)}^{p(x)} dp(x) = \int_0^x -10.13 \times 10^{18} dx$$

$$J_p(x) = -q D_p \frac{dp(x)}{dx}$$

$$20 = 1.6 \times 10^{-19} \times 12.33 \times \frac{dp(x)}{dx}$$

$$\frac{dp(x)}{dx} = -10.13 \times 10^{18}$$

$$\mu_p(400 \text{ K}) = \mu_p(300 \text{ K}) \left(\frac{400}{300} \right)^{-3/2}$$

$$\mu_p(300 \text{ K}) = 550 \text{ cm}^2/\text{V.s}$$

$$\mu_p(400 \text{ K}) = ?$$

$$\mu_p \propto T^{-3/2}$$

$$\frac{\mu_p(400 \text{ K})}{\mu_p(300 \text{ K})} = \left(\frac{400}{300} \right)^{-3/2}$$

$$\mu_p(400 \text{ K}) = \left(\frac{3}{4} \right)^{3/2} \times 550 = 357.23 \text{ cm}^2/\text{V.s}$$

$$V_T = kT \Rightarrow V_T \propto T$$

$$\frac{V_T(400 \text{ K})}{V_T(300 \text{ K})} = \frac{400}{300}$$

$$\Rightarrow V_T(400 \text{ K}) = \frac{4}{3} \times 0.0259 \text{ V} = 0.0345 \text{ V}$$

$$D_p(400 \text{ K}) = \mu_p(400 \text{ K}) \cdot V_T(400 \text{ K}) = 357.23 \times 0.0345 \text{ cm}^2/\text{sec}$$

$$D_p = 12.33 \text{ cm}^2/\text{sec}$$

$$p(x) - p(0) = -10.13 \times 10^{18} x$$

$$p(x) = p(0) - 10.13 \times 10^{18} x$$

$$p(0.05 \text{ cm}) = 3 \times 10^{18} - 10.13 \times 10^{18} \times 0.05 = 2.5 \times 10^{18} \text{ cm}^{-3}$$

MSQ The correct form of the continuity equation for a semiconductor under non-equilibrium is:

✓ A. $\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n}{\partial x} + G_n - R_n$

✓ B. $\frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} + G_p - R_p$

✓ C. $\frac{\partial n}{\partial t} = n\mu_n \frac{\partial E}{\partial x} + \mu_n E \frac{\partial n}{\partial x} + D_n \frac{\partial^2 n}{\partial x^2} + G_n - R_n$

(Here, $n = n(x)$, $E = E(x)$, and both drift and diffusion contributions are included)

✓ D. $\frac{\partial p}{\partial t} = -p\mu_p \frac{\partial E}{\partial x} - \mu_p E \frac{\partial p}{\partial x} + D_p \frac{\partial^2 p}{\partial x^2} + G_p - R_p$

(Here, $p = p(x)$, $E = E(x)$, and both drift and diffusion contributions are included)

$$J_n(x) = n(x) q \mu E(x) + q D_n \frac{dn(x)}{dx}$$

$$\frac{1}{q} \frac{dJ_n(x)}{dx} = \mu E(x) \frac{dn(x)}{dx} + \mu n(x) \frac{dE(x)}{dx} + D_n \frac{d^2 n(x)}{dx^2}$$

Fermi-Dirac Distribution:

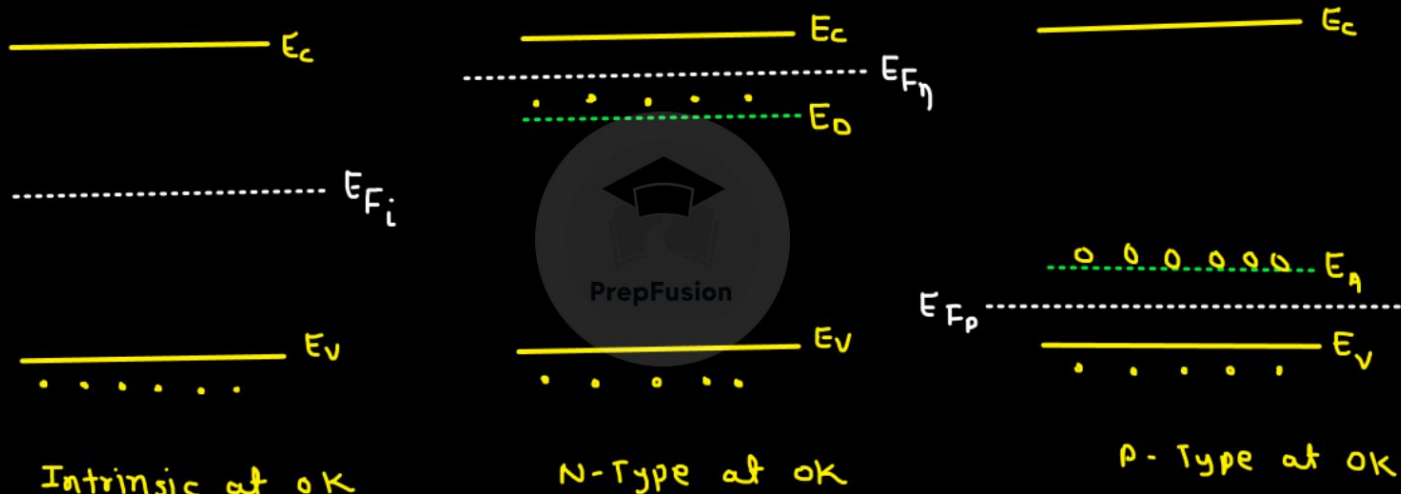
It tells us the probability of finding an electron in a particular energy state at a certain temperature.

[NOT level]

$$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}$$

- E = energy of the state
- E_F = Fermi energy level
- k = Boltzmann constant
- T = temperature

at 0K



↳ Effective Density of States:-

As you move higher in the conduction band, the density of states increases, and as you move lower in the valence band, the density of states also increases. To simplify calculations, we define the **effective density of states**, which represents the total number of energy states per unit volume in the conduction or valence band that are effectively available for electrons or holes to occupy. It essentially tells us how many electrons can occupy the conduction band and how many holes can occupy the valence band at a given temperature.

Effective density of the states in C.B. = $N_C \text{ cm}^{-3}$

" " " " " "

" " " " " "

$$N_C \propto (m_n^*)^{3/2}$$

$$N_C \propto T^{3/2}$$

$$N_C \propto (m_n^* T)^{3/2}$$

$$N_V \propto (m_p^*)^{3/2}$$

$$N_V \propto T^{3/2}$$

$$N_V \propto (m_p^* T)^{3/2}$$

$n e^-$

$$V.B. = N_V \text{ cm}^{-3}$$

↳ holes

$$N_C = N_{C0} (m_n^* T)^{3/2}$$

$$N_V = N_{V0} (m_p^* T)^{3/2}$$

Total no. of electrons per unit volume: $N_c \cdot f(E_c)$

$$n = N_c e^{-(E_c - E_F)/kT}$$

Total no. of holes per unit volume: $N_v \cdot [1 - f(E_v)]$

$$p = N_v e^{-(E_F - E_v)/kT}$$

Remⁿ

Formula Summary:-

$$(i) f(E) = \frac{1}{1 + e^{(E - E_F)/kT}} \approx e^{-(E - E_F)/kT}$$

$$(ii) \text{Probability of finding } e^- \text{ in c.b.} \approx e^{-(E_c - E_F)/kT}$$

$$(iii) \text{Probability of finding hole in v.b.} \approx e^{-(E_F - E_v)/kT}$$

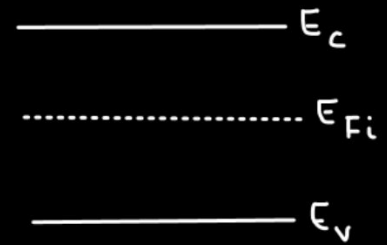
$$(iv) n = N_c e^{-(E_c - E_F)/kT} \quad ; \quad N_c = N_{c0} (m_n^* T)^{3/2}$$

$$N_v = N_{v0} (m_p^* T)^{3/2}$$

$$(v) p = N_v e^{-(E_F - E_v)/kT}$$

Remⁿ $E_{Fi} = \frac{E_c + E_v}{2} - \frac{kT}{2} \ln \frac{N_c}{N_v}$ ✓

$$E_{Fi} = E_{mid-gap} - \frac{kT}{2} \ln \frac{N_c}{N_v}$$



Case-1

$N_c = N_v$ OR $M_n^* = M_p^*$
OR NOT mentioned

$$E_{Fi} = \frac{E_c + E_v}{2}$$

$$E_{Fi} = \frac{E_c + E_v}{2} - \frac{3kT}{4} \ln \left[\frac{M_n^*}{M_p^*} \right]$$

Case-2 M_n^* and M_p^* are given.

$$E_{Fi} = \frac{E_c + E_v}{2} - \frac{kT}{2} \ln \left[\frac{M_n^*}{M_p^*} \right]^{3/2}$$

(ii) N-Type s/c:-

Fermi level of N-Type s/c = E_{Fn}

$$n = N_c e^{-(E_c - E_{Fn})/kT}$$

$$-kT \ln \left[\frac{n}{N_c} \right] = (E_c - E_{Fn})$$

$$E_c - E_{Fn} = kT \ln \left(\frac{N_c}{n} \right)$$
 ✓

$$p = N_v e^{-(E_{Fn} - E_v)/kT}$$

$$E_{Fn} - E_v = kT \ln \left(\frac{N_v}{p} \right)$$
 ✓

N-Type s/c:-

$$n = N_c e^{-(E_c - E_{Fn})/kT} \quad \text{--- (1)}$$

Intrinsic s/c

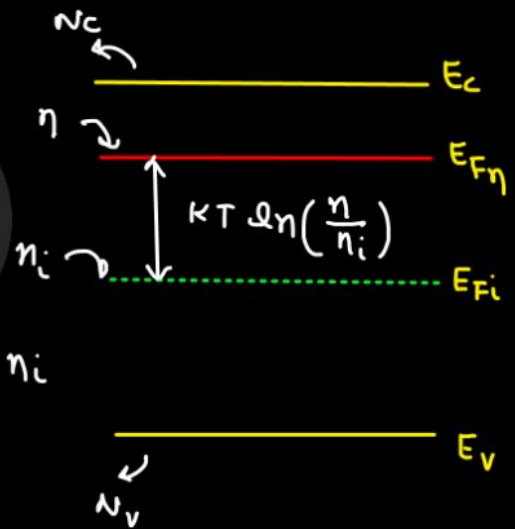
$$n_i = N_c e^{-(E_c - E_{Fi})/kT}$$

$$\frac{n}{n_i} = e^{(E_{Fn} - E_{Fi})/kT}$$

Rem^m

$$E_{Fn} - E_{Fi} = kT \ln \left(\frac{n}{n_i} \right)$$

$$E_{Fn} - E_{Fi} = ?$$



$$n > n_i$$

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YT Pla

(ii) P-Type s/c:-

$$p = N_v \cdot e^{-(E_{Fp} - E_v)/kT}$$

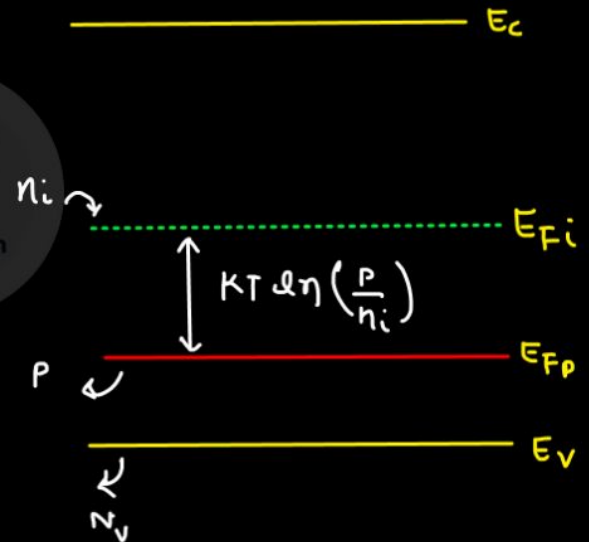
$$E_{Fp} - E_v = kT \ln \left(\frac{N_v}{p} \right)$$

Rem^m

$$E_{Fi} - E_{Fp} = kT \ln \left(\frac{p}{n_i} \right)$$

$$n = N_c e^{-(E_c - E_{Fp})/kT}$$

$$E_c - E_{Fp} = kT \ln \left(\frac{N_c}{n} \right)$$



$$p > n_i$$

NOTE:- We have studied:

$$N_c = N_{c0} (M_n^* T)^{3/2}$$

$$N_v = N_{v0} (M_p^* T)^{3/2}$$

$$n_i^2 = A_0 T^3 e^{-E_g/kT}$$

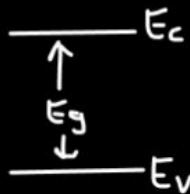
Proof:-

Mass-action law; $n_p = n_i^2$

$$n_p = n_i^2$$

$$N_c \cdot e^{-(E_c - E_F)/kT} \cdot N_v e^{-(E_F - E_v)/kT} = n_i^2$$

$$n_i^2 = N_c N_v e^{-E_g/kT}$$



$$E_v - E_c = -E_g$$

$$\underbrace{N_{c0}}_{\substack{M_n^{*3/2} \\ T^{3/2}}} \cdot \underbrace{N_{v0}}_{\substack{M_p^{*3/2} \\ T^{3/2}}} e^{-E_g/kT} = n_i^2$$

$$n_i^2 = A_0 T^3 e^{-E_g/kT}$$

↳ Formulae to Remember:-

$$\hookrightarrow f(E) = \frac{1}{1 + e^{(E - E_F)/kT}} \approx e^{-(E - E_F)/kT}$$

$$\hookrightarrow n = N_c e^{-(E_c - E_F)/kT} ; \hookrightarrow p = N_v e^{-(E_F - E_v)/kT}$$

$$\hookrightarrow E_{Fi} = \frac{E_c + E_v}{2} - \frac{kT}{2} \ln\left(\frac{N_c}{N_v}\right) ; \hookrightarrow N_c = N_{c0} (M_n^* T)^{3/2}$$

$$N_v = N_{v0} (M_p^* T)^{3/2}$$

$$\hookrightarrow E_{Fn} - E_{Fi} = kT \ln\left(\frac{n}{n_i}\right)$$

$$\hookrightarrow E_{Fi} - E_{Fp} = kT \ln\left(\frac{p}{n_i}\right)$$

$$\hookrightarrow n_i^2 = N_c N_v e^{-E_g/kT}$$

When a p-type and n-type semiconductor are joined together, a **p-n junction** is formed. Initially, holes from the p-side diffuse towards the n-side and electrons from the n-side diffuse towards the p-side due to the concentration difference. As they cross the junction, they recombine with each other, leaving behind immobile charged ions near the junction. These uncovered ions form the **depletion region**, which is depleted of mobile charge carriers. The region now contains fixed negative acceptor ions on the p-side and fixed positive donor ions on the n-side. This charge separation creates an **electric field** directed from the n-side to the p-side. This field opposes further diffusion of carriers. As the diffusion current (due to carriers moving from high to low concentration) increases, the electric field produces an opposite **drift current**. At equilibrium, the **diffusion current equals the drift current**, resulting in **zero net current** across the junction. The potential difference developed due to this internal field is called the **built-in potential** (V_{bi}). Thus, under equilibrium, the p-n junction maintains balance between drift and diffusion, and no external current flows.

Immobile

↓

not Movable

⊕ ⊖

Built-in potential:-

$$V_{bi} = V_T \ln \left(\frac{p_{p0}}{n_{p0}} \right)$$

$$V_{bi} = V_T \ln \left(\frac{n_{n0}}{p_{n0}} \right)$$

$$qV_{bi} = kT \ln \left(\frac{N_A}{n_i} \right) + kT \ln \left(\frac{N_D}{n_i} \right)$$

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

$$V_{bi} = V_T \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

★ For neutral P-side:-

$$E_{Fi} - E_{Fp} = kT \ln \left(\frac{p}{n_i} \right)$$

★ For neutral N-side:-

$$E_{Fn} - E_{Fi} = kT \ln \left(\frac{n}{n_i} \right)$$

For depletion region:- [under equilibrium]

**

Under eqll:-
 $n p = n_i^2$

$$E_{Fn} = E_{Fp} = E_F$$

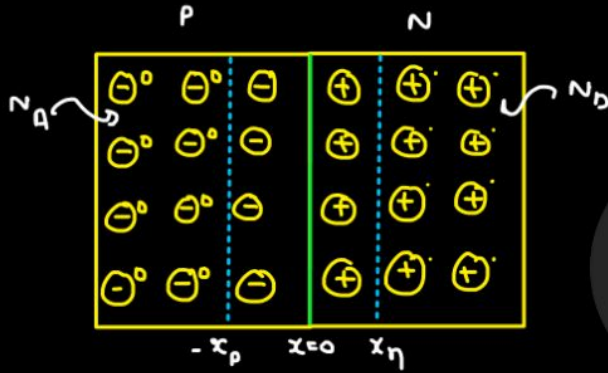
$$E_{Fi} - E_F = kT \ln \left(\frac{p_d}{n_i} \right) \quad \text{--- (1)} \quad p_d : \text{hole concn in dep. region}$$

$$E_F - E_{Fi} = kT \ln \left(\frac{n_d}{n_i} \right) \quad \text{--- (2)}$$

$$0 = kT \ln \left(\frac{p_d n_d}{n_i^2} \right)$$

$$p_d n_d = n_i^2$$

* charge neutrality:- [Assumption: → charge carrier inside the depletion region will be always ignored.]



↳ Total depletion width $w = x_n + x_p$

$$w = \frac{N_A}{N_D} x_p + x_p$$

$$x_p = \frac{N_D}{N_A + N_D} w$$

$$x_n = \frac{N_A}{N_A + N_D} w$$

$(-x_p < x < 0)$ Total negative charge = $(0 < x < x_n)$ Total positive charge

$$N_A \cdot A \cdot x_p = N_D \cdot A \cdot x_n$$

$$N_A x_p = N_D x_n$$

$$\frac{N_A}{N_D} = \frac{x_n}{x_p} \Rightarrow N_A : N_D = x_n : x_p \Rightarrow x_p \propto \frac{1}{N_A}$$

* Maximum E-field occurs at the Junction:-

$$|\epsilon_{max}| = \frac{q N_A x_{p0}}{\epsilon} = \frac{q N_D x_{n0}}{\epsilon}$$

↳ Built-in potential in terms of max-E field:-

$$V_{bi} = \frac{1}{2} w \epsilon_0 \epsilon_{max}$$

$$\epsilon_{max} = \frac{2 V_{bi}}{w}$$

$$V_{bi} = \frac{1}{2} w \frac{q N_A}{\epsilon} x_{p0}$$

$$x_{p0} = \frac{w \cdot N_D}{N_A + N_D}$$

$$\frac{q \epsilon}{2} V_{bi} = \frac{w^2 \cdot N_A N_D}{N_A + N_D}$$

* Important formulae Summary:- ; $N_A x_p = N_D x_n$

(i) $|\epsilon_{max}| = \frac{q N_A x_p}{\epsilon} = \frac{q N_D x_n}{\epsilon}$

(ii) $V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right) = \frac{1}{2} W \epsilon_{max} \Rightarrow \epsilon_{max} = \frac{2 V_{bi}}{W}$

(iii) $W = \sqrt{\frac{2\epsilon}{q} \left[\frac{1}{N_A} + \frac{1}{N_D} \right] V_{bi}}$

$x_p = \frac{W N_D}{N_A + N_D}$; $x_n = \frac{W N_A}{N_A + N_D}$; $x_n + x_p = W$

* Important formulae:-

① Ideal Diode current Equation:-

$$I = I_0 (e^{V_a/V_T} - 1)$$

I_0 : Reverse saturation current = $\frac{A q D_p p_{n0}}{L_p} + \frac{A q D_n n_{p0}}{L_n}$

② non ideal Diode current equation:-

$$I_{diode} = I_{ideal} + I_{recomb}$$

$$= I_0 [e^{V_a/\eta V_T} - 1]$$

η : Ideality factor

$\eta = 1 \text{ or } 2$

③ Depletion width:-

$$W = \sqrt{\frac{2\epsilon}{q} \left[\frac{1}{N_A} + \frac{1}{N_D} \right] (V_{bi} - V_F)} \quad V_F \approx V_a$$

④ Carrier concentration:-

$$\hookrightarrow p_n(x_n) = p_{n0} e^{V_a/V_T}$$

$$\Delta p_n(x_n) = p_{n0} [e^{V_a/V_T} - 1]$$

$$\Delta p_n(x) = p_{n0} [e^{V_a/V_T} - 1] e^{-(x-x_n)/L_p}$$

$$\hookrightarrow n_p(x_p) = n_{p0} e^{V_a/V_T}$$

$$\Delta n_p(x_p) = n_{p0} [e^{V_a/V_T} - 1]$$

$$\Delta n_p(x) = n_{p0} [e^{V_a/V_T} - 1] e^{(x+x_p)/L_n}$$

⑤ Maximum E-field at the Junction:-

$$E_{max} = \frac{q N_D x_n}{\epsilon} = \frac{q N_A x_p}{\epsilon}$$

$$x_n = \frac{W N_A}{N_A + N_D}$$

Reverse saturation Current:-

$$I_{ideal} = I_0 [e^{V_a/V_T} - 1]$$

$$I_0 = \frac{A q D_p p_{n0}}{L_p} + \frac{A q D_n n_{p0}}{L_n}$$

$$I_0 = \frac{A q D_p}{L_p} \times \frac{n_i^2}{N_D} + \frac{A q D_n}{L_n} \frac{n_i^2}{N_A}$$

$$n_i = A T^3 e^{-E_g/KT}$$

$$\begin{aligned} n_i \uparrow &\Rightarrow I_0 \uparrow \\ E_g \uparrow &\Rightarrow n_i \downarrow \Rightarrow I_0 \downarrow \\ T \uparrow &\Rightarrow I_0 \uparrow \\ N_A \uparrow, N_D \uparrow &\Rightarrow I_0 \downarrow \end{aligned}$$

* Temperature dependence of I_0 :-

$$(I_0)_{T_2} = (I_0)_{T_1} \cdot (2)^{(T_2-T_1)/10}; \quad \text{For every } 10^\circ\text{C}/10\text{K rise in Temp, } I_0 \text{ gets doubled.}$$

Q. At 300K, I_0 is 10^{-10} Amp. find Reverse sat. current at 325K.

$$\hookrightarrow (I_0)_{325K} = 10^{-10} (2)^{25/10} = 5.65 \times 10^{-10} \text{ Amp.}$$

$$V(x) \propto - \int_{-\infty}^x E(x) \cdot dx$$

* Uniform doping:-

$$p_v(x) \propto x^0$$

$$E(x) \propto x^1$$

$$V(x) \propto x^2$$

↳ $\boxed{\text{Abrupt } J^n}$

$$W = \sqrt{\frac{2\epsilon}{q} \left[\frac{1}{N_A} + \frac{1}{N_D} \right] V_{bi}}$$

$$\boxed{W \propto (V_{bi})^{1/2}} \quad \text{we}$$

* Non-uniform doping

$$p_v(x) \propto x^\eta ; \quad \eta \neq 0$$

$$E(x) \propto x^{\eta+1}$$

$$V(x) \propto x^{\eta+2}$$

PrepFusion

$$\boxed{\text{Linearly graded } J^\eta}$$

$$V_{bi} \propto W^3$$

$$\boxed{W \propto (V_{bi})^{1/3}} \quad \text{we}$$

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YT PI

$$\checkmark \quad \boxed{n \cdot p = n_i^2 e^{V_a/V_T}}$$

; when $V_a = 0 \downarrow$

$$\boxed{n \cdot p = n_i^2}$$

at the edge of depletion region;

(i) $x = x_n$: N-side

$$n_n(x_n) \cdot p_n(x_n) = n_i^2 e^{V_a/V_T}$$

$$N_D \cdot p_n(x_n) = n_i^2 e^{V_a/V_T}$$

$$p_n(x_n) = \frac{n_i^2}{N_D} e^{V_a/V_T}$$

$$\boxed{p_n(x_n) = p_{n0} e^{V_a/V_T}}$$

(ii) $x = -x_p$: P side

$$n_p(-x_p) \cdot p_p(-x_p) = n_i^2 e^{V_a/V_T}$$

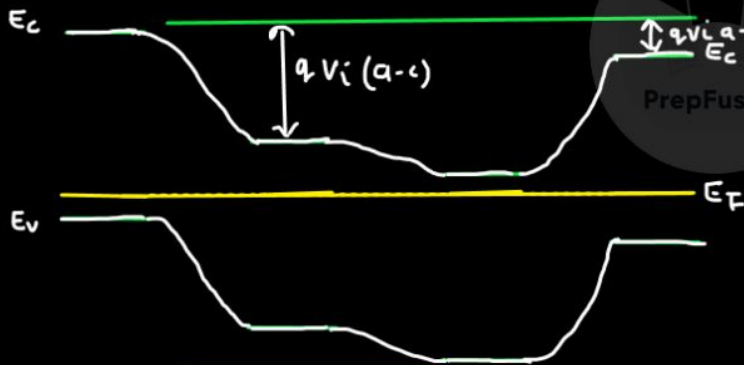
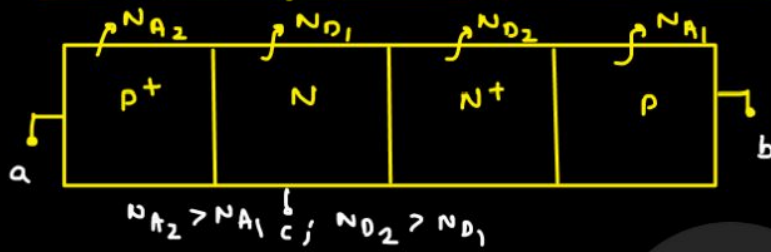
$$n_p(-x_p) \cdot N_A = n_i^2 e^{V_a/V_T}$$

$$n_p(-x_p) = \frac{n_i^2}{N_A} e^{V_a/V_T}$$

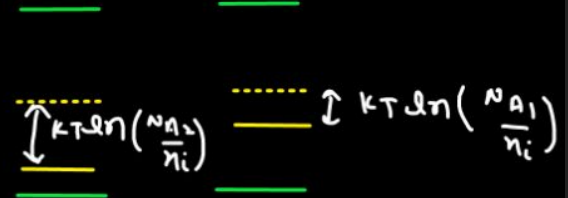
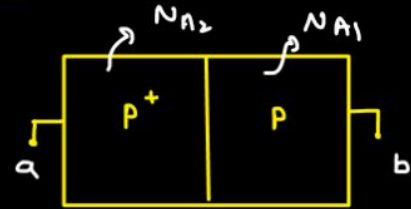
$$\boxed{n_p(-x_p) = n_{p0} e^{V_a/V_T}}$$

PrepFusion

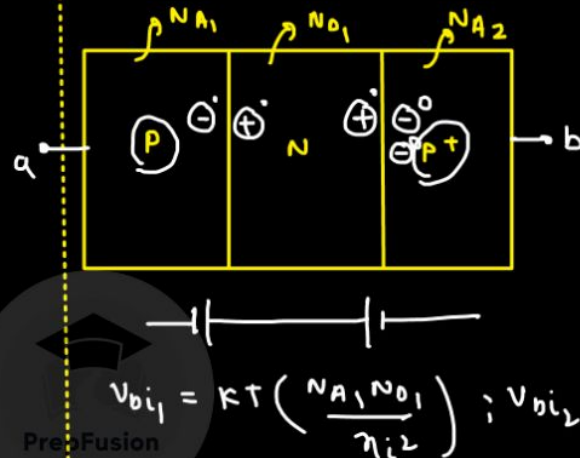
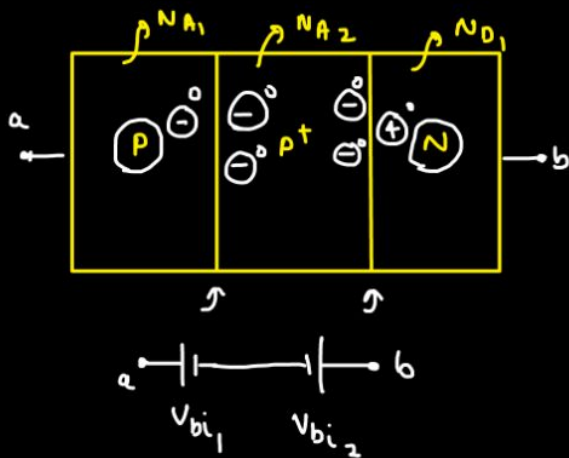
Understanding by Energy Band Diagram:-



$$kT \ln \left(\frac{N_{A2}}{N_{A1}} \right)$$



$$(qV_i)_{a-b} = kT \ln \left(\frac{N_{A2}}{N_{A1}} \right)$$



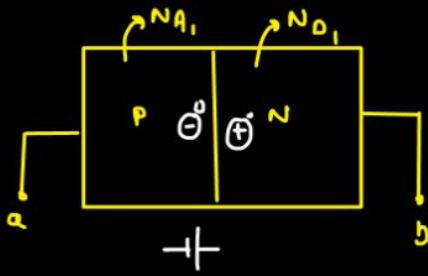
$$(V_{bi})_{ab} = |V_{bi1} - V_{bi2}|$$

$$V_{bi1} = kT \ln \left(\frac{N_{A2}}{N_{A1}} \right); V_{bi2} = kT \ln \left(\frac{N_{D1} N_{A2}}{n_i^2} \right)$$

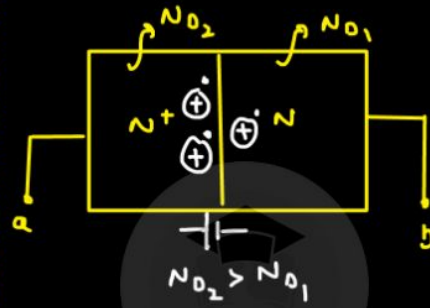
$$|(V_{bi})_{ab}| = kT \ln \left(\frac{N_{A2}}{N_{A1}} \right)$$

$$|(V_{bi})_{ab}| = kT \ln \left(\frac{N_{A1} N_{D1}}{n_i^2} \right)$$

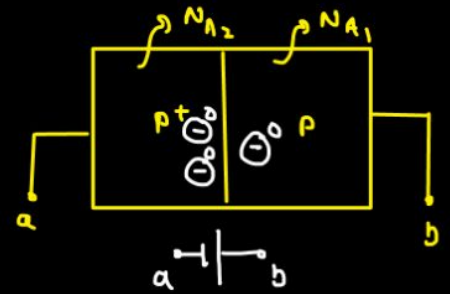
* Some Discussion on Built-in potential:-



$$|(V_{bi})_{ab}| = kT \ln \left(\frac{N_{A1} N_{D1}}{n_i^2} \right)$$



$$|(V_{bi})_{ab}| = kT \ln \left(\frac{N_{A2} N_{D1}}{n_i^2} \right)$$



$$|(V_{bi})_{ab}| = kT \ln \left(\frac{N_{A1} N_{D2}}{n_i^2} \right)$$

BJT

Emitter: Region that injects carriers.

Base: Thin region that controls carriers.

Collector: Region that collects carriers.

Emitter doping should be highest, and base width should be lowest.

✓ **Width:-** collector > emitter > Base ; why collector width highest? ✓

✓ **Doping:-** Emitter > Base > collector ; why base doping higher than collector? ✓

Why base doping should be less than emitter doping:

If base doping is low, fewer majority carriers exist in the base, so injected carriers from the emitter have a much lower chance of recombining. This allows more carriers to reach the collector, making the collector current strongly controlled by the emitter-junction forward bias.

Emitter J^n :- $J_E : E-B$

Collector J^n :- $J_C : C-B$

* Emitter Injection Efficiency (γ) :-

$$\gamma = \frac{I_{pE}}{I_{pE} + I_{nE}}$$

* Base Transport Factor (b) :-

$$b = \frac{I_{pC}}{I_{pE}}$$

$$\therefore \gamma b = \frac{I_{pC}}{I_{pE} + I_{nE}} = \frac{I_C}{I_E} = \alpha \rightarrow \underline{\text{common Base current gain}}$$

$$\hookrightarrow I_E = \underbrace{I_{PE}} + \underbrace{I_{NE}} = I_0 [e^{V_F/nV_T} - 1]$$

$$I_{PE} = \frac{Aq D_p p_{n0}}{L_p} [e^{V_F/nV_T} - 1]$$

$$= \frac{Aq D_p n_i^2}{N_D L_p} [e^{V_F/nV_T} - 1]$$

✓

$$I_{PE} = \frac{Aq D_B n_i^2}{N_B W_B} [e^{V_F/nV_T} - 1]$$

N_B : Majority carrier concn in Base

W_E : Width of the neutral emitter region

W_B : Width of the neutral base region

W_C : Width of the neutral collector region

D_E : Diffusivity of minority carriers in the emitter

D_B : Diffusivity of minority carriers in the base

D_C : Diffusivity of minority carriers in the collector

N_E : Majority carrier concn in Emitter

$$I_{NE} = \frac{Aq D_n n_{p0}}{L_n} [e^{V_F/nV_T} - 1] = \frac{Aq D_n n_i^2}{N_A L_n} [e^{V_F/nV_T} - 1]$$

$$I_{NE} = \frac{Aq D_E n_i^2}{N_E W_E} [e^{V_F/nV_T} - 1]$$

$$I_E = I_{PE} + I_{NE} = Aq n_i^2 \left[\frac{D_B}{N_B W_B} + \frac{D_E}{N_E W_E} \right] [e^{V_F/nV_T} - 1]$$

$$I_E \approx I_C \approx I_0 [e^{V_F/nV_T} - 1]$$

PNP:- $*I_C = I_{sc} [e^{V_{EB}/nV_T} - 1]$

NPN:- $I_C = I_{sc} [e^{V_{BE}/nV_T} - 1]$

$$I_E = I_C + I_B$$

$$I_C = \alpha I_E$$

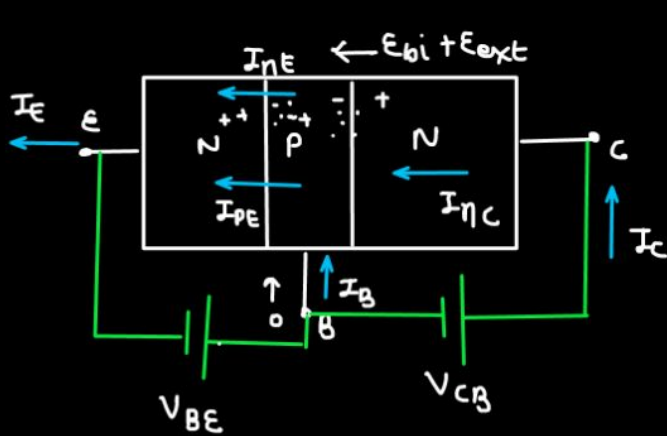
$$= \alpha \underbrace{I_0} [e^{V_F/nV_T} - 1]$$

$$\hookrightarrow I_C = I_{sc} [e^{V_F/nV_T} - 1]$$

$$I_0 = I_{SE}$$

$$\alpha I_0 = I_{sc}$$

Given GATE



$$\gamma = \frac{I_{nE}}{I_{nE} + I_{pE}}$$

$$b = \frac{I_{nC}}{I_{nC} + I_{pC}}$$

$$\alpha = \gamma b$$

$$I_E = I_{nE} + I_{pE}$$

$$I_{nE} = \frac{A q D_n n_{p0}}{L_n} [e^{V_F / n V_T} - 1]$$

$$I_{nE} = \frac{A q D_B n_i^2}{N_B W_B} [e^{V_F / n V_T} - 1]$$

$$I_{pE} = \frac{A q D_E n_i^2}{N_E W_E} [e^{V_F / n V_T} - 1]$$

$$I_E = A q n_i^2 \left[\frac{D_B}{N_B W_B} + \frac{D_E}{N_E W_E} \right] [e^{V_F / n V_T} - 1]$$

* Relationship b/w α and β :-

From eqⁿ (1) and (2)

$$I_C = \alpha I_E$$

$$I_C = \alpha (I_C + I_B)$$

$$I_C = \alpha I_C + \alpha I_B$$

$$I_C (1 - \alpha) = \alpha I_B$$

$$I_C = \frac{\alpha}{1 - \alpha} I_B \quad (4)$$

conclusion:-

$$\frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\alpha \approx 1 (< 1)$$

$$\beta \gg 1$$

$$I_E = I_C + I_B \quad (1)$$

$$I_C = \alpha I_E + I_{B0} \quad (2)$$

$$I_C = \beta I_B + I_{C0} \quad (3)$$

from (3) and (4)

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\frac{1}{\beta} = \frac{1 - \alpha}{\alpha} \Rightarrow \frac{1}{\beta} = \frac{1}{\alpha} - 1$$

$$\Rightarrow \frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\alpha = \frac{\beta}{\beta + 1}$$

↳ Relation b/w I_{CE0} and I_{CB0} :

$$I_C = \alpha I_E + I_{CB0} \quad \text{--- (1)}$$

$$I_C = \beta I_B + I_{CE0} \quad \text{--- (2)}$$

From eqⁿ (1)

$$I_C = \alpha I_E + I_{CB0}$$

$$I_C = \frac{\beta(I_C + I_B)}{\beta + 1} + I_{CB0}$$

$$\cancel{\beta I_C} + I_C = \cancel{\beta I_C} + \beta I_B + (\beta + 1) I_{CB0}$$

$$I_C = \beta I_B + (\beta + 1) I_{CB0} \quad \text{--- (3)}$$

$$I_{CE0} = (\beta + 1) I_{CB0}$$

$$I_{CE0} > I_{CB0}$$

★ Emitter Injection Efficiency:- [PNP]

$$\gamma = \frac{I_{PE}}{I_{PE} + I_{NE}} = \frac{1}{1 + \frac{I_{NE}}{I_{PE}}}$$

$$I_{NE} = \frac{q A D_E n_i^2}{N_E W_E} [e^{V_F / \eta V_T} - 1]$$

$$I_{PE} = \frac{q A D_B n_i^2}{N_B W_B} [e^{V_F / \eta V_T} - 1]$$

$$\gamma = \frac{1}{1 + \frac{D_E N_B W_B}{D_B N_E W_E}}$$

★ Base Transport factor (b):-

$$b = \frac{I_{PC}}{I_{PE}} = 1 - \frac{1}{2} \left[\frac{W_B}{L_B} \right]^2$$

$$L_B = \sqrt{D_B \tau_B}$$

W_B : width of neutral Base region

L_B : Actual Diffusion length of minority carriers in Base region

↳ Conclusion: $I_E = I_C + I_B$ ✓

When $J_E: \nabla B$ and $J_C: \nabla B$ or zero bias

① $I_C = \alpha I_E + I_{CE0}$

$I_C = \beta I_B + I_{CE0}$

② $\frac{1}{\alpha} = \frac{1}{\beta} + 1$

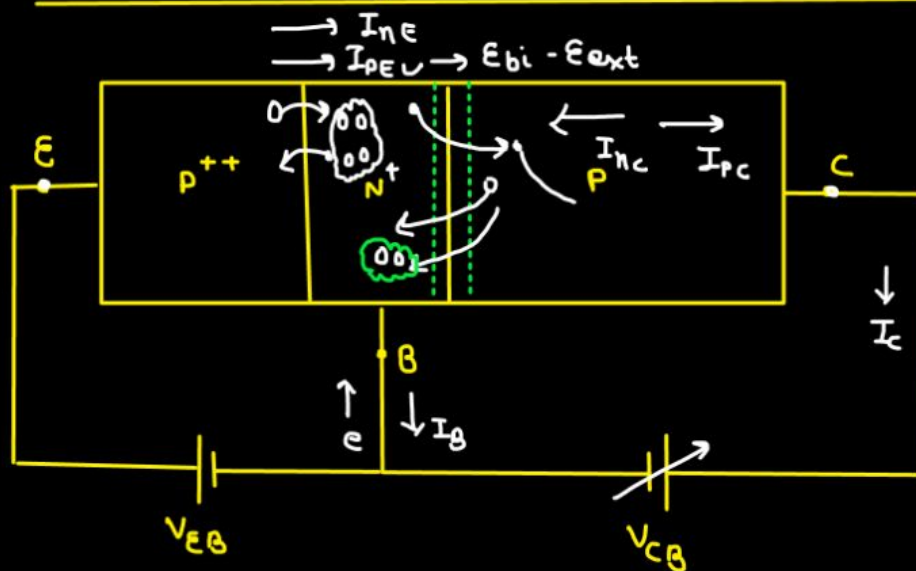
③ $I_{CE0} = (\beta + 1) I_{CB0}$

④ $I_E = I_{PE} + I_{nE} = q n_i^2 \left[\frac{D_E}{w_E N_E} + \frac{D_B}{w_B N_B} \right] [e^{V_F / n V_T} - 1]$

⑤ $I_C = I_S [e^{V_F / n V_T} - 1] \underbrace{\quad}_{I_0}$

⑥ $\gamma = \frac{1}{1 + \frac{D_E N_B w_B}{N_E w_E D_B}} ; b = 1 - \frac{1}{2} \left[\frac{w_B}{L_B} \right]^2 ; \alpha = \gamma b$

* Forward Biasing the collector Junction: → saturation region =



$J_E: \nabla B$

$J_C: \nabla B$

↓ $I_C = I_{pC} - I_{nC}$

$I_C = f(V_{CB})$

★ $J_C: \nabla B \uparrow \uparrow \Rightarrow I_C \downarrow \downarrow$

$I_C = f(V_{EC})$

$V_{CB} \uparrow \uparrow \Rightarrow$ Base gets saturated with holes $\Rightarrow$ recombination $\uparrow$

↳ $V_{CB} \uparrow \Rightarrow I_B \uparrow$ and $I_C \downarrow \Rightarrow I_C \neq \beta I_B$ $\downarrow$
 $I_B \uparrow$

In a PNP transistor operating in saturation, both the emitter-base and collector-base junctions are forward biased. The emitter junction injects many holes into the base, and the collector junction also injects holes from its side. With both junctions forward biased, the depletion regions collapse and the base becomes heavily filled with excess minority carriers. Because the base now contains a large number of holes, recombination in the base increases, and therefore the base current rises. $I_B \uparrow$

The collector-base junction, although forward biased, still has a very weak electric field. Because of this weak field, only a small portion of emitter-injected holes reach the collector. Meanwhile, the increased forward bias at the collector junction causes more electrons to diffuse from the base into the collector, and this electron current flows opposite to the hole current coming from the emitter. As the forward bias at the collector increases, this opposing electron current becomes larger, so the net collector current decreases. Thus, the usual relation $I_C = \beta I_B$ does not hold in saturation.

$$\hookrightarrow I_C \neq \beta I_B \rightarrow \text{saturation} \quad I_C = \beta I_B \rightarrow \text{active}$$

Because most injected carriers recombine in the base and only a very small fraction reach the collector terminal, the transistor loses its current-gain behavior. Under saturation, the device effectively behaves like two forward-biased diodes, and in circuit analysis we replace it with a small constant $V_{CE(sat)}$ rather than treating it as a controlled amplifier.

$$\hookrightarrow (V_{CE})_{sat} \text{ OR } (V_{EC})_{sat} = \text{Fixed value}$$

$$\begin{array}{ccc} (I_C)_{\text{active}} & > & (I_C)_{\text{sat.}} \\ \underbrace{\phantom{(I_C)_{\text{active}}}} & & \underbrace{\phantom{(I_C)_{\text{sat.}}}} \\ \downarrow & & \downarrow \\ \text{Amplification} & & \text{Switch (ON)} \\ I_C = \beta I_B & & \end{array}$$

Because most injected carriers recombine in the base and only a very small fraction reach the collector terminal, the transistor loses its current-gain behavior. Under saturation, the device effectively behaves like two forward-biased diodes, and in circuit analysis we replace it with a small constant $V_{CE(sat)}$ rather than treating it as a controlled amplifier.

↳ $(V_{CE})_{sat}$ OR $(V_{EC})_{sat}$ = Fixed value

$(I_C)_{active} > (I_C)_{sat}$
 ↓ ↓
 Amplification Switch (ON)
 $I_C = \beta I_B$

* What If both Junctions are Reverse biased:-

$J_E : RB, J_C : RB$

$I_C \approx 0$ [OR very small reverse sat. current]

	Mode of working	J_E	J_C	Use
✓	① Active	F.B.	R.B.	Amplifier
↳	② Saturation	F.B.	F.B.	ON switch
↳	③ Cut-off	R.B.	R.B.	OFF switch

When doping becomes extremely high, the donor atoms sit so close to each other that their originally discrete donor energy levels begin to **split and spread out**, forming a small **donor band**. As the doping increases further, the donor atoms come even closer, causing this donor band to **widen and move upward** toward the conduction band. Eventually the donor band **merges with the conduction band**, and at that moment the separation between the valence band and conduction band effectively becomes smaller. In other words, the **bandgap has shrunk** because the conduction-band edge has been pulled downward by the merging donor band.

↳ At moderate or High doping :- $n_i^2 = A_0 T^3 e^{-E_g/kT}$

↳ At very High doping :- $(n_i^2)_D = A_0 T^3 e^{-(E_g - \Delta E_g)/kT}$
 $= \underbrace{n_i^2}_{\text{at moderate doping}} e^{\Delta E_g/kT}$

MOSFET

↳ pN Jⁿ diode symbol:-



↳ Important Points:-

- (i) Forward Bias $\rightarrow$ less Depletion width, More current, Diffusion $\uparrow$
- (ii) Reverse Bias $\rightarrow$ More Depletion width, less current, Drift $\uparrow$

$$(W_{dep})_{eq.} = \sqrt{\frac{2\epsilon_{si}}{q} \left[\frac{1}{N_A} + \frac{1}{N_D} \right] V_{bi}} \quad ; \quad (W_{dep})_{f.b.} = \sqrt{\frac{2\epsilon_{si}}{q} \left[\frac{1}{N_A} + \frac{1}{N_D} \right] (V_{bi} - V_a)}$$

$$\hookrightarrow (W_{dep})_{R.B.} = \sqrt{\frac{2\epsilon_{si}}{q} \left[\frac{1}{N_A} + \frac{1}{N_D} \right] [V_{bi} + V_a]}$$

(i) If $V_{GS} < V_T \Rightarrow$ No current flow

(ii) If $V_{GS} > V_T$ and $V_{DS} > 0$

$\downarrow$ $\downarrow$
 Inversion current
 layer formed flows

↳ At low value of V_{DS} ($V_{DS} \sim 0.05V$ to $0.1V$) & $V_{GS} > V_T$:-

$$Q_{in} = C_{ox} (V_{GS} - V_T) \quad ; \quad Q_{in} \neq f(V_{DS})$$

$\hookrightarrow$ MOSCAP

$$\hookrightarrow Q_{in} = C_{ox}' (V_{GS} - V_T) WL$$

$C_{ox}' \rightarrow$ capacitance / area
 F/cm^2

MOSFET:-

$$Q_{in} = C_{ox} (V_{GS} - V_T) WL$$

$C_{ox} \Rightarrow F/cm^2$

$$I_D = \frac{Q_{in}}{\tau}$$

$$\tau = \frac{L}{v_d}$$

L : channel length
 v_d : Drift velocity of electron

$$I_D = \frac{C_{ox}(V_{GS} - V_T)WL}{L^2} \mu_n v_{DS}$$

$$\tau = \frac{L}{\mu_n E}$$

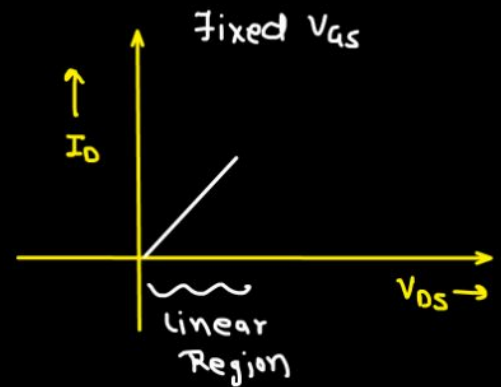
$$\tau = \frac{L^2}{\mu_n v_{DS}}$$

$$E = v_{DS}/L$$

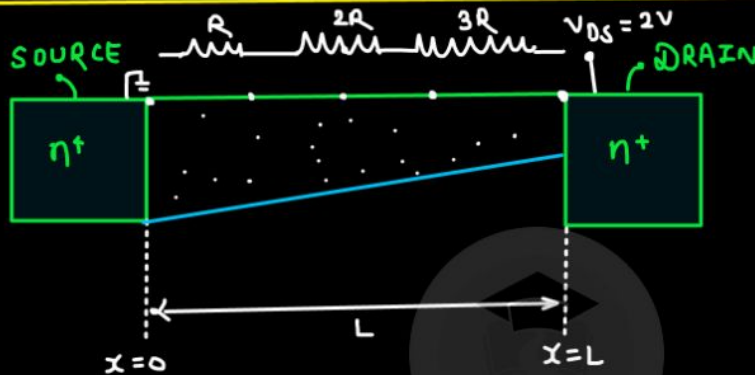
$$I_D = \frac{\mu_n C_{ox} W}{L} (V_{GS} - V_T) v_{DS} : \text{Linear Region}$$

Linear Relation b/w I_D and v_{DS}

↳ For low v_{DS}



↳ Effect of v_{DS} on channel and Inversion charge:-



Distribution of v_{DS}

$$v(x)|_{x=L} = v_{DS} = 2V$$

$$v(x)|_{x=0} = 0V$$

$v(x)$: Potential distribution of v_{DS}

When there was no v_{DS} (OR VERY LOW v_{DS}):-

$$Q_{in} \propto (V_{GS} - V_T)$$

$$Q_{in} = C_{ox} (V_{GS} - V_T) WL$$

Inversion Charge when there is significant V_{DS} :-

$$Q_{in} = C_{ox} [V_{GS} - V_T - V(x)] WL$$

at $x=L$:- $V(x=L) = V_{DS}$; $Q_{in}(x=L) = C_{ox} [V_{GS} - V_T - V_{DS}] WL$

$V(x=0) = 0$; $Q_{in}(x=0) = C_{ox} [V_{GS} - V_T] WL \rightarrow \text{unchanged}$

Now, when $V_{DS} = (V_{DS})_{sat}$ $\Rightarrow Q_{in}(x=L) = 0$

$$C_{ox} [V_{GS} - V_T - (V_{DS})_{sat}] = 0$$

$$\Rightarrow (V_{DS})_{sat} = V_{GS} - V_T \quad \checkmark$$

↳ For turning ON the MOSFET:-

$$V_{GS} > V_T, \quad V_{DS} > 0$$

↳ Achieving the Saturation:-

$$V_{GS} > V_T ; \quad V_{DS} = V_{GS} - V_T ; \quad \underbrace{V_{DS} > V_{GS} - V_T}_{\text{Saturation}}$$

↳ Current Eqⁿ in Linear region:-

$$I_D = \frac{\mu_n C_{ox} W}{L} (V_{GS} - V_T) V_{DS} : V_{DS} \text{ is very low} =$$

$$V_{DS} \ll V_{GS} - V_T$$

$$I_D = \frac{Q_{in}}{\tau} ; \quad \begin{matrix} Q_{in} \neq f(x) \\ \tau \neq f(x) \end{matrix}$$

$$\int_0^L dx = [x]_0^L = L$$

$$\int_0^L dV(x) = [V(x)]_0^L = V(L) - V(0) \\ = V_{DS} - 0 \\ = V_{DS}$$

$$\int_0^L V(x) \cdot dV(x) = \left[\frac{V^2(x)}{2} \right]_0^L = \frac{V^2(L)}{2} - \frac{V^2(0)}{2} = \frac{V_{DS}^2}{2}$$

$$I_D = \frac{\mu_n C_{ox} W}{L} \left\{ (V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right\} ; V_{DS} < V_{GS} - V_T$$

$$I_D = f(V_{GS}, V_{DS}) : \text{linear, sub-linear}$$

$$I_D = f(V_{GS}) ; I_D \neq f(V_{DS}) : \text{Saturation}$$

saturation current

$$(I_D)_{\text{sat}} = \frac{\mu_n C_{ox} W}{2L} (V_{GS} - V_T)^2$$

$$(V_{DS})_{\text{sat}} = V_{GS} - V_T$$

↳ MOSFET (NMOS) current Eqⁿ [Linear & sub-Linear Region]:-

$$I_D = \frac{\mu_n C_{ox} W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right] : \text{linear, sub-linear}$$

↳ for Saturation Region:-

saturation is achieved at $V_{DS} = (V_{DS})_{\text{sat}} = V_{GS} - V_T$

$$(I_D)_{\text{sat}} = \frac{\mu_n C_{ox} W}{2L} (V_{GS} - V_T)^2 : \text{saturation}$$

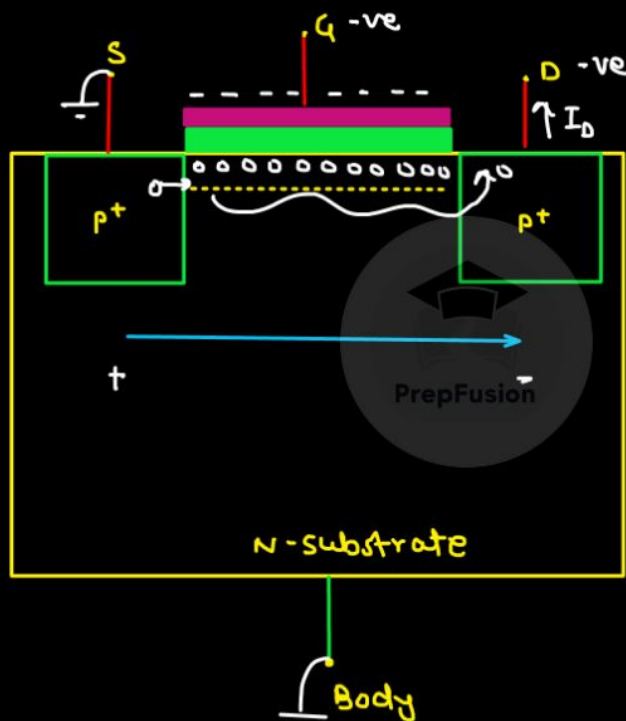
↳ $V_{GS} > V_T$, $V_{DS} > 0$: NMOS ON

↳ $V_{DS} > V_{GS} - V_T$: NMOS sat

↳ $V_{DS} < V_{GS} - V_T$: NMOS linear, sub-linear

$V_{GS} < V_T$: $I_D = 0$; NMOS cut off

* pMOS:- [p-channel Enhancement Type MosFET]



$$V_{DS} = V_D - V_S = -ve$$

$$V_{SD} = V_S - V_D = +ve$$

$$pMOS \begin{cases} V_{SD} = +ve & , V_{DS} = -ve \\ V_{SG} = +ve & , V_{GS} = -ve \\ V_T = -ve \end{cases}$$

$$NMOS \begin{cases} V_{DS} = +ve \\ V_{GS} = +ve & , V_T = +ve \end{cases}$$

current Eqⁿ:-

$$\text{Linear / Sub-linear} \rightarrow I_D = \frac{\mu_p C_{ox} W}{L} \left[(V_{SG} - |V_T|) V_{SD} - \frac{V_{SD}^2}{2} \right]$$

$$\text{Saturation} \rightarrow (I_D)_{sat} = \frac{\mu_p C_{ox} W}{2L} (V_{SG} - |V_T|)^2$$

	NMOS	pMOS
$I_D = 0$ Cut-off $\rightarrow$	$V_{GS} < V_T$	$V_{SG} < V_T $
Saturation $\rightarrow$	$V_{DS} \geq V_{GS} - V_T$	$V_{SD} \geq V_{SG} - V_T $
Current $\rightarrow$ [Linear / Sub-linear]	$I_D = \frac{\mu_n C_{ox} W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$	$I_D = \frac{\mu_p C_{ox} W}{L} \left[(V_{SG} - V_T) V_{SD} - \frac{V_{SD}^2}{2} \right]$
Current $\rightarrow$ [Saturation]	$(I_D)_{sat} = \frac{\mu_n C_{ox} W}{2L} (V_{GS} - V_T)^2$	$(I_D)_{sat} = \frac{\mu_p C_{ox} W}{2L} (V_{SG} - V_T)^2$
Higher Potential	Drain	Source
low Potential	Source	Drain
Current flow $\rightarrow$ direction	Drain to Source [Inside Mos]	Source to Drain [Inside Mos]

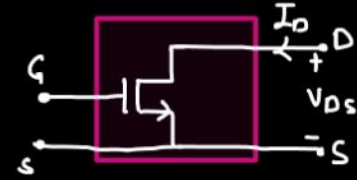
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YT Playl:

Ideal case:- (Saturation)

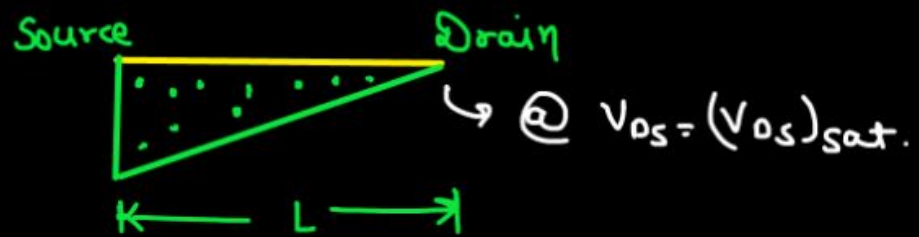
$$(I_D)_{sat} = \frac{\mu_n C_{ox} W}{2L} (V_{GS} - V_T)^2 \Rightarrow \text{constant for fixed } V_{GS} \text{ (Independent of } V_{DS})$$

Drain-Source ON Resistance:- (Saturation)

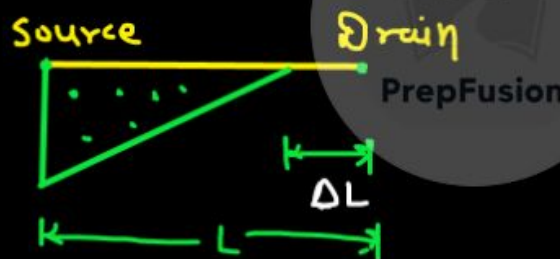
$$\frac{1}{r_{ds}} = \frac{\partial I_D}{\partial V_{DS}} = 0 \Rightarrow r_{ds} = \infty$$



Practically:-



Further increasing V_{DS} :-



$$(I_D)_{sat} = \frac{\mu_n C_{ox} W}{2(L - \Delta L)} (V_{GS} - V_T)^2$$



$$[(I_D)_{sat.}]_{CLM} = \underbrace{[(I_D)_{sat.}]_{ideal}} [1 + \lambda V_{DS}]$$

$\lambda \rightarrow$ process technology parameter (V^{-1})

Drain - Source ON resistance [Saturation] [CLM is present] :-

$$r_{ds} = \frac{1}{\frac{\partial I_D}{\partial V_{DS}}} = \frac{1}{\lambda [(I_D)_{sat.}]_{ideal}}$$

if $\lambda = 0 \Rightarrow$ NO CLM $\Rightarrow r_{ds} = \infty$
ideal case

Early voltage -

$$[(I_D)_{sat.}]_{CLM} = [(I_D)_{sat.}]_{ideal} [1 + \lambda V_{DS}]$$

$$= 0$$

$$\Rightarrow 1 + \lambda V_{DS} = 0$$

$$V_{DS} = -\frac{1}{\lambda}$$

$\Rightarrow$ The value of V_{DS} @ which the drain current becomes zero, That is known as early voltage.

$$V_A = \frac{1}{\lambda}$$

@ Higher V_{GS} , you get Higher value of $\frac{\partial I_D}{\partial V_{GS}}$

and that eventually gives more gain.

But, while using MOS as an amplifier, you can't keep $V_{GS}(dc)$ value to be very High. Why?

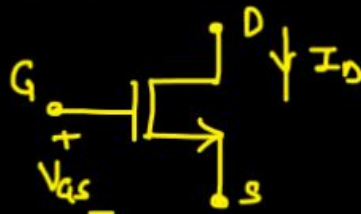
- (i) More power consumption
- (ii) MOS will fall out of saturation.

Concept of Transconductance:- (g_m)



$$g_m = \frac{\partial I_{out}}{\partial V_{in}}$$

considering NMOS:-



$$g_m = \frac{\partial I_D}{\partial V_{GS}}$$

Transconductance for Triode region:-

$$I_D = \frac{\mu_n C_{ox} W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}}$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \frac{\mu_n C_{ox} W}{L} [V_{DS} - 0]$$

$$(g_m)_{\text{Triode}} = \frac{\mu_n C_{ox} W}{L} \times V_{DS}$$

Transconductance for Saturation region:-

$$I_D = \frac{\mu_n C_{ox} W}{2L} (V_{GS} - V_T)^2 \quad \text{--- (1)}$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}}$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \frac{\mu_n C_{ox} W}{L} (V_{GS} - V_T) \quad \text{--- (2)}$$

From eqⁿ (1)

$$\frac{\partial I_D}{\partial V_{GS} - V_T} = \frac{\mu_n C_{ox} W}{L} (V_{GS} - V_T) = g_m$$

$$g_m = \frac{2 I_D}{V_{GS} - V_T} \quad \text{--- (3)}$$

$$I_D = \frac{\mu_n C_{ox} W}{2L} (V_{GS} - V_T)^2$$

$$2 \frac{\mu_n C_{ox} W}{L} I_D = \left(\frac{\mu_n C_{ox} W}{L} \right)^2 (V_{GS} - V_T)^2$$

$$\sqrt{2 \frac{\mu_n C_{ox} W}{L} I_D} = \frac{\mu_n C_{ox} W}{L} (V_{GS} - V_T) = g_m$$

$$g_m = \sqrt{2 \times \frac{\mu_n C_{ox} W}{L} I_D} \quad \text{--- (3)}$$

$$\checkmark (I_D)_{sat.} = \frac{\mu_n C_{ox} W}{2L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

$$(I_D)_{linear} = \frac{\mu_n C_{ox} W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

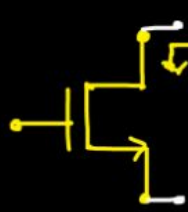
For saturation region:-

$$r_{ds} = \frac{\partial V_{DS}}{\partial I_D} = \frac{1}{\frac{\partial I_D}{\partial V_{DS}}}$$

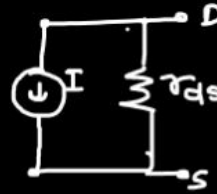
$$r_{ds} = \frac{1}{\lambda [I_D]_{sat-ideal}} \quad \text{practical}$$

$$r_{ds} = \infty \quad \text{ideal}$$

MOS in saturation region:-



practical
ideal



$$r_{ds} = \frac{1}{\lambda(I_D)_{sat} - ideal}$$



$$I = f(V_{GS})$$

⇒



For Triode region:-

$$I_D = \frac{\mu_n C_{ox} W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

For deep Triode region:-

$$I_D = \frac{\mu_n C_{ox} W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

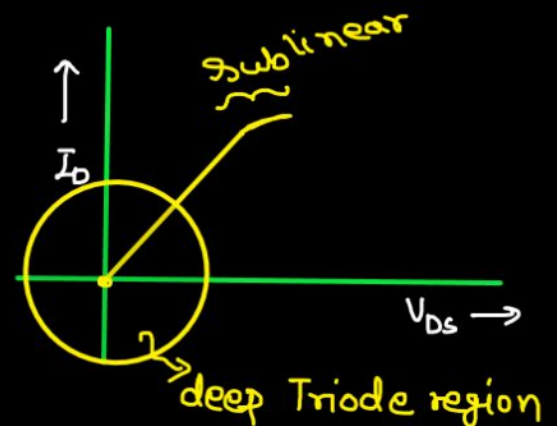
$$R_{on} = \frac{\partial V_{DS}}{\partial I_D} = \frac{1}{\partial I_D / \partial V_{DS}}$$

For deep Triode region,

V_{DS} is very small

$$\left[V_{DS} < \frac{(V_{GS} - V_T)}{2} \right]$$

$$V_{DS} \approx 0$$



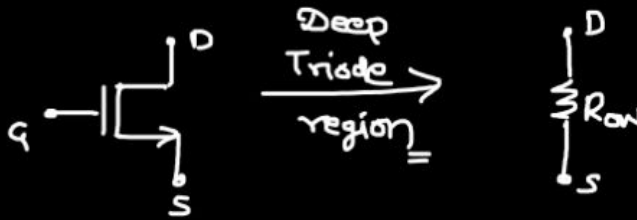
$$I_D = \frac{\mu_n C_{ox} W}{L} (V_{GS} - V_T) V_{DS}$$

$$\frac{\partial I_D}{\partial V_{DS}} = \frac{\mu_n C_{ox} W}{L} (V_{GS} - V_T)$$

$$R_{on} = \frac{1}{\frac{\partial I_D}{\partial V_{DS}}} = \frac{1}{\frac{\mu_n C_{ox} W}{L} (V_{GS} - V_T)}$$

PrepFusion

deep MOS in Triode region



* What happens with Threshold voltage of NMOS when body is at Negative bias?

+ve $V_{GS} > V_T \rightarrow$ Inversion layer ✓
S D +ve

Threshold voltage when $V_B = -ve$, $V_{SB} > 0$, $V_{SB} = +ve$

$$\phi_s = 2\phi_F + V_{SB}$$

$$V_{GB} = V_{FB} + \sqrt{\frac{2\epsilon_{si} q N_A (2\phi_F + V_{SB})}{C_{ox}}} + (2\phi_F + V_{SB})$$

$$\begin{aligned} V_{GB} - V_{SB} &= V_G - V_B - V_S + V_B \\ &= V_G - V_S \\ &= V_{GS} \end{aligned}$$

$$V_{GS} = V_{FB} + \sqrt{\frac{2\epsilon_{si} q N_A (2\phi_F + V_{SB})}{C_{ox}}} + 2\phi_F$$

$$V_T = V_{FB} + \underbrace{\sqrt{\frac{2\epsilon_{si} q N_A (2\phi_F)}{C_{ox}}}}_{V_{T0}} + 2\phi_F + \underbrace{\sqrt{\frac{2\epsilon_{si} q N_A (2\phi_F + V_{SB})}{C_{ox}}}}_{+}$$

$$V_T = V_{T0} + \frac{\sqrt{2\epsilon_{si} q N_A}}{C_{ox}} [\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}]$$

$$V_T = V_{T0} + \gamma [\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}] \quad : \text{NMOS} =$$

Gate to Source
Potential =

$$\gamma = \frac{\sqrt{2\epsilon_{si} q N_A}}{C_{ox}} \rightarrow \text{unit: } V^{1/2}$$

Body effect coefficient



PMOS:- Body: +ve, $V_S = 0$, $V_B = +ve$, $V_{BS} > 0$

$V_T = -ve$

$$V_T = V_{T0} - \gamma [\sqrt{2\phi_F + V_{BS}} - \sqrt{2\phi_F}]$$

Negative
positive

Q. An n-channel enhancement MOSFET is fabricated on a uniformly doped p-type silicon substrate whose acceptor concentration is $1 \times 10^{16} \text{ cm}^{-3}$. The intrinsic carrier concentration of silicon at 300 K is $1.5 \times 10^{10} \text{ cm}^{-3}$. The gate oxide thickness is 10 nm, and the permittivity of the gate oxide is $3.45 \times 10^{-13} \text{ F/cm}$. The permittivity of silicon is $11.7 \epsilon_0$, where $\epsilon_0 = 8.85 \times 10^{-14} \text{ F/cm}$. When the substrate (body) is kept at 0 V with respect to the source, the measured threshold voltage of the MOSFET is 0.60 V. Now the substrate is biased at a negative potential such that the source-to-body voltage becomes $V_{SB} = 0.40 \text{ V}$.

Determine the gate-to-source voltage required to reach strong inversion (i.e., just to turn ON the MOSFET) under this body-bias condition. Ans. $\rightarrow 0.636 \text{ Volt}$

↳

$$V_{T0} = 0.6 \text{ Volt}, \quad V_{SB} = 0.4 \text{ V}, \quad \text{NMOS} =$$

$$V_S - V_B = 0.4$$

$$V_B = -0.4 \text{ V} \rightarrow \text{Threshold voltage } \uparrow$$

$$V_T = V_{T0} + \gamma \left[\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F} \right]$$

$$V_T = 0.6 + 0.166 \left[\sqrt{1.094} - \sqrt{0.694} \right]$$

$$\phi_F = 0.0259 \ln \left(\frac{10^{16}}{1.5 \times 10^{16}} \right)$$

$$= 0.347 \text{ V}$$

$$V_T = 0.6353 \text{ V}$$



$$\gamma = \frac{\sqrt{2\epsilon_{Si} q N_A}}{C_{ox}} t_{ox}$$

$$= \frac{\sqrt{2 \times 11.7 \times 8.85 \times 10^{-14} \times 1.6 \times 10^{-19} \times 10^{16}}}{3.45 \times 10^{-13}} \times 10^{-6}$$

$$\gamma = 0.166 \text{ V}^{1/2}$$

Key Point

For Ge, energy band gap is given by,

$$E_g(T) = [0.785 - 2.23 \times 10^{-4} T] \text{ eV}$$

Semiconductor	$E_g(0^\circ\text{K})$	$E_g(300^\circ\text{K})$
Ge	0.785 eV	0.72 eV
Si	1.21 eV	1.1 eV

1.20 The longest wavelength that can be absorbed by silicon, which has the band gap of 1.12 eV, is $1.1 \mu\text{m}$. If the longest wavelength that can be absorbed by another material is $0.87 \mu\text{m}$, then the band gap of this material is

- (A) 1.416 eV (B) 0.886 eV
(C) 0.854 eV (D) 0.706 eV

1.20 (A)

Given :

- (i) $\lambda_1 = 1.1 \mu\text{m}$ when $E_{g_1} = 1.12 \text{ eV}$
(ii) $\lambda_2 = 0.87 \mu\text{m}$ when $E_{g_2} = ?$

Method 1

Band gap (in eV) and wavelength (in μm) are related by following equation,

$$E_g = \frac{1.24}{\lambda (\mu\text{m})} \text{ eV} = \frac{1.24}{0.87} \text{ eV}$$

$$E_g = 1.425 \text{ eV} \approx 1.416 \text{ eV}$$

Hence, the correct option is (A).

Method 2

Since, $E_g \propto \frac{1}{\lambda (\mu\text{m})}$

So,
$$\frac{E_{g_1}}{E_{g_2}} = \frac{\lambda_2}{\lambda_1}$$

$$\frac{1.12}{E_{g_2}} = \frac{0.87}{1.1}$$

$$E_{g_2} = 1.416 \text{ eV}$$

Hence, the correct option is (A).

Method 3

Since, $E_g \propto \frac{1}{\lambda (\mu\text{m})}$

Hence, for $\lambda_1 > \lambda_2$,

$$E_{g_2} > E_{g_1}$$

Only option (A) satisfies this condition.

Hence, the correct option is (A).

1.24 Under low level injection assumption, the injected minority carrier current for an extrinsic semiconductor is essentially the

- (A) diffusion current.
(B) drift current.
(C) recombination current.
(D) induced current.

1.24 (A)

Low level injection is a process which injects minority carrier concentration into an extrinsic semiconductor.

Generally, doping is used to increase majority carrier concentration and low level injection is used to increase minority carrier concentration. Assume uniformly doped n -type semiconductor, hence

$$n = N_D$$

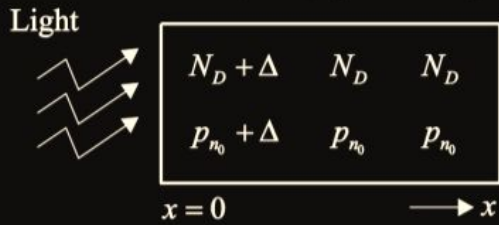
N_D	N_D	N_D	N_D	N_D
p_{n_0}	p_{n_0}	p_{n_0}	p_{n_0}	p_{n_0}

$x = 0 \quad \quad \quad \longrightarrow x$

Let minority carriers be p_n , then

$$p_n|_{x=0} = p_{n_0} = \frac{n_i^2}{N_D}$$

When light is illuminated at one side of semiconductor, the covalent bonds are broken. So, the electron-hole (e^- - h) pair are generated.



If Δ number of e^- and holes are generated, then total number of e^- becomes $N_D + \Delta$, where N_D is very large compared to Δ .

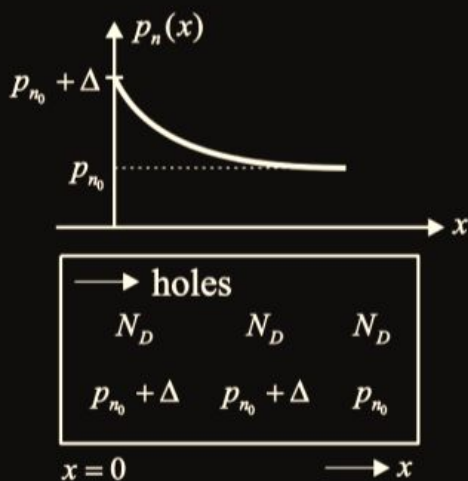
So, $N_D + \Delta \approx N_D$

So, the change in majority carrier is negligible. Compared to minority carrier Δ is very large. Hence, we cannot neglect Δ as compared to minority carrier.

At $x = 0$,

Total number of minority carrier = $p_n + \Delta$.

As x increases, the minority carrier will be reduced to p_{n_0} .

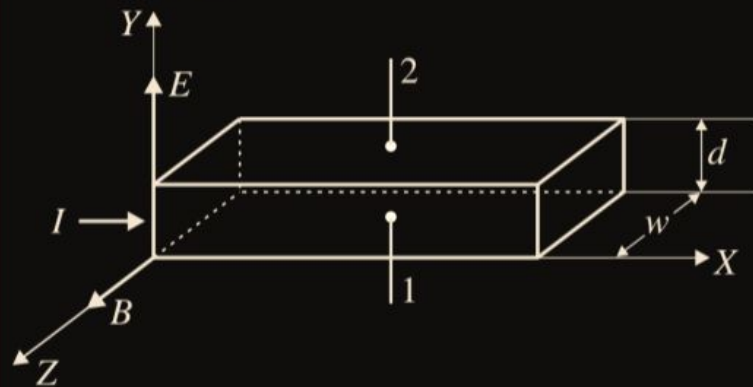


To establish equilibrium condition, the holes will move from higher concentration to the lower concentration, which is called diffusion. Because of low level injection, large number of minority carrier will be created and they will move from higher concentration to lower concentration due to diffusion process and the current resulting due to diffusion process is called diffusion current.

Hence, the correct option is (A).

Key Point

1. If a specimen (metal or semiconductor) carrying a current I is placed in a transverse magnetic field, an electric field is induced in the direction perpendicular to both I and B . This phenomenon is known as the **Hall effect**.



Hall voltage is given by,

$$V_H = E \times d = Bvd = \frac{BJd}{\rho} = \frac{BI}{\rho w}$$

Hall coefficient is given by,

$$R_H = \frac{1}{\rho} = \frac{V_H w}{BI}$$

2. Hall effect is used to determine,
 - (i) The type of semiconductor
 - (ii) Carrier concentration of specimen
 - (iii) Conductivity

1.25 (B)

According to Lorentz's law, at equilibrium

$$q\vec{E} + q(\vec{v} \times \vec{B}) = 0$$

$$\vec{E} + \vec{v} \times \vec{B} = 0$$

$$\vec{E} = -\vec{v} \times \vec{B}$$

$$\vec{E} = \vec{B} \times \vec{v}$$

Hence, the correct option is (B).

- 1.28 Silicon is doped with boron to a concentration of $4 \times 10^{17} \text{ atom/cm}^3$. Assume the intrinsic carrier concentration of silicon to be $1.5 \times 10^{10} / \text{cm}^3$ and the value of $\frac{kT}{q}$ to be 25 mV at 300°K .

Compared to undoped silicon, the Fermi level of doped silicon

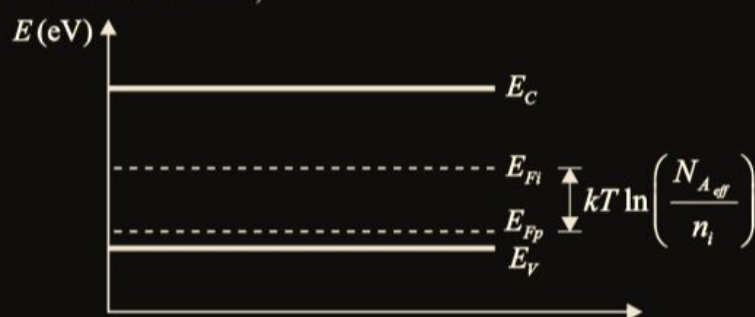
- (A) goes down by 0.13 eV.
 (B) goes up by 0.13 eV.
 (C) goes down by 0.427 eV.
 (D) goes up by 0.427 eV.

1.28 (C)

Given :

- (i) Boron is an acceptor atom so, silicon doped with boron will be of p -type semiconductor.
 (ii) Doping concentration,
 $N_{A_{\text{eff}}} = N_A = 4 \times 10^{17} \text{ atom/cm}^3$
 (iii) Intrinsic carrier concentration,
 $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$
 (iv) $\frac{kT}{q} = 25 \text{ mV at } 300^\circ \text{K}$

Energy band diagram of a p -type semiconductor is shown below,



Position of Fermi level in p -type semiconductor with respect to intrinsic semiconductor is given by,

$$E_{F_i} - E_{F_p} = kT \ln \frac{N_{A_{\text{eff}}}}{n_i} \text{ eV}$$

$$E_{F_i} - E_{F_p} = 25 \times 10^{-3} \ln \left[\frac{4 \times 10^{17}}{1.5 \times 10^{10}} \right]$$

$$E_{F_i} - E_{F_p} = 0.427 \text{ eV}$$

$$E_{F_p} = E_{F_i} - 0.427 \text{ eV}$$

Hence, from above expression we can see that compared to undoped silicon, Fermi level of doped silicon goes down by 0.427 eV.

Hence, the correct option is (C).

Key Point

1. If we increase the doping level, fermi level moves away from intrinsic fermi level.
2. If we increase the temperature, fermi level moves towards the intrinsic fermi level.
3. Shift in the position of fermi level with respect to intrinsic fermi level in n -type semiconductor is given by,

$$E_{F_n} - E_{F_i} = kT \ln \left(\frac{N_{D_{\text{eff}}}}{n_i} \right)$$

4. Probability of electron occupying donor energy level = $\frac{1}{1 + \frac{1}{g} e^{\left(\frac{E_D - E_F}{kT} \right)}}$
5. Probability of hole occupying acceptor energy level = $\frac{1}{1 + \frac{1}{g} e^{\left(\frac{E_F - E_A}{kT} \right)}}$

where,

g = Degeneracy factor

E_D = Donor energy level

E_F = Fermi energy level

k = Boltzmann constant

$$= 8.625 \times 10^{-5} \text{ eV/}^\circ \text{K}$$

E_A = Acceptor energy level

1.30

(A)

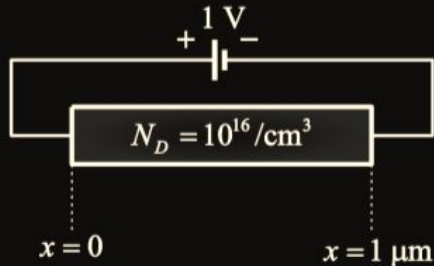
The Einstein's relation is given by,

$$\frac{D_p}{\mu_p} = \frac{D_n}{\mu_n} = V_T$$

$$\text{Unit of } \frac{\mu_n}{D_n} = \text{Unit of } \frac{1}{V_T} = (\text{Volt})^{-1}$$

Hence, the correct option is (A).

The silicon sample with unit cross-sectional area shown below is in thermal equilibrium. The following information is given : $T = 300^0 \text{ K}$, electronic charge $= 1.6 \times 10^{-19} \text{ C}$, thermal voltage $= 26 \text{ mV}$ and electron mobility $= 1350 \text{ cm}^2/\text{V}\cdot\text{s}$.



1.31 The magnitude of the electric field at $x = 0.5 \mu\text{m}$ is

- (A) 1 kV/cm (B) 5 kV/cm
(C) 10 kV/cm (D) 26 kV/cm

1.32 The magnitude of the electron drift current density at $x = 0.5 \mu\text{m}$ is

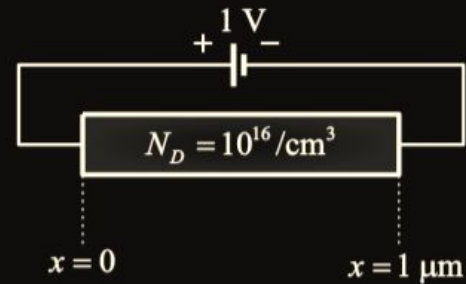
- (A) $2.16 \times 10^4 \text{ A/cm}^2$
(B) $1.08 \times 10^4 \text{ A/cm}^2$
(C) $4.32 \times 10^3 \text{ A/cm}^2$
(D) $6.48 \times 10^2 \text{ A/cm}^2$

1.31

(C)

Given :

- (i) $T = 300^0 \text{ K}$
(ii) Electronic charge, $q = 1.6 \times 10^{-19} \text{ C}$
(iii) Thermal voltage, $V_T = 26 \text{ mV}$
(iv) Electron mobility, $\mu_n = 1350 \text{ cm}^2/\text{V}\cdot\text{s}$
(v) Donor concentration, $N_D = 10^{16} / \text{cm}^3$



Method 1

The sample is in thermal equilibrium. Hence, at any distance from $x = 0$ (end of the bar) electric field will be constant.

$$E = \frac{V}{d} = \frac{1}{1 \times 10^{-6}} = 10^6 \text{ V/m}$$

$$E = 10 \text{ kV/cm}$$

Hence, the correct option is (C).

Method 2

Conductivity in n -type semiconductor is given by,

$$\sigma_n = N_D q \mu_n = 10^{16} \times 1.6 \times 10^{-19} \times 1350$$

$$\sigma_n = 2.16 \text{ mho/cm}$$

Length, $l = 1 \mu\text{m} = 10^{-4} \text{ cm}$

Area of cross-section, $A = \text{unit}$

Resistance is given by,

$$R = \frac{1}{\sigma} \frac{l}{A} = \frac{1}{2.16} \times \frac{10^{-4}}{A}$$

Current is given by,

$$I = \frac{V}{R} = \frac{2.16 \times A}{10^{-4}} \quad [V = 1 \text{ V}]$$

Current density is given by,

$$J = \frac{I}{A} = \frac{2.16 \times A}{10^{-4} \times A} = 2.16 \times 10^4 \text{ A/cm}^2$$

From ohm's law, $J = \sigma E$

$$E = \frac{J}{\sigma} = \frac{2.16 \times 10^4}{2.16} = 10^4 \text{ V/cm}$$

$$E = 10 \text{ kV/cm}$$

So, electric field is constant throughout the bar.
Hence, the correct option is (C).

1.32 (A)

Electron drift current density is given by,

$$J_n = \sigma_n E$$

where, σ_n = Electron conductivity

$$J_n = N_D q \mu_n E \quad [\sigma_n = N_D q \mu_n]$$

$$J_n = 10^{16} \times (1.6 \times 10^{-19}) \times 1350 \times (10 \times 10^3)$$

$$J_n = 2.16 \times 10^4 \text{ A/cm}^2$$

Hence, the correct option is (A).

Key Point

Concept of complete impurity ionization :

- (i) For n -type semiconductor, the complete impurity ionization is a process in which all donor atoms present in donor states (E_D) donate an electron to conduction band.

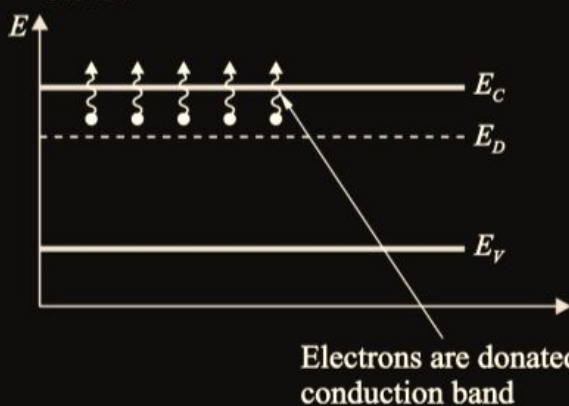


Fig. Energy band diagram of complete ionization of donor states

- (ii) For p -type semiconductor, the complete impurity ionization is a process in which all acceptor atoms present in acceptor states (E_A) accept an electron from valence band.

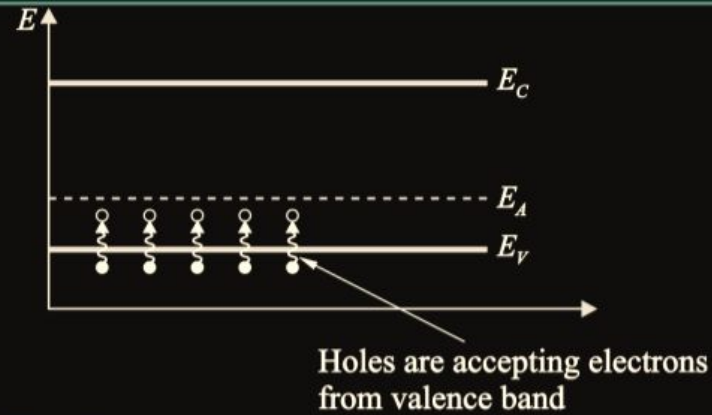


Fig. Energy band diagram of complete ionization of acceptor states

- 1.35** Assume electronic charge $q = 1.6 \times 10^{-19} \text{ C}$, $kT/q = 25 \text{ mV}$ and electron mobility $\mu_n = 1000 \text{ cm}^2/\text{V-s}$. If the concentration gradient of electrons injected into a p -type silicon sample is $1 \times 10^{21} / \text{cm}^4$, the magnitude of electron diffusion current density (in A/cm^2) is

1.35 4000

Given :

- (i) $q = 1.6 \times 10^{-19} \text{ C}$, $\frac{kT}{q} = 25 \text{ mV}$
 (ii) $\mu_n = 1000 \text{ cm}^2/\text{V-sec}$
 (iii) $\frac{dn}{dx} = 1 \times 10^{21} / \text{cm}^4$

Einstein's relation is given by,

$$\frac{D_n}{\mu_n} = V_T$$

$$D_n = \mu_n V_T = 1000 \times 25 \times 10^{-3}$$

$$D_n = 25 \text{ cm}^2/\text{sec}$$

Electron diffusion current density is given by,

$$J_n = q D_n \frac{dn}{dx}$$

$$J_n = 1.6 \times 10^{-19} \times 25 \times 10^{21}$$

$$J_n = 4 \times 10^3 \text{ A/cm}^2$$

Hence, the magnitude of electron diffusion current density is **4000 A/cm²**.

1.36 A thin p -type silicon sample is uniformly illuminated with light which generates excess carriers. The recombination rate is directly proportional to **[Set - 03]**

- (A) the minority carrier mobility.
- (B) the minority carrier recombination life time.
- (C) the majority carrier concentration.
- (D) the excess minority carrier concentration.

1.36 (D)

Let n_0 and p_0 are concentrations of electrons and holes in equilibrium.

When light is exposed then more covalent bond will be broken.

Let, after illumination of light, number of generated electron is Δn and number of generated hole is Δp .

$$\Delta n = \Delta p$$

Hence, electron concentration becomes,

$$n = n_0 + \Delta n$$

and hole concentration becomes, $p = p_0 + \Delta p$

Recombination rate is given by,

$$g = \frac{\Delta n}{\tau_n} = \frac{\Delta p}{\tau_p} \quad \dots (i)$$

where, Δn = Excess minority carrier concentration, Δp = Excess majority carrier concentration,

τ_n = Mean life time of excess minority electrons in p -type semiconductor,

τ_p = Mean life time of holes.

Hence, the recombination rate is directly proportional to the excess minority carrier concentration.

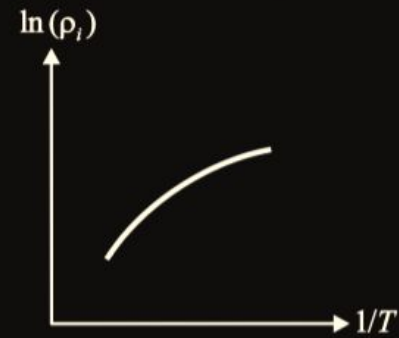
Hence, the correct option is (D).

$$E_{\min} (\text{eV}) = \frac{1.24}{\lambda_{\max} (\mu\text{m})}$$

$$\lambda_{\max} (\mu\text{m}) = \frac{1.24}{E_{\min} (\text{eV})} = \frac{1.24}{1.42} = 0.874 \mu\text{m}$$

1.39

In the figure, $\ln(\rho_i)$ is plotted as a function of $\frac{1}{T}$, where ρ_i is the intrinsic resistivity of silicon, T is the temperature, and the plot is almost linear.



The slope of the line can be used to estimate

[Set - 04]

- (A) band gap energy of silicon (E_g).
- (B) sum of electron and hole mobility in silicon ($\mu_n + \mu_p$).
- (C) reciprocal of the sum of electron and hole mobility in silicon $(\mu_n + \mu_p)^{-1}$.
- (D) intrinsic carrier concentration of silicon (n_i).

Intrinsic concentration of a semiconductor is given by,

$$n_i^2 = A_0 T^3 e^{-E_{g_0}/kT}$$

$$n_i = \sqrt{A_0} T^{3/2} e^{-E_{g_0}/2kT} \quad \dots (i)$$

Intrinsic conductivity of a semiconductor is given by,

$$\sigma_i = q n_i (\mu_n + \mu_p)$$

Intrinsic resistivity of a semiconductor is given by,

$$\rho_i = \frac{1}{\sigma_i} = \frac{1}{q n_i (\mu_n + \mu_p)} \quad \dots (ii)$$

From equation (i) and equation (ii),

$$\rho_i = \frac{1}{q (\mu_n + \mu_p) \sqrt{A_0} T^{3/2} e^{-E_{g_0}/2kT}}$$

$$\rho_i = \frac{1}{q \sqrt{A_0} (\mu_n + \mu_p)} T^{\frac{-3}{2}} e^{E_{g_0}/2kT}$$

$$\ln(\rho_i) = \ln \left[\frac{1}{q \sqrt{A_0} (\mu_n + \mu_p)} \right] - \frac{3}{2} \ln T + \frac{E_{g_0}}{2kT}$$

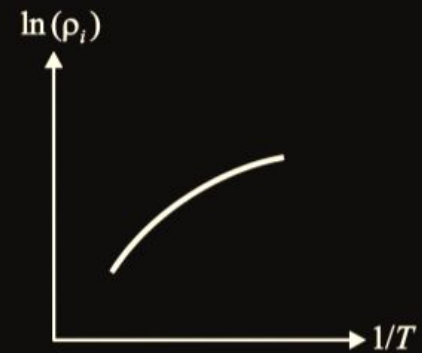
In above expression, with variation in temperature, variation in term $\frac{3}{2} \ln T$ is very small. Hence, it is considered as constant.

$$\text{Let, } \ln \left[\frac{1}{q \sqrt{A_0} (\mu_n + \mu_p)} \right] - \frac{3}{2} \ln T = C$$

$$\text{Then, } \ln(\rho_i) = \frac{E_{g_0}}{2k} \left(\frac{1}{T} \right) + C$$

Compare above equation with $y = mx + C$,

$$y = \ln \rho_i, \quad m = \frac{E_{g_0}}{2k}, \quad x = \frac{1}{T}$$



$$\text{Slope} = m = \frac{dy}{dx} = \frac{E_{g_0}}{2k}$$

Hence, slope can be used to estimate the band gap E_g .

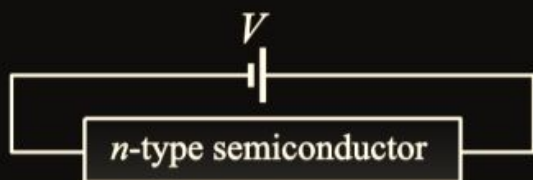
Hence, the correct option is (A).

Key Point

- (i) As temperature increases, E_g decreases and electron hole pair generation increases and hence σ increases, therefore ρ_i decreases.
- (ii) Similarly, as temperature decreases, E_g increases and electron hole pair generation decreases, hence σ decreases and therefore ρ_i increases.

So, ρ_i increase as $\frac{1}{T}$ increases.

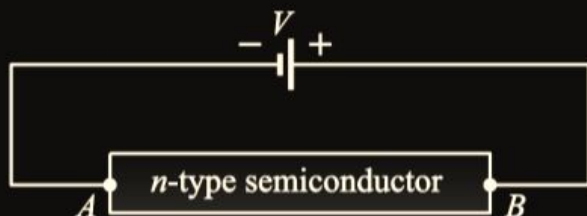
1.42 An n -type semiconductor having uniform doping is biased as shown in the figure.
[Set - 04]



If E_C is the lowest energy level of the conduction band, E_V is the highest energy level of the valance band and E_F is the Fermi level, which one of the following represents the energy band diagram for the biased n -type semiconductor?

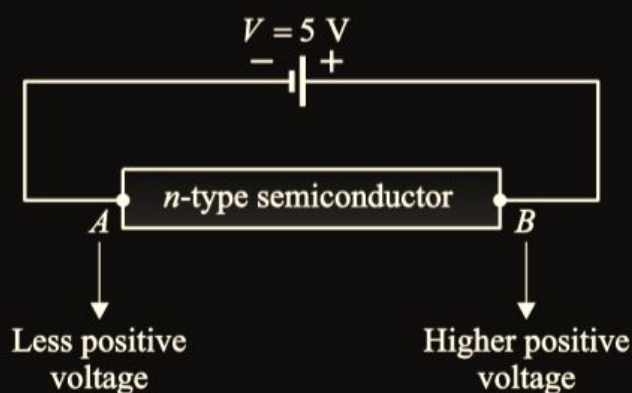
1.42 (D)

Given circuit is shown below,

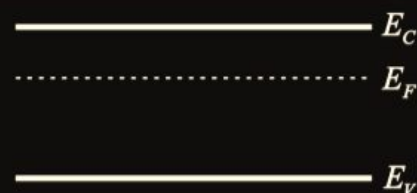


Let $V = 5 \text{ V}$, then terminal A will be at less positive voltage i.e. 0 volt and terminal B will be at higher positive voltage i.e. 5 volt. Hence, it looks like, terminal A will be less reverse biased and terminal B will be more reverse biased.

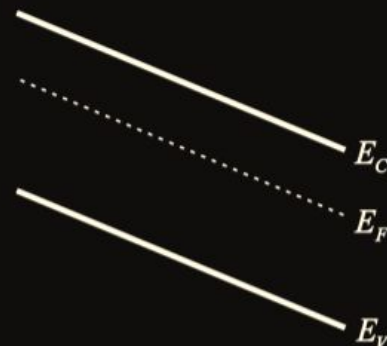
As reverse bias will increase, energy levels will move down. Therefore, height of energy level towards terminal A will be higher and height of energy level toward terminal B will be lower.



Energy band diagram, before application of voltage V , is shown below,



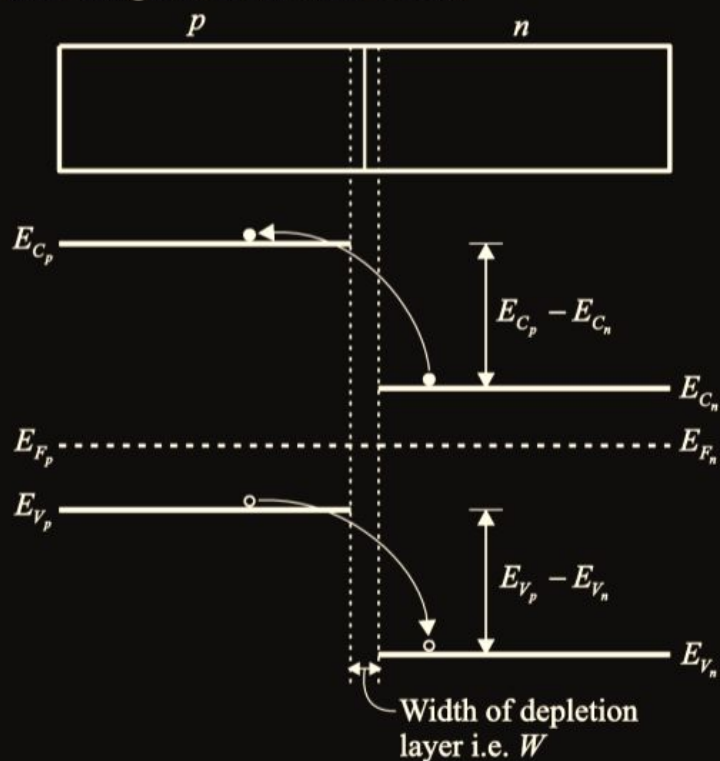
Energy band diagram after application of voltage V is shown below,



Hence, the correct option is (D).

Key Point

An open circuited p - n diode with its energy band diagram is shown below,



where, $E_{C_p} - E_{C_n}$ and $E_{V_p} - E_{V_n}$ are height of potential barrier.

$$(E_{C_p} - E_{C_n}) \propto \text{built-in potential i.e. } V_{bi}$$

$$(E_{V_p} - E_{V_n}) \propto \text{built-in potential i.e. } V_{bi}$$

When forward biased is applied to p - n diode, then height of potential barrier and width of

depletion layer will decrease therefore, E_{C_p} will move down and E_{C_n} will move up. Hence, $E_{C_p} - E_{C_n}$ will decrease and in same way $E_{V_p} - E_{V_n}$ will also decrease. Therefore, electron from E_{C_n} will jump to E_{C_p} and hole from E_{V_p} will jump to E_{V_n} .

1.46 The energy band diagram and the electron density profile $n(x)$ in a semiconductor are shown in the figures.

Assume that $n(x) = 10^{15} e^{\left(\frac{q\alpha x}{kT}\right)} \text{ cm}^{-3}$, with $\alpha = 0.1 \text{ V/cm}$ and x expressed in cm.

Given $\frac{kT}{q} = 0.026 \text{ V}$, $D_n = 36 \text{ cm}^2 \text{ s}^{-1}$

and $\frac{D}{\mu} = \frac{kT}{q}$. The electron current density (in A/cm^2) at $x = 0$ is [Set - 02]

1.46 (C)

Given :

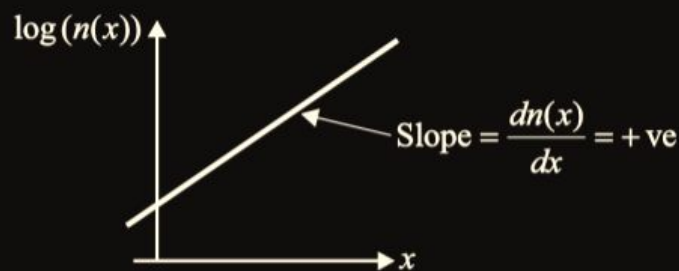
(i) $n(x) = 10^{15} e^{\left(\frac{q\alpha x}{kT}\right)} \text{ cm}^{-3}$

(ii) $\alpha = 0.1 \text{ V/cm}$, $\frac{kT}{q} = 0.026 \text{ V}$

(iii) $D_n = 36 \text{ cm}^2 \text{ s}^{-1}$, $\frac{D}{\mu} = \frac{kT}{q}$

(iv) $\mu_n = \frac{D_n}{kT/q} = \frac{D_n}{V_T} = \frac{36}{0.026} \text{ cm}^2/\text{V-sec}$

$$n(x) = 10^{15} e^{\left(\frac{q\alpha x}{kT}\right)} \text{ cm}^{-3} \quad \dots \text{ (i)}$$



$$\frac{dn(x)}{dx} = 10^{15} \times \frac{q\alpha}{kT} e^{q\alpha x/kT}$$

$$\left. \frac{dn(x)}{dx} \right|_{x=0} = 10^{15} \times e^0 \times \frac{0.1}{0.026}$$

$$\left. \frac{dn(x)}{dx} \right|_{x=0} = 10^{15} \times \frac{100}{26} = \frac{10^{17}}{26}$$

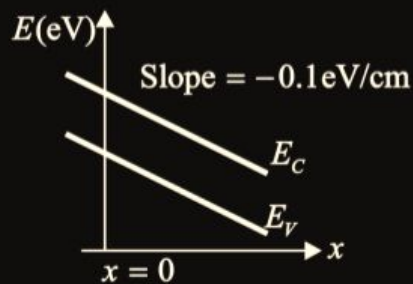
For non-uniform semiconductor, the electron are moving from higher concentration to lower concentration. Hence, electron diffusion current density is given by,

$$J_{n_1}(x)\Big|_{x=0} = qD_n \frac{dn(x)}{dx}\Big|_{x=0}$$

$$J_{n_1}(0) = 1.6 \times 10^{-19} \times 36 \times \frac{10^{17}}{26}$$

$$J_{n_1}(0) = \frac{1.6 \times 36}{26} \times 10^{-2}$$

$$J_{n_1}(0) = 2.21 \times 10^{-2} \text{ A/cm}^2 \quad \dots \text{ (ii)}$$



In above figure, slope of energy bands indicates applied electric field. Hence,

Electric field, $E(x) = -0.1 \text{ V/cm}$

Electron drift current density is given by,

$$J_{n_2}(x)\Big|_{x=0} = n(x)q\mu_n E(x)\Big|_{x=0}$$

From equation (i),

$$n(0) = 10^{15} / \text{cm}^3$$

$$\text{Hence, } J_{n_2}(0) = 10^{15} \times 1.6 \times 10^{-19} \times \frac{36}{0.026} \times (-0.1)$$

$$J_{n_2}(0) = -2.21 \times 10^{-2} \text{ A/cm}^2 \quad \dots \text{ (iii)}$$

From equation (ii) and (iii), total electron current density at $x=0$ is given by,

$$J_n(0) = J_{n_1}(0) + J_{n_2}(0)$$

$$J_n(0) = 2.21 \times 10^{-2} + (-2.21 \times 10^{-2})$$

$$J_n(0) = 0 \text{ A/cm}^2$$

Hence, the correct option is (C).

1.51 A bar of Gallium Arsenide (GaAs) is doped with Silicon such that the Silicon atoms occupy Gallium and Arsenic sites in the GaAs crystal. Which one of the following statements is true? [Set - 01]

- (A) Silicon atoms act as *p*-type dopants in Arsenic sites and *n*-type dopants in Gallium sites.
- (B) Silicon atoms act as *n*-type dopants in Arsenic sites and *p*-type dopants in Gallium sites.
- (C) Silicon atoms act as *p*-type dopants in Arsenic as well as Gallium sites.
- (D) Silicon atoms act as *n*-type dopants in Arsenic as well as Gallium sites.

1.51 (A)

Given : GaAs is doped with Si
i.e. Si atom is doped into GaAs.

Case 1 :

Si+Ga (G-III) : Since Ga is G-III element, So number of valence electron in Ga is 3 but it requires 4 valence electrons to be as a part of compound semi-conductor.

Ga will take one electron from Si. Therefore, Si will donate one electron to GaAs crystal. Since Si is donating the electron, Hence Si is acting as *n*-type dopant.

Case 2 :

Si+As (G-V) : Since As is G-V element, So number of valence electron in As is 5 but it requires only 4 valence electron to be as a part of compound semiconductor.

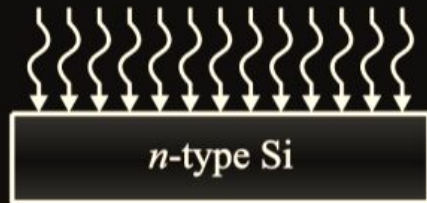
So, As will loose one electron and Si accepts one electron.

Due to this there is a vacancy created in crystal that is treated as hole. Since, Si is able to create a hole in crystal it means Si is acting as *p*-type dopant.

[Characteristic of compound semiconductor is each elements present in compound semiconductor should have 4 valence electrons.]

Hence, the correct option is (A).

1.49 Consider a silicon sample at $T = 300^\circ \text{K}$, with a uniform donor density $N_D = 5 \times 10^{16} \text{ cm}^{-3}$, illuminated uniformly such that the optical generation rate is $= 1.5 \times 10^{20} \text{ cm}^{-3} \text{ s}^{-1}$ throughout the sample. The incident radiation is turned off at $t = 0$. Assume low-level injection to be valid and ignore surface effects. The carrier lifetimes are $\tau_{p0} = 0.1 \mu\text{s}$ and $\tau_{n0} = 0.5 \mu\text{s}$.



The hole concentration at $t = 0$ and the hole concentration at $t = 0.3 \mu\text{s}$, respectively, are

[Set - 01]

- (A) $1.5 \times 10^{13} \text{ cm}^{-3}$ and $7.47 \times 10^{11} \text{ cm}^{-3}$.
- (B) $1.5 \times 10^{13} \text{ cm}^{-3}$ and $8.23 \times 10^{11} \text{ cm}^{-3}$.
- (C) $7.5 \times 10^{13} \text{ cm}^{-3}$ and $3.73 \times 10^{11} \text{ cm}^{-3}$.
- (D) $7.5 \times 10^{13} \text{ cm}^{-3}$ and $4.12 \times 10^{11} \text{ cm}^{-3}$.

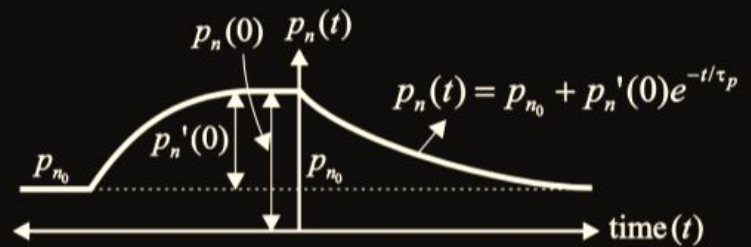
1.49 (A)

Given :

- (i) $T = 300^\circ \text{K}$
- (ii) Donor density, $N_D = 5 \times 10^{16} / \text{cm}^3$
- (iii) Optical generation rate,
 $G_{opt} = 1.5 \times 10^{20} / \text{cm}^3 \cdot \text{s}$,
- (iv) Hole carrier life time, $\tau_{p0} = 0.1 \mu\text{s}$
- (v) Electron carrier life time, $\tau_{n0} = 0.5 \mu\text{s}$

Consider a n -type semiconductor having ' p_{n0} ' hole concentration at equilibrium and illuminated by light.

$$p_n(0) = p_n'(0) + p_{n0} \quad \dots (i)$$



where, $p_n(0)$ is total hole concentration at $t = 0$, after illumination.

p_{n0} is thermally generated hole concentration before illumination.

$p_n'(0)$ is excess hole concentration at $t = 0$, due to illumination.

$p_n(t)$ = Hole concentration at any time t , after illumination is stopped and it is given by,

$$p_n(t) = p_{n0} + p_n'(0)e^{-t/\tau_p} \quad \dots (ii)$$

Since, $p_n'(0) = G_{opt} \times \tau_{p0}$

$$p_n'(0) = (1.5 \times 10^{20} \times 0.1 \times 10^{-6}) / \text{cm}^3$$

$$p_n'(0) = 1.5 \times 10^{13} / \text{cm}^3$$

From mass action law,

$$np = n_i^2$$

where, $p = p_{n0}$ and $n = N_D$

$$\text{Hence, } p_{n0} = \frac{n_i^2}{N_D}$$

$$p_{n_0} = \frac{n_i^2}{N_D} = \left[\frac{(1.5 \times 10^{10})^2}{5 \times 10^{16}} \right] / \text{cm}^3$$

$$p_{n_0} = 0.45 \times 10^4 / \text{cm}^3$$

- (a) From equation (i), At $t = 0$ sec, hole concentration is given by,

$$p_n(0) = p_n'(0) + p_{n_0}$$

$$p_n(0) = (1.5 \times 10^{13} + 0.45 \times 10^4) / \text{cm}^3$$

$$p_n(0) \approx (1.5 \times 10^{13}) / \text{cm}^3$$

- (b) From equation (ii), At $t = 0.3 \mu\text{sec}$, hole concentration is given by,

$$p_n(0.3) = p_{n_0} + p_n'(0)e^{-0.3 \times 10^{-6} / \tau_{p_0}}$$

$$p_n(0.3) = [0.45 \times 10^4 + 1.5 \times 10^{13} \times e^{\left(\frac{-0.3 \times 10^{-6}}{0.1 \times 10^{-6}} \right)}] / \text{cm}^3$$

$$p_n(0.3) \approx 7.46 \times 10^{11} / \text{cm}^3$$

Hence, the correct option is (A).

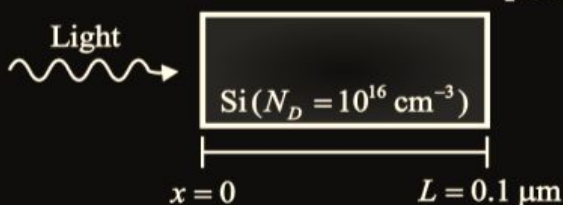
- 1.53** As shown, a uniformly doped Silicon (Si) bar of length $L = 0.1 \mu\text{m}$ with a donor concentration $N_D = 10^{16} \text{ cm}^{-3}$ is illuminated at $x = 0$ such that electron and hole pairs are generated at the rate of

$$G_L = G_{L_0} \left(1 - \frac{x}{L} \right), 0 \leq x \leq L,$$

where $G_{L_0} = 10^{17} / \text{cm}^3 \text{ s}$.

Hole life time is 10^{-4} s electronic charge $q = 1.6 \times 10^{-19} \text{ C}$, hole diffusion coefficient $D_p = 100 \text{ cm}^2 / \text{s}$ and low level injection condition prevails. Assuming a linearly decaying steady state excess hole concentration that goes to 0 at $x = L$, the magnitude of the diffusion current density at $x = L/2$, in A/cm^2 , is _____.

[Set - 01]



Given :

- (i) Electron-hole pairs generation rate,

$$G_L = G_{L_0} \left(1 - \frac{x}{L} \right)$$

- (ii) $G_{L_0} = 10^{17} / \text{cm}^3 \text{ s}$

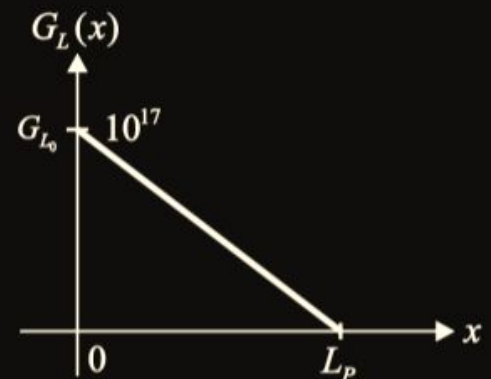
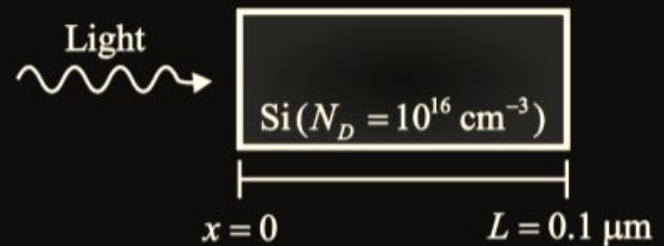
- (iii) Length of the bar, $L = 0.1 \times 10^{-4} \text{ cm}$

- (iv) Hole diffusion coefficient,

$$D_p = 100 \text{ cm}^2 / \text{sec}$$

- (v) Hole life time, $\tau_p = 10^{-4} \text{ sec}$,

- (vi) Electronic charge, $q = 1.6 \times 10^{-19} \text{ C}$



Hole diffusion current density is given by,

$$J_p = -qD_p \left. \frac{dp}{dx} \right|_{x=L/2} \dots (i)$$

Concentration of hole in Si bar is given by,

$$p = G_L \times \tau_p$$

$$p = G_{L_0} \left(1 - \frac{x}{L}\right) \times 10^{-4} = 10^{17} \left(1 - \frac{x}{L}\right) \times 10^{-4}$$

$$p = 10^{13} \left(1 - \frac{x}{L}\right) / \text{cm}^3$$

Hence, $\frac{dp}{dx} = 10^{13} \left(0 - \frac{1}{L}\right) = \frac{-10^{13}}{L}$

$$\frac{dp}{dx} = \frac{-10^{13}}{0.1 \times 10^{-4}} = -10^{18} / \text{cm}^4 \quad \dots \text{(ii)}$$

Since, hole concentration gradient is independent of length of Si bar hence, current density will be independent of the length of the silicon bar.

From equations (i) and (ii),

$$J_p = -1.6 \times 10^{-19} \times 100 \times (-10^{18})$$

$$J_p = 16 \text{ A/cm}^2$$

Hence, the magnitude of the diffusion current density at $x = L/2$ is **16 A/cm²**.

1.55 A single crystal intrinsic semiconductor is at a temperature of 300 K with effective density of states for holes twice that of electrons. The thermal voltage is 26 mV.

The intrinsic Fermi level is shifted from mid-bandgap energy level by

- (A) 9.01 meV (B) 13.45 meV
(C) 18.02 meV (D) 26.90 meV

1.55 (A)

Given : At $T = 300 \text{ K}$, the effective density of states for holes is twice that of electrons i.e., $N_v = 2N_c$ and thermal voltage is 26 mV.

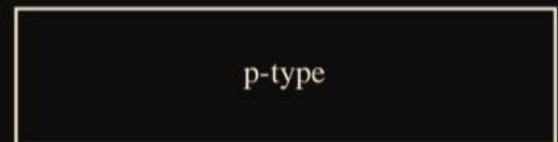
Location of Fermi level is given by,

$$E_F = \frac{E_C + E_V}{2} - \frac{KT}{2} \ln \frac{N_c}{N_v}$$

The difference of Fermi level from mid band gap energy level is

$$E_d = \frac{-KT}{2} \ln \left(\frac{1}{2}\right) = 9.01 \text{ meV}$$

1.56 The energy band diagram of a p -type semiconductor bar of length L under equilibrium condition (i.e. the Fermi energy levels E_F is constant) is shown in the figure. The valance band E_V is sloped since doping is non-uniform along the bar. The difference between the energy levels of the valance band at the two edges of the bar is Δ .



If the charge of an electron is q then the magnitude of the electric field developed inside this semiconductor bar is

- (A) $\frac{\Delta}{qL}$ (B) $\frac{2\Delta}{qL}$
(C) $\frac{\Delta}{2qL}$ (D) $\frac{3\Delta}{2qL}$

1.56 (A)

From questions it is clear that, change in energy level / difference in energy level of E_C and E_V is Δ .



$$E = -\frac{dV}{dx}$$

$$\therefore E_C = -qV$$

$$E = +\frac{1}{q} \frac{dE_C}{dx} \quad \dots \text{(i)}$$

dE_c = Change in energy level = Δ

1.58 (C)

dx = Length of bar = L

Hence, equation (i) becomes as

$$E = \frac{+\Delta}{qL}$$

Hence, the correct option is (A).

1.58 Consider a long rectangular bar of direct bandgap p -type semiconductor. The equilibrium hole density is 10^{17} cm^{-3} the intrinsic carrier concentration is 10^{10} cm^{-3} . Electron and hole diffusion lengths are $2 \mu\text{m}$ and $1 \mu\text{m}$, respectively.

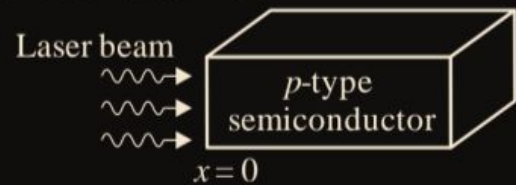
The left side of the bar ($x=0$) is uniformly illuminated with a laser having photon energy greater than the bandgap of the semiconductor. Excess electron-hole pairs are generated ONLY at $x=0$ because of the laser. The steady state electron density at $x=0$ is 10^{14} cm^{-3} due to laser illumination. Under these conditions and ignoring electric field, the closest approximation (among the given options) of the steady state electron density at $x=2 \mu\text{m}$, is

- (A) $0.63 \times 10^{13} \text{ cm}^{-3}$
- (B) $3.7 \times 10^{14} \text{ cm}^{-3}$
- (C) $0.37 \times 10^{14} \text{ cm}^{-3}$
- (D) 10^3 cm^{-3}

Given :

- (i) p -type semiconductor
- (ii) Equilibrium hole density,
 $p_0 = 10^{17} \text{ cm}^{-3}$
- (iii) Intrinsic carrier concentration,
 $n_i = 10^{10} \text{ cm}^{-3}$
- (iv) Electron diffusion length, $L_n = 2 \mu\text{m}$
- (v) Hole diffusion length, $L_p = 1 \mu\text{m}$
- (vi) Excess electron density at $x=0$,
 $n_p(0) = 10^{14} \text{ cm}^{-3}$

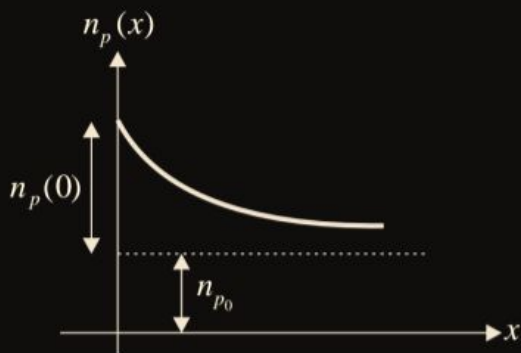
Consider a bar of direct bandgap p -type semiconductor with light is illuminated at left side of the bar at $x=0$.



From **Mass action law**, $np = n_i^2$

Electron or minority carrier concentration is given by,

$$n_{p_0} = \frac{n_i^2}{p_0} = \frac{(10^{10})^2}{10^{17}} = \frac{10^{20}}{10^{17}}$$
$$n_{p_0} = 10^3 \text{ cm}^{-3}$$



Electron concentration at any distance 'x' is given by,

$$n_p(x) = n_{p0} + n_p(0)e^{-x/L_n}$$

$$n_p(x) = 10^3 + 10^{14}e^{-x/L_n}$$

Electron concentration (density) at $x = 2 \mu\text{m}$ is,

$$n_p(x=2) = 10^3 + 10^{14}e^{-2/2}$$

$$n_p(x=2) = 10^3 + 10^{14}e^{-1}$$

$$n_p(x=2) = 10^3 + 10^{14} \times 0.37$$

$$n_p(x=2) \cong 0.37 \times 10^{14} \text{ cm}^{-3}$$

Hence, the correct option is (C).

1.59 In a non-degenerate bulk semiconductor with electron density $n = 10^{16} \text{ cm}^{-3}$, the value of $E_C - E_{Fn} = 200 \text{ meV}$, where E_C and E_{Fn} denote the bottom of the conduction band energy and electron Fermi level energy, respectively. Assume thermal voltage as 26 meV and the

intrinsic carrier concentration is 10^{10} cm^{-3} . For $n = 0.5 \times 10^{16} \text{ cm}^{-3}$, the closest approximation of the value of $(E_C - E_{Fn})$, among the given options, is

(A) 174 meV (B) 182 meV

(C) 226 meV (D) 218 meV

Given :

(i) Electron density, $n_1 = 10^{16} \text{ cm}^{-3}$

(ii) $(E_C - E_{Fn})_1 = 200 \text{ meV}$

(iii) Intrinsic carrier concentration, $n_i = 10^{10} \text{ cm}^{-3}$

(iv) We have to find $(E_C - E_{Fn})_2$ for electron density, $n_2 = 0.5 \times 10^{16} \text{ cm}^{-3}$

Since, electron density for n -type semiconductor is given by,

$$n = N_c e^{\frac{-(E_C - E_{Fn})}{kT}}$$

Where, n = Electron density, N_c = Effective density of states at the edge of conduction band.

$$n_1 = N_c e^{\frac{-(E_C - E_{Fn})_1}{kT}} \quad \dots(i)$$

$$n_2 = N_c e^{\frac{-(E_C - E_{Fn})_2}{kT}} \quad \dots(ii)$$

Dividing equations (i) and (ii),

$$\frac{n_1}{n_2} = \frac{N_c e^{\frac{-(E_C - E_{Fn})_1}{kT}}}{N_c e^{\frac{-(E_C - E_{Fn})_2}{kT}}}$$

$$\frac{10^{16}}{0.5 \times 10^{16}} = e^{\frac{(E_C - E_{Fn})_2 - (E_C - E_{Fn})_1}{kT}}$$

$$2 = e^{\frac{(E_C - E_{Fn})_2 - 200 \times 10^{-3}}{26 \times 10^{-3}}}$$

$$(E_C - E_{Fn})_2 - 200 \times 10^{-3} = 26 \ln 2$$

$$(E_C - E_{Fn})_2 = 26 \ln 2 + 200 = 218 \text{ meV}$$

Hence, the correct option is (D).

1.64 For non-degenerately doped n - type silicon, which one of the following plots represents the temperature (T) dependence of free electron concentration (n) ?

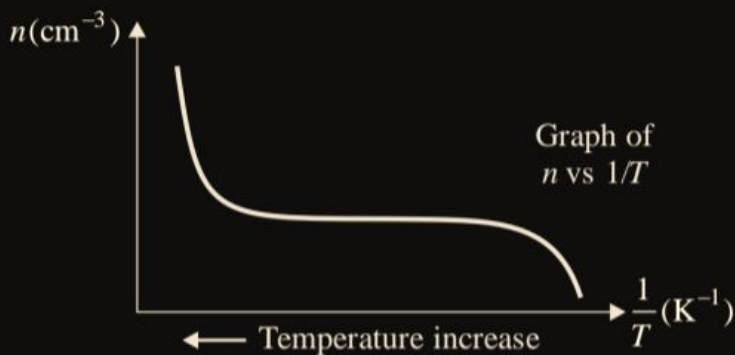
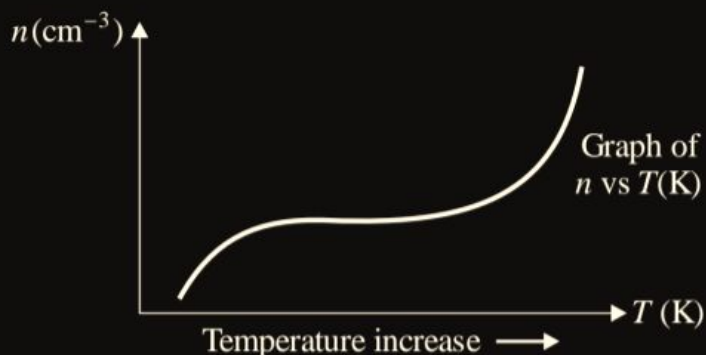
1.64 (A)

In a non degenerately doped n type semiconductor.

- (i) At low temperature : At low temperature negligible intrinsic electron hole pairs exist and the donor electrons is bounded to donor atoms.
- (ii) As the temperature is raised these electrons are donated to conduction band and at about 100 K all donor atoms are ionized.

This temperature range is called ionization region.

Based on this the graph of dependence of free electrons concentration (n) and temperature (T) is



Hence, the correct option is (A).

1.66 A non-degenerate n-type semiconductor has 5% neutral dopant atoms. Its Fermi level is located at 0.25 eV below the conduction band (E_C) and the donor energy level (E_D) has a degeneracy of 2. Assuming the thermal voltage to be 20 mV , the difference between E_C and E_D (in eV , rounded off to two decimal places) is _____.

1.66 0.177

Given : $E_C - E_F = 0.25\text{ eV}$

Degeneracy of donor energy levels = 2

Percentage of neutral dopant atoms in non-degenerate semiconductor = 5%

Now, degeneracy factor

$N_D^o = 5\%$ of N_D (Non ionized donor atom)

$$N_D^o = 0.05 N_D$$

$N_D^+ = 95\%$ of N_D (Ionized donor atom)

$$N_D^+ = 0.95 N_D$$

$$N_D = N_D^o + N_D^+$$

N_D^o = Remaining filled energy state at donor energy level

N_D^+ = Unfilled energy state at E_D .

Due to degeneracy factor effective level of energy state at donor energy level is

$$N_D = N_D^o + 2N_D^+$$

Fraction of state that is filled is given by,

$$f(E_D) = \frac{N_D^o}{N_D^o + 2N_D^+}$$

$$f(E_D) = \frac{0.05N_D}{0.05N_D + (2 \times 0.95)N_D}$$

$$f(E_D) = \frac{0.05}{0.05 + 1.9}$$

$$f(E_D) = \frac{0.05}{1.95}$$

$$f(E_D) = \frac{5}{195} \quad \dots(i)$$

$$\text{Now, } f(E_D) = \frac{1}{1 + e^{\frac{(E_D - E_F)}{KT}}} \quad \dots(ii)$$

From equation (i) and (ii),

$$\frac{1}{1 + e^{\frac{(E_D - E_F)}{KT}}} = \frac{5}{195}$$

$$e^{\frac{E_D - E_F}{KT}} = 38$$

$$E_D - E_F = KT \ln(38)$$

$$E_D - E_F = 20 \times 10^{-3} \ln(38)$$

$$E_D - E_F = 72.7 \text{ meV}$$

$$E_D - E_F = 0.0727 \text{ eV}$$

$$\text{Now, } E_C - E_D = (E_C - E_F) - (E_D - E_F)$$

$$[\text{Given } E_C - E_F = 0.25 \text{ eV}]$$

$$= 0.25 - 0.0727 = 0.1773 \text{ eV}$$

Hence, the difference between E_C and E_D is **0.177 eV**.

Pn junction

2.9 The diffusion potential across a p - n junction

- (A) decreases with increasing doping concentration.
- (B) increases with decreasing band gap.
- (C) does not depend on doping concentration.
- (D) increases with increase in doping concentrations.

2.9 (D)

The built-in potential or diffusion potential of p - n junction diode is given by,

$$V_{bi} = V_T \ln \left[\frac{N_A N_D}{n_i^2} \right] \text{ V}$$

Since, $n_i^2 = A_0 T^3 e^{\frac{-E_{g_0}}{kT}}$

Hence, $V_{bi} = V_T \ln \left[\frac{N_A N_D}{A_0 T^3 e^{\frac{-E_{g_0}}{kT}}} \right]$

$$V_{bi} = V_T \left[\left(\ln \frac{N_A N_D}{A_0} \right) - 3 \ln T + \frac{E_{g_0}}{kT} \right]$$

$$V_{bi} = \frac{E_{g_0}}{q} + V_T \ln \left(\frac{N_A N_D}{A_0} \right) - 3V_T \ln(T)$$

Hence, from above equation V_{bi} will increase if

- (i) doping concentration will increase.
- (ii) band gap will increase.

Hence, the correct option is (D).

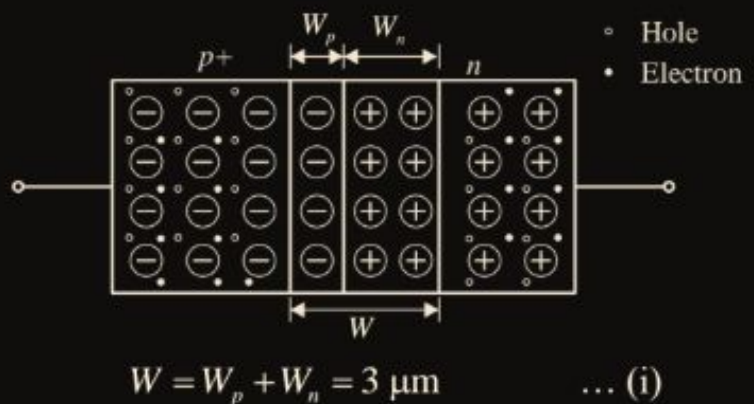
2.15 In an abrupt p - n junction, the doping concentrations on the p -side and n -side are $N_A = 9 \times 10^{16} / \text{cm}^3$ and $N_D = 1 \times 10^{16} / \text{cm}^3$ respectively. The p - n junction is reverse biased and the total depletion width is $3 \mu\text{m}$. The depletion width on the p -side is

- (A) $2.7 \mu\text{m}$ (B) $0.3 \mu\text{m}$
- (C) $2.25 \mu\text{m}$ (D) $0.75 \mu\text{m}$

2.15 (B)

Given :

- (i) Doping concentration on p -side, $N_A = 9 \times 10^{16} / \text{cm}^3$
- (ii) Doping concentration on n -side, $N_D = 1 \times 10^{16} / \text{cm}^3$
- (iii) Total depletion width, $W = W_p + W_n = 3 \mu\text{m}$



From charge neutrality equation,

$$N_A W_p = N_D W_n$$

$$W_n = \frac{N_A}{N_D} W_p$$

$$W_n = \frac{9 \times 10^{16}}{1 \times 10^{16}} \times W_p = 9 W_p \quad \dots (ii)$$

From equation (i) and (ii),

$$W_p + 9W_p = 3 \mu\text{m}$$

$$10W_p = 3 \mu\text{m}$$

$$W_p = 0.3 \mu\text{m}$$

Hence, the correct option is (B).

2.16 Consider an abrupt p - n junction. Let V_{bi} be the built-in potential of this junction and V_R be the applied reverse bias. If the junction capacitance (C_j) is 1 pF for $V_{bi} + V_R = 1$ V, then for $V_{bi} + V_R = 4$ V, C_j will be

- (A) 4 pF (B) 2 pF
(C) 0.25 pF (D) 0.5 pF

2.16 (D)

Given :

- (i) For $(V_{bi} + V_R)_1 = 1$ V, $C_1 = 1$ pF
(ii) For $(V_{bi} + V_R)_2 = 4$ V, $C_2 = ?$

Method 1

For a p - n junction, junction capacitance is given by,

$$C_T = C_j = \frac{C_{j_0}}{\left(1 + \frac{V_R}{V_{bi}}\right)^{\frac{1}{n}}}$$

where, $n = 2$ (for abrupt p - n junction)

V_{bi} = Built-in potential

V_R = Reverse bias voltage

C_{j_0} = Junction capacitance at zero reverse bias voltage

$$C_j \propto \frac{1}{\sqrt{V_{bi} + V_R}}$$

Hence,
$$\frac{C_{j_2}}{C_{j_1}} = \sqrt{\frac{(V_{bi} + V_R)_1}{(V_{bi} + V_R)_2}}$$

$$\frac{C_{j_2}}{1} = \sqrt{\frac{1}{4}}$$

$$C_{j_2} = \frac{1}{2} = 0.5 \text{ pF}$$

Hence, the correct option is (D).

Method 2

For a p - n junction, junction capacitance is given by,

$$C_T = C_j = \frac{C_{j_0}}{\left(1 + \frac{V_R}{V_{bi}}\right)^{\frac{1}{n}}}$$

where, $n = 2$ (for abrupt p - n junction)

$$C_j = \frac{C_{j_0}}{\sqrt{1 + \frac{V_R}{V_{bi}}}} = \frac{C_{j_0} \times \sqrt{V_{bi}}}{\sqrt{V_R + V_{bi}}}$$

For $V_{bi} + V_R = 1$ V,

$$C_{j_1} = \frac{C_{j_0} \times \sqrt{V_{bi}}}{\sqrt{V_{bi} + V_R}}$$

$$1 = \frac{C_{j_0} \times \sqrt{V_{bi}}}{\sqrt{V_{bi} + V_R}} \quad [\text{Given : } C_j = 1 \text{ pF}]$$

$$C_{j_0} \sqrt{V_{bi}} = 1 \quad \dots(i)$$

For $V_{bi} + V_R = 4$ V,

$$C_{j_2} = \frac{C_{j_0} \times \sqrt{V_{bi}}}{\sqrt{V_{bi} + V_R}}$$

From equation (i),

$$C_{j_2} = \frac{1}{\sqrt{4}} = \frac{1}{2} = 0.5 \text{ pF}$$

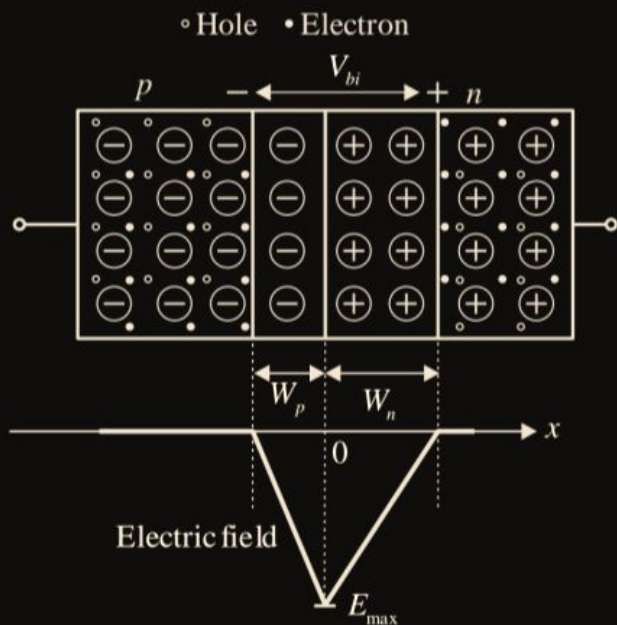
Hence, the correct option is (D).

2.20 In a p^+n junction diode under reverse bias, the magnitude of electric field is maximum at

- (A) the edge of the depletion region on the p -side.
- (B) the edge of the depletion region on the n -side.
- (C) the p^+n junction.
- (D) the centre of the depletion region on the n -side.

2.20 (C)

p^+n junction means p^+ -side is highly doped and n -side is lightly doped comparatively, so most of the width of depletion region is lying in n -layer i.e. $W_n > W_p$.



In the depletion region, on p -side there is $-ve$ charge due to acceptor ions, while on n -side there is $+ve$ charge due to donor ions.

At the junction, charge q changes sharply from $-ve$ to $+ve$. Hence, E is maximum at the junction.

The maximum electric field in $p-n$ junction is given by,

$$E_{\max} = \frac{-qN_D W_n}{\epsilon} = \frac{-qN_A W_p}{\epsilon}$$

$$E_{\text{peak}} = |E_{\max}| = \frac{qN_D W_n}{\epsilon} = \frac{qN_A W_p}{\epsilon}$$

Hence, maximum E occurs at the p^+n junction.

Hence, the correct option is (C).

Key Point

Peak electric field is given by,

$$E_{\text{peak}} = |E_{\max}| = \frac{qN_D W_n}{\epsilon} = \frac{qN_A W_p}{\epsilon}$$

2.21 A p^+n junction has a built-in potential of 0.8 V. The depletion layer width at a reverse bias of 1.2 V is $2 \mu\text{m}$. For a reverse bias of 7.2 V, the depletion layer width will be

- (A) $4 \mu\text{m}$ (B) $4.9 \mu\text{m}$
- (C) $8 \mu\text{m}$ (D) $12 \mu\text{m}$

Given :

- (i) Built-in potential, $V_{bi} = 0.8 \text{ V}$
 (ii) $W_1 = 2 \text{ } \mu\text{m}$, $V_{RB_1} = 1.2 \text{ V}$
 (iii) $V_{RB_2} = 7.2 \text{ V}$

For abrupt p - n junction, junction potential is given by,

$$V_j = \frac{qNW^2}{2\varepsilon}$$

Hence, $W \propto \sqrt{V_j}$

where, $V_j = V_{bi} + V_R$

V_{bi} = Built-in potential

V_R = Reverse bias voltage

$$\frac{W_1}{W_2} = \sqrt{\frac{V_{j1}}{V_{j2}}} = \sqrt{\frac{V_{bi} + V_{RB_1}}{V_{bi} + V_{RB_2}}}$$

$$W_2 = 2 \times \sqrt{\frac{0.8 + 7.2}{0.8 + 1.2}} = 4 \text{ } \mu\text{m}$$

Hence, the correct option is (A).

2.25 Compared to p - n junction with $N_A = N_D = 10^{14} / \text{cm}^3$, which one of the following statement is TRUE for a p - n junction with $N_A = N_D = 10^{20} / \text{cm}^3$?

- (A) Reverse breakdown voltage is lower and depletion capacitance is lower.
 (B) Reverse breakdown voltage is higher and depletion capacitance is lower.
 (C) Reverse breakdown voltage is lower and depletion capacitance is higher.
 (D) Reverse breakdown voltage is higher and depletion capacitance is higher.

Given : $N_{A_1} = N_{D_1} = 10^{14} / \text{cm}^3$

$$N_{A_2} = N_{D_2} = 10^{20} / \text{cm}^3$$

(i) Breakdown voltage

$$\propto \frac{1}{\text{Doping Concentration}}$$

Hence, when doping concentration is increased then breakdown voltage will decrease.

(ii) Depletion capacitance, $C_T = \frac{\varepsilon A}{W}$

When we increase doping concentration then depletion width W will decrease and finally capacitance will increase.

Hence, the correct option is (C).

2.26 A silicon p - n junction is forward biased with a constant current at room temperature. When the temperature is increased by 10°C , the forward bias voltage across the p - n junction

(A) increases by 60 mV.
 (B) decreases by 60 mV.
 (C) increases by 25 mV.
 (D) decreases by 25 mV.

For a p - n junction diode, rate of change of voltage is given by,

$$\frac{dV}{dT} = -2.3 \text{ mV}/^\circ\text{C}$$

Hence, for 10°C rise in temperature,

$$\Delta V = -2.3 \times 10 = -23 \text{ mV} \approx -25 \text{ mV}$$

Negative sign shows the decrease in voltage.

Hence, the correct option is (D).

2.28 The doping concentrations on the p -side and n -side of a silicon diode are

$$1 \times 10^{16} \text{ cm}^{-3} \text{ and } 1 \times 10^{17} \text{ cm}^{-3}$$

respectively. A forward bias of 0.3 V is applied to the diode. At $T = 300 \text{ K}$, the intrinsic carrier concentration of silicon

$$n_i = 1.5 \times 10^{10} \text{ cm}^{-3} \text{ and } \frac{kT}{q} = 26 \text{ mV}.$$

The electron concentration at the edge of the depletion region on the p -side is

[Set - 01]

- (A) $2.3 \times 10^9 \text{ cm}^{-3}$
- (B) $1 \times 10^{16} \text{ cm}^{-3}$
- (C) $1 \times 10^{17} \text{ cm}^{-3}$
- (D) $2.25 \times 10^6 \text{ cm}^{-3}$

Given : Doping concentration on p -side,

$$N_A = 10^{16} \text{ cm}^{-3},$$

Forward bias voltage, $V_A = 0.3 \text{ V}$

Doping concentration on n -side,

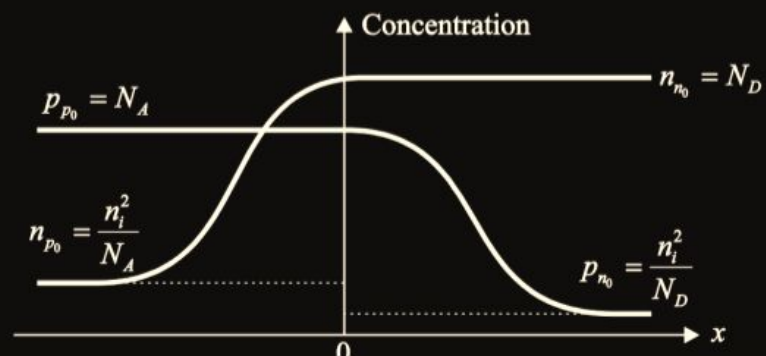
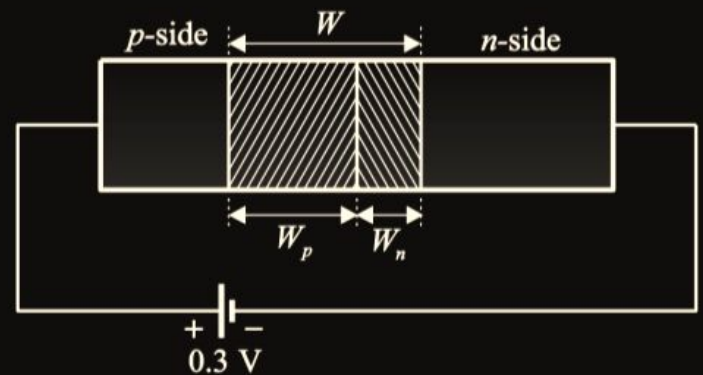
$$N_D = 10^{17} \text{ cm}^{-3}$$

Intrinsic carrier concentration,

$$n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$$

$$\frac{kT}{q} = V_T = 26 \text{ mV}$$

n -side is heavily doped. Hence, depletion region penetrates less on n -side.



where, n_{p_0} = Minority carrier i.e. electron concentration on p -side,
 p_{p_0} = Majority carrier i.e. hole concentration on p -side
 n_{n_0} = Majority carrier i.e. electron concentration on n -side
 p_{n_0} = Minority carrier i.e. hole concentration on n -side.

From boundary condition, electron concentration on p -side, is given by,

$$n_p(x) = n_{p_0} e^{V_A/V_T}$$

Concentration of electron on p -side at the edge of depletion layer, is given by,

$$n_p(0) = n_{p_0} e^{V_A/V_T} \quad \dots (i)$$

From Mass action law,

$$n_{p_0} \times p_{p_0} = n_i^2$$

$$\text{Hence, } n_{p_0} = \frac{n_i^2}{p_{p_0}} = \frac{n_i^2}{N_A} \quad \dots (ii)$$

From equation (i) and (ii),

$$n_p(0) = \frac{n_i^2}{N_A} e^{V_A/V_T}$$

$$n_p(0) = \frac{(1.5 \times 10^{10})^2}{10^{16}} \times e^{0.3/0.026}$$

$$n_p(0) = 2.25 \times 10^4 \times e^{0.3/0.026}$$

$$n_p(0) = 2.3 \times 10^9 \text{ /cm}^3$$

Hence, the correct option is (A).

Key Point

The Hole concentration at the edge of depletion region on n -side is given by,

$$p_n(x) = p_{n_0} e^{V_A/V_T}$$

where, p_{n_0} = Hole concentration at thermal equilibrium.

2.31 The donor and Acceptor impurities in an abrupt junction silicon diode are $1 \times 10^{16} \text{ cm}^{-3}$ and $5 \times 10^{18} \text{ cm}^{-3}$, respectively. Assume that the intrinsic carrier concentration in silicon $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ at 300 K $\frac{kT}{q} = 26 \text{ mV}$ and the permittivity of silicon $\epsilon_{\text{Si}} = 1.04 \times 10^{-12} \text{ F/cm}$. The built-in potential and the depletion width of the diode under thermal equilibrium conditions, respectively, are **[Set - 03]**

- (A) 0.7 V and $1 \times 10^{-4} \text{ cm}$
 (B) 0.86 V and $1 \times 10^{-4} \text{ cm}$
 (C) 0.7 V and $3.3 \times 10^{-5} \text{ cm}$
 (D) 0.86 V and $3.3 \times 10^{-5} \text{ cm}$

2.31 (D)

Given :

Donor impurity, $N_D = 1 \times 10^{16} \text{ /cm}^3$

Acceptor impurity, $N_A = 5 \times 10^{18} \text{ /cm}^3$

Intrinsic carrier concentration,

$$n_i = 1.5 \times 10^{10} \text{ /cm}^3$$

$$T = 300 \text{ K}, V_T = \frac{KT}{q} = 26 \text{ mV}$$

Permittivity of silicon,

$$\epsilon_{\text{Si}} = 1.04 \times 10^{-12} \text{ F/cm}$$

Built-in potential is given by,

$$V_{bi} = V_T \ln \left[\frac{N_A N_D}{n_i^2} \right] \text{ V}$$

$$V_{bi} = 0.026 \ln \left[\frac{5 \times 10^{18} \times 1 \times 10^{16}}{(1.5 \times 10^{10})^2} \right]$$

$$V_{bi} = 0.859 \text{ V} \approx 0.86 \text{ V}$$

Built-in potential can be written as,

$$V_{bi} = \frac{qNW^2}{2\epsilon_{Si}}$$

where, $N = N_A \parallel N_D$

$$\frac{1}{N} = \frac{1}{N_A} + \frac{1}{N_D}$$

As $N_D \ll N_A$,

Hence, $N \cong N_D$

From equation (i),

$$0.86 = \frac{1.6 \times 10^{-19} \times 1 \times 10^{16} \times W^2}{2 \times 1.04 \times 10^{-12}}$$

$$W = 3.34 \times 10^{-5} \text{ cm} = 0.334 \text{ } \mu\text{m}$$

Hence, the correct option is (D).

2.33 For a silicon diode with long p and n regions, the acceptor and donor impurity concentrations are $1 \times 10^{17} \text{ cm}^{-3}$ and $1 \times 10^{15} \text{ cm}^{-3}$, respectively. The lifetime of electrons in p region and holes in n region are both $100 \text{ } \mu\text{s}$. The electron and hole diffusion coefficient are $49 \text{ cm}^2/\text{s}$ and $36 \text{ cm}^2/\text{s}$, respectively. Assume $kT/q = 26 \text{ mV}$, the intrinsic carrier concentration is $1 \times 10^{10} \text{ cm}^{-3}$ and $q = 1.6 \times 10^{-19} \text{ C}$. When a forward voltage of 208 mV is applied across the diode, the hole current density (in nA/cm^2) injected from p region to n region is _____. **[Set - 01]**

Given :

- (i) Acceptor impurity, $N_A = 1 \times 10^{17} \text{ cm}^{-3}$
- (ii) Donor impurity, $N_D = 1 \times 10^{15} \text{ cm}^{-3}$
- (iii) Electron life time on p -side, $\tau_n = 100 \text{ } \mu\text{s}$
- (iv) Hole life time on n -side, $\tau_p = 100 \text{ } \mu\text{s}$
- (v) Electron diffusion constant, $D_n = 49 \text{ cm}^2/\text{s}$
- (vi) Hole diffusion constant, $D_p = 36 \text{ cm}^2/\text{s}$
- (vii) Intrinsic carrier concentration, $n_i = 1 \times 10^{10} \text{ cm}^{-3}$ and $q = 1.6 \times 10^{-19} \text{ C}$
- (viii) Forward voltage drop = 208 mV

Diode current is given by,

$$I_D = I_0 (e^{V/V_T} - 1) \quad \dots \text{ (i)}$$

$$\text{where, } I_0 = \frac{AqD_p p_{n0}}{L_p} + \frac{AqD_n n_{p0}}{L_n} \quad \dots \text{ (ii)}$$

The current density is given by,

$$J = \frac{I_D}{A} \quad \dots \text{ (iii)}$$

Hence, from equation (i), (ii) and (iii),

$$J = \frac{qD_p p_{n0}}{L_p} (e^{V/V_T} - 1) + \frac{qD_n n_{p0}}{L_n} (e^{V/V_T} - 1)$$

Electron diffusion current density injected from n -region to p -region is given by,

$$J_n = \frac{qD_n n_{p0}}{L_n} (e^{V/V_T} - 1)$$

Hole diffusion current density injected from p -region to n -region is given by,

$$J_p = \frac{qD_p p_{n0}}{L_p} (e^{V/V_T} - 1)$$

From mass action law,

$$p_{n_0} = \frac{n_i^2}{N_D}$$

So, $J_p = \frac{qn_i^2 D_p}{N_D L_p} (e^{V/V_T} - 1) \dots \text{(iv)}$

where, q = Electronic charge, n_i = Intrinsic carrier concentration in silicon, L_p = Mean diffusion length of hole, V = Forward voltage applied across diode

$$V_T = \frac{kT}{q} = 26 \text{ mV}$$

Diffusion length is given by,

$$L_p = \sqrt{\tau_p D_p} = \sqrt{100 \times 10^{-6} \times 36} = 0.06 \text{ cm}$$

Put all the values in equation (iv),

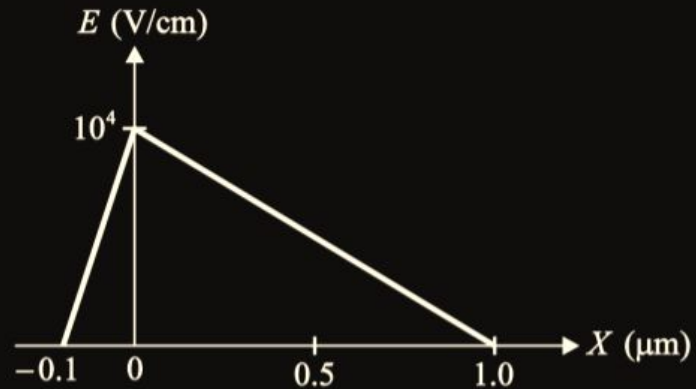
$$J_p = \frac{1.6 \times 10^{-19} \times (10^{10})^2 \times 36}{10^{15} \times 6 \times 10^{-2}} \times \left(e^{\frac{208}{26}} - 1 \right)$$

$$J_p = 28.6 \times 10^{-9} \text{ A/cm}^2 = 28.6 \text{ nA/cm}^2$$

Hence, the hole current density injected from p region to n region is **28.6 nA / cm²**.

2.34 The electric field profile in the depletion region of a p - n junction in equilibrium is shown in the figure. Which one of the following statements is NOT TRUE?

[Set - 03]



- (A) The left side of the junction is n -type and the right side is p -type.
- (B) Both the n -type and p -type depletion regions are uniformly doped.
- (C) The potential difference across the depletion region is 700 mV.
- (D) If the p -type region has a doping concentration of 10^{15} cm^{-3} , then the doping concentration in the n -type region will be 10^{16} cm^{-3} .

2.34 (C)

Given : $E_{\max} = 10^4 \text{ V/cm} = 10^6 \text{ V/m}$

$$W = 1 - (-0.1) \mu\text{m} = 1.1 \times 10^{-6} \text{ m}$$

From Poisson's equation, $\frac{d^2V}{dx^2} = \frac{-\rho_V}{\epsilon}$

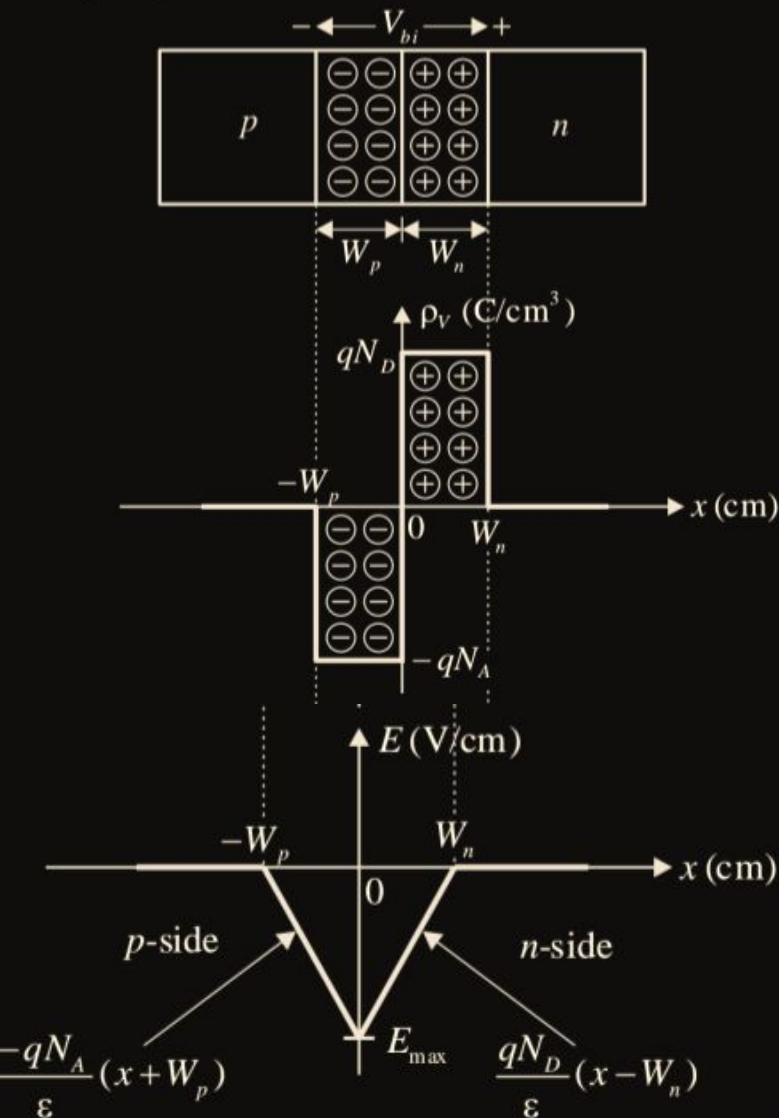
Since, $E = \frac{-dV}{dx}$

Hence, $\frac{dE}{dx} = \frac{\rho_V}{\epsilon}$

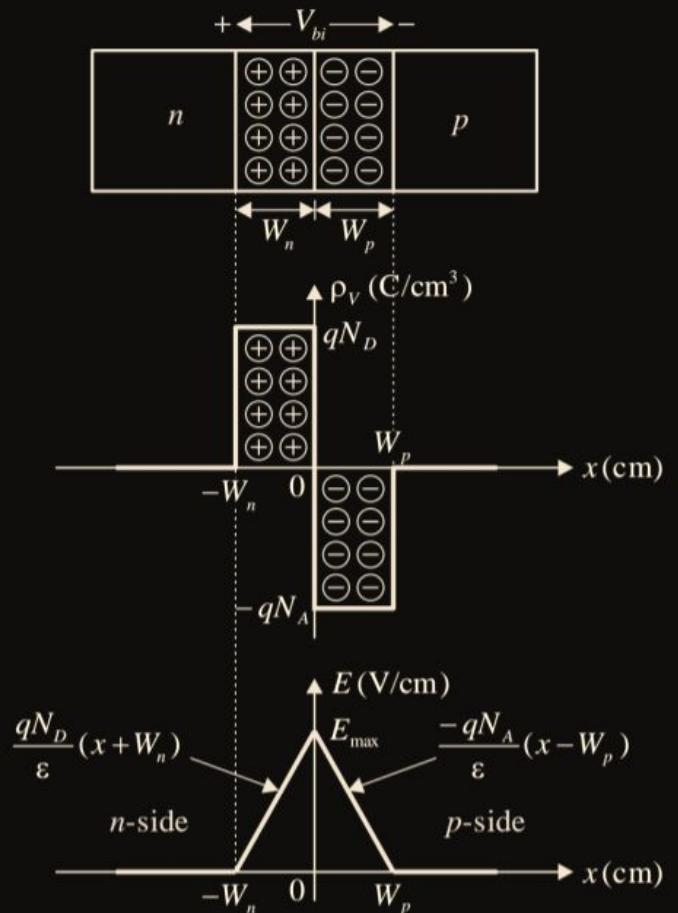
$$E = \int \frac{\rho_V}{\epsilon} dx \Rightarrow E \propto \int \rho_V$$

where, $\rho_V = \text{Charge density.}$

For p - n junction diode :



For n - p junction diode :



So, given electric field profile is of n - p junction.

(i) **Option (A)** is true because left side of junction is n -type and right side is p -type.

(ii) **Option (B)** is also true because $\rho_V = \text{constant. i.e. uniformly doped.}$

(iii) **For option (C),**
Potential difference V is given by,

$$V = -\int E dx$$

$$V = -(\text{Area under } E\text{-}x \text{ curve})$$

$$V = -\frac{1}{2} \times (W_p - W_n) \times E_{\max}$$

$$V = -\frac{1}{2} \times 1.1 \times 10^{-4} \times 10^4 = -550 \text{ mV}$$

So, option (C) is not true.

(iv) **For option (D),**

From charge neutrality equation,

$$N_A W_p = N_D W_n$$

$$10^{15} \times 1 \times 10^{-4} = N_D \times 0.1 \times 10^{-4}$$

$$N_D = 10^{16} \text{ cm}^{-3}$$

So, option (D) is also true.

Hence, the correct option is (C).

2.35 Consider a silicon p - n junction with a Built-in potential (for zero bias) can be written as,
 10^{17} cm^{-3} on the p -side and a uniform donor doping concentration of 10^{16} cm^{-3} on the n -side. No external voltage is applied to the diode. Given :

$$kT/q = 26 \text{ mV}, n_i = 1.5 \times 10^{10} \text{ cm}^{-3},$$

$$\epsilon_{Si} = 12\epsilon_0, \epsilon_0 = 8.85 \times 10^{-14} \text{ F/m and}$$

$q = 1.6 \times 10^{-19} \text{ C}$. The charge per unit junction area (nC cm^{-2}) in the depletion region on the p -side is _____.

2.35 - 4.89

Given :

Acceptor concentration, $N_A = 10^{17} \text{ cm}^{-3}$

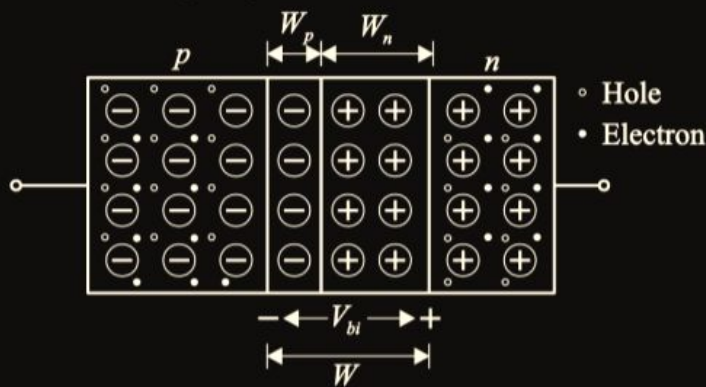
Donor concentration, $N_D = 10^{16} \text{ cm}^{-3}$

$$\frac{kT}{q} = 26 \text{ mV}, n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$$

$$\epsilon_{Si} = 12\epsilon_0, q = 1.6 \times 10^{-19} \text{ C}$$

$$\epsilon_0 = 8.85 \times 10^{-14} \text{ F/m} = 8.85 \times 10^{-16} \text{ F/cm}$$

An unbiased p - n junction is shown below,



Built-in potential of p - n junction diode is given by,

$$V_{bi} = V_T \ln \frac{N_A N_D}{n_i^2} \text{ V}$$

$$V_{bi} = 26 \times 10^{-3} \ln \frac{10^{16} \times 10^{17}}{(1.5 \times 10^{10})^2} = 0.757 \text{ V}$$

$$V_{bi} = \frac{qNW^2}{2\epsilon}$$

where, $N = N_A \parallel N_D$

Depletion width of p - n junction diode is given by,

$$W = \sqrt{\frac{2\epsilon}{q} \left[\frac{1}{N_A} + \frac{1}{N_D} \right] V_{bi}}$$

$$W = \sqrt{\frac{2 \times 8.854 \times 10^{-16} \times 12 \left[\frac{1}{10^{16}} + \frac{1}{10^{17}} \right]}{1.6 \times 10^{-19}}} \times 0.757$$

$$W = 3.325 \times 10^{-6} \text{ cm}$$

$$\text{Now, } W_p = \frac{N_D W}{N_A + N_D} = \frac{3.325 \times 10^{-6} \times 10^{16}}{10^{16} + 10^{17}}$$

$$W_p = 3.023 \times 10^{-7} \text{ cm}$$

Charge per unit junction area in the depletion layer on the p -side is given by,

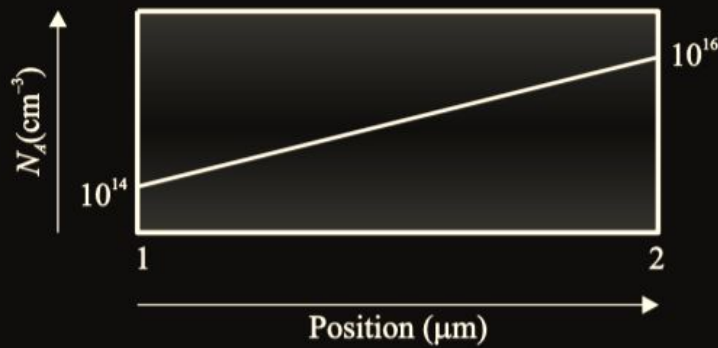
$$\frac{Q}{A} = -qN_A W_p$$

$$\frac{Q}{A} = -1.6 \times 10^{-19} \times 10^{17} \times 3.023 \times 10^{-7}$$

$$\frac{Q}{A} = -4.89 \text{ nC/cm}^2$$

Hence, the charge per unit junction area in the depletion region on the p -side is -4.89 nC/cm^2 .

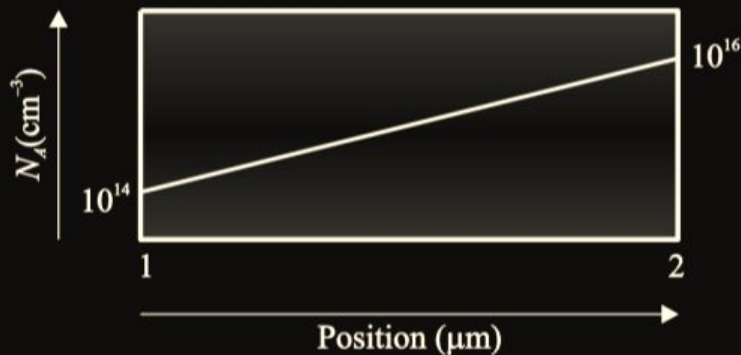
2.36 The figure below shows the doping distribution in a p -type semiconductor in log scale. [Set - 01]



The magnitude of the electric field (in kV/cm) in the semiconductor due to non-uniform doping is _____.

2.36 1.198

Given :



The current density equation is given by,

$$J = J_{drift} + J_{Diffusion}$$

$$J = -qD_p \frac{dp}{dx} + q\mu_p pE$$

For isolated semiconductor, $J = 0$

$$\text{Hence, } 0 = -qD_p \frac{dp}{dx} + q\mu_p pE$$

$$qD_p \frac{dp}{dx} = q\mu_p pE$$

$$D_p \frac{dp}{dx} = \mu_p pE$$

$$E = \frac{D_p}{p\mu_p} \frac{dp}{dx} \quad \dots (i)$$

From Einstein's relation,

$$\frac{D_p}{\mu_p} = V_T \quad \dots (ii)$$

Hence, from equations (i) and (ii),

$$E = \frac{V_T}{p} \frac{dp}{dx}$$

Since, $p \cong N_A$

$$E = \frac{V_T}{N_A} \frac{dN_A}{dx}$$

$$\text{Hence, } E = V_T \frac{d}{dx} \ln[N_A(x)]$$

where, $V_T = 26 \text{ mV}$

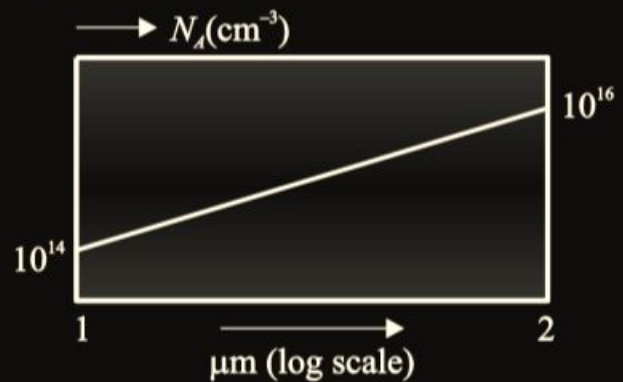
Units are in log scale, hence

$$\ln(10^{14}) = 32.23$$

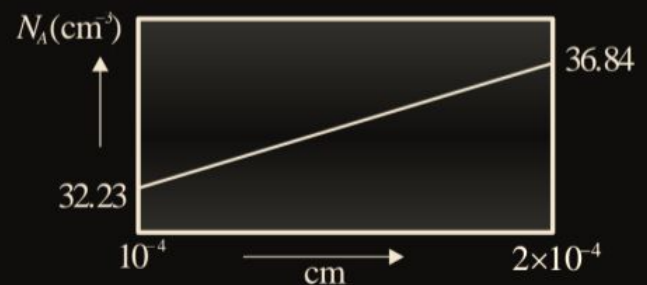
$$\text{and } \ln(10^{16}) = 36.84$$

$$x_1 = 1 \text{ } \mu\text{m} = 10^{-4} \text{ cm}$$

$$x_2 = 2 \text{ } \mu\text{m} = 2 \times 10^{-4} \text{ cm}$$



Hence, above figure can be formed as given below,



$$E = 0.026 \left[\frac{36.84 - 32.23}{2 \times 10^{-4} - 10^{-4}} \right]$$

$$E = 1.198 \text{ kV/cm}$$

Hence, the magnitude of the electric field is **1.198 kV/cm**.

Key Point

Due to non-uniform doping, to maintain equilibrium, internal electric field is generated. The generated electric field opposes the diffusion of carrier due to concentration gradient and thus, there will be no current flow inside the semiconductor.

$n_1 =$ Doping concentration on n^+ -side $= N_{D_1}$

$n_2 =$ Doping concentration on n -side $= N_{D_2}$

Hence, from equation (ii),

$$V_{bi} = V_T \ln \frac{N_{D_1}}{N_{D_2}}$$

2.42 A p - n step junction diode with a contact potential of 0.65 V has a depletion width of $1 \mu\text{m}$ at equilibrium. The forward voltage (in volts, correct to two decimal places) at which this width reduces to $0.6 \mu\text{m}$ is _____.

2.42 0.42

Given :

- (i) Contact potential $= 0.65 \text{ V}$ at depletion width $W_1 = 1 \mu\text{m}$.
- (ii) For $W_2 = 0.6 \mu\text{m}$, forward voltage $V_A = ?$

Junction potential is given by,

$$V_j = \frac{qN_A W^2}{2\epsilon}$$

$$V_j \propto W^2 \Rightarrow \frac{V_{j_1}}{V_{j_2}} = \left(\frac{W_1}{W_2} \right)^2$$

Then,
$$\frac{V_0}{V_0 - V_a} = \left(\frac{W_1}{W_2} \right)^2$$

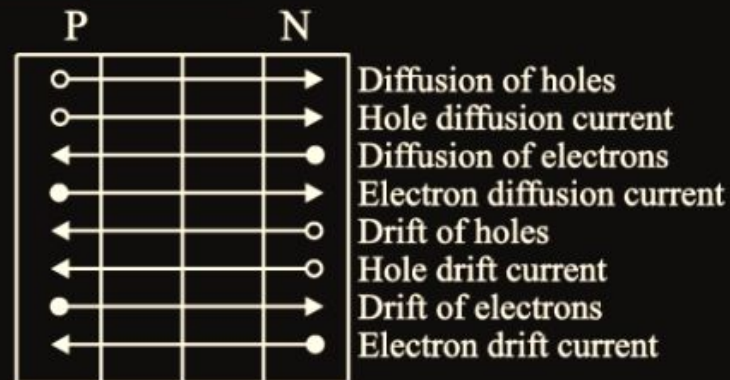
$$\frac{0.65}{0.65 - V_a} = \left(\frac{1 \mu\text{m}}{0.6 \mu\text{m}} \right)^2 = \left(\frac{5}{3} \right)^2$$

$$\frac{0.65}{0.65 - V_a} = \frac{25}{9}$$

$$5.85 = 16.25 - 25V_a$$

$$V_a = \frac{16.25 - 5.85}{25} = 0.416 \text{ V} \approx 0.42 \text{ V}$$

Hence, the value of forward voltage is **0.42 V**.



Energy band gap and built in potential of a p - n junction is inversely proportional to wavelength,

i.e.
$$E_g, V_g \propto \frac{1}{\lambda}$$

2.46 In an ideal pn junction with an ideality factor of 1 at $T = 300 \text{ K}$, the magnitude of reverse-bias voltage required to set 75% of its reverse saturation current, rounded off to 2 decimal places, is _____ mV.

$$k = 1.38 \times 10^{-23} \text{ JK}^{-1},$$

$$h = 6.625 \times 10^{-34} \text{ J-s}, q = 1.602 \times 10^{-19} \text{ C}$$

Equation of diode current

$$I_D = I_0 (e^{V/\eta V_T} - 1) \quad \dots (i)$$

Given, $\eta = 1$

$$I_D = -75\% \text{ of } I_0$$

(Since, current flows from n to p in reverse bias, -ve sign will be considered)

$$I_D = -\frac{3}{4} I_0$$

By using equation (i),

$$-\frac{3}{4} I_0 = I_0 (e^{V/1 \times 0.026} - 1)$$

$$-0.75 I_0 + I_0 = I_0 e^{V/0.026}$$

$$I_0 e^{\frac{V}{0.026}} = 0.25 I_0$$

$$\frac{V}{0.026} = -1.386$$

$$V = -36 \text{ mV}$$

Taking magnitude, $|V| = 36 \text{ mV}$

Hence, the magnitude of reverse bias voltage is **36 mV**.

2.47 Consider the recombination process via bulk traps in a forward biased pn homojunction diode. The maximum recombination rate is $U_{\max}$. If the electron and the hole capture cross sections are equal, which one of the following is FALSE?

- (A) $U_{\max}$ occurs at the edges of the depletion region in the device.
- (B) $U_{\max}$ depends exponentially on the applied bias.
- (C) With all other parameters unchanged, $U_{\max}$ increases if the thermal velocity of carrier increases.

The maximum recombination rate is given by,

$$U_{\max} = \frac{1}{2} V_{th} n_i \sigma_0 e^{V/2V_T} \cdot N_t$$

where, n_i = Intrinsic concentration,

σ_0 = capture cross section,

V = applied voltage,

V_T = thermal voltage,

V_{th} = thermal velocity,

N_t = recombination centers in bulk of semiconductors

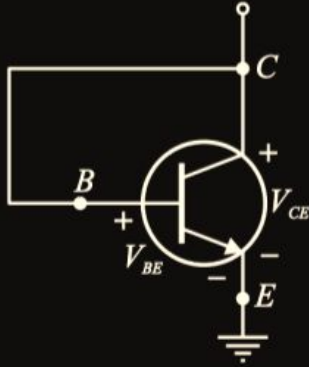
$$\sigma_n = \sigma_p = \sigma_0$$

In Homojunction $p-n$ diode we find that current in a pn junction can only exist if there is recombination or generation of electron and holes somewhere throughout the structure. The ideal diode equation is a result of the recombination and generation in the quasi-neutral regions (including recombination at the contacts) whereas recombination and generation in the depletion region yield enhanced leakage or photocurrents.

Hence, the correct option is (A).

Key Point

BJT will behave as a diode if collector is shorted to base i.e. $V_{CB} = 0$ V. This type of BJT is referred as **Diode connected transistor**.



From figure, $V_{CB} = 0$ V, $V_{CE} = V_{BE}$

$$I_C = I_{C_0} (e^{\frac{V_{CE}}{V_T}} - 1) = I_{C_0} (e^{\frac{V_{BE}}{V_T}} - 1)$$

- 3.18** The DC current gain (β) of a BJT is 50. Assuming that the emitter injection efficiency is 0.995, the base transport factor is
- (A) 0.980 (B) 0.985
(C) 0.990 (D) 0.995

3.18 (B)

Given : $\beta_{dc} = 50$, $\gamma = 0.995$

$$\alpha_{dc} = \frac{\beta_{dc}}{\beta_{dc} + 1} = \frac{50}{51}$$

Base transport factor is given by,

$$\beta^* = \frac{\alpha_{dc}}{\text{Emitter injection efficiency } (\gamma)}$$

$$\beta^* = \frac{50}{51 \times 0.995} = 0.985$$

Hence, the correct option is (B).

Key Point

- (i) For non-uniformly doped BJT, order of doping is,
 $N_E \gg N_C \gg N_B$
- (ii) If emitter injection efficiency is unity then,
 - (a) $N_E \gg N_B$ and $N_B < N_C$
 - (b) $N_E \gg N_B$ and $N_B = N_C$
 - (c) $N_E \gg N_B$ and $N_B > N_C$
- (iii) Emitter injection efficiency is independent of doping concentration of base and collector.

- 3.21** An increase in the base recombination of a BJT will increase [Set - 02]
- (A) the common emitter dc current gain β .
- (B) the breakdown voltage BV_{CEO} .
- (C) the unity-gain cutoff frequency f_T .
- (D) the transconductance g_m .

3.21 (B)

1. Common emitter dc current gain (β) :

If base recombination of BJT will increase then recombination of emitter carrier entering into base region will increase and hence number of carrier going from emitter region to collector region will decrease and hence collector current I_C will decrease. Therefore,

$$\alpha \left(= \frac{I_C}{I_E} \right) \text{ will decrease and hence,}$$

$$\beta \left(= \frac{\alpha}{1 - \alpha} = \frac{I_C}{I_B} \right) \text{ will decrease.}$$

2. Breakdown voltage BV_{CEO} :

BV_{CEO} is collector to emitter breakdown voltage with base terminal open.

BV_{CBO} is collector to base breakdown voltage with emitter terminal open.

The reverse leakage current is flowing due to the minority charge carriers of base (electrons in $n-p-n$) and collector (holes in $n-p-n$). If reverse voltage across CB junction increases, the charge carriers are generated due to avalanche mechanism (i.e. cumulative process of collision of charge carriers with covalent bonds).

If recombination in base increases, higher breakdown voltage will be required across CB junction to generate charge carriers hence breakdown voltage BV_{CBO} or BV_{CEO} increases.

3. Unity gain cutoff frequency (f_T):

Unity gain cutoff frequency is given by,

$$f_T = h_{fe} f_\beta = \frac{g_m}{2\pi(C_e + C_c)}$$

where, f_β = Cutoff frequency

C_e = Diffusion capacitance across emitter base junction.

C_c = Transition capacitance across collector base junction.

Transconductance, $g_m = \frac{I_C}{V_T}$

$$\text{Hence, } f_T = \frac{I_C}{2\pi V_T (C_e + C_c)}$$

When recombination in base region of BJT will increase, then I_C will decrease and hence, f_T will decrease.

4. Transconductance (g_m) :

Transconductance g_m is given by,

$$g_m = \frac{\text{Output current}}{\text{Input voltage}}$$

For common base configuration, g_m is given by,

$$g_m = \frac{I_C}{V_{BE}}$$

Hence, when recombination in base region of BJT will increase then I_B will increase and hence, I_C will decrease and therefore, transconductance g_m will decrease.

Hence, the correct option is (B).

Key Point

- (i) As recombination takes place in base, i.e. covalent bond is formed, the energy is released in the form of heat which is absorbed by another covalent bond near base terminal and the generated electron is attracted from base terminal by emitter base supply, so base current is flowing due to recombination in base region.
- (ii) If temperature increases, the recombination of charge carriers in base decreases but generation of charge carriers increases as more covalent bonds are broken due to thermal energy. So, I_B decreases. Hence, β increases, with increase in temperature.

3.22 Consider two BJTs biased at the same collector current with area

$$A_1 = 0.2 \mu\text{m} \times 0.2 \mu\text{m}$$

$$\text{and } A_2 = 300 \mu\text{m} \times 300 \mu\text{m}$$

Assuming that all other device parameters are identical, $kT/q = 26 \text{ mV}$, the intrinsic carrier concentration is $1 \times 10^{10} \text{ cm}^{-3}$ and $q = 1.6 \times 10^{-19} \text{ C}$, the difference between the base-emitter voltages (in mV) of the two BJTs (i.e. $V_{BE_1} - V_{BE_2}$) is _____. **[Set - 04]**

3.22 380.28

Given : $A_1 = 0.2 \mu\text{m} \times 0.2 \mu\text{m}$

$$A_2 = 300 \mu\text{m} \times 300 \mu\text{m}$$

$$\frac{kT}{q} = 26 \text{ mV}$$

Intrinsic carrier concentration,

$$n_i = 1 \times 10^{10} / \text{cm}^3 \text{ and } q = 1.6 \times 10^{-19} \text{ C}$$

Collector current of BJT is given by,

$$I_C = I_0 \left[e^{\frac{V_{BE}}{\eta V_T}} - 1 \right]$$

$$I_C = I_0 \exp \left[\frac{V_{BE}}{\eta V_T} \right] \quad \left[e^{\frac{V_{BE}}{\eta V_T}} \gg 1 \right]$$

$$\text{Let, } I_{C_1} = I_{0_1} \exp \left[\frac{V_{BE_1}}{\eta V_T} \right] \quad \dots \text{ (i)}$$

$$I_{C_2} = I_{0_2} \exp \left[\frac{V_{BE_2}}{\eta V_T} \right] \quad \dots \text{ (ii)}$$

Since, device parameters are identical

$$I_{C_1} = I_{C_2}$$

$$I_{0_1} \exp \left[\frac{V_{BE_1}}{\eta V_T} \right] = I_{0_2} \exp \left[\frac{V_{BE_2}}{\eta V_T} \right]$$

$$V_{BE_1} - V_{BE_2} = \eta V_T \ln \left[\frac{I_{0_2}}{I_{0_1}} \right] \quad \dots \text{ (iii)}$$

Reverse saturation current for $p-n$ junction diode is given by,

$$I_0 = Aq \left[\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A} \right] n_i^2$$

All parameter except area is identical for both BJT, hence

$$\frac{I_{0_2}}{I_{0_1}} = \frac{A_2}{A_1} \quad \dots \text{ (iv)}$$

From equation (iii) and (iv),

$$V_{BE_1} - V_{BE_2} = \eta V_T \ln \left(\frac{A_2}{A_1} \right)$$

where, $\eta = 1$ (for all transistors)

$$V_{BE_1} - V_{BE_2} = 26 \ln \left[\frac{300 \times 300}{0.2 \times 0.2} \right] \text{ mV}$$

$$V_{BE_1} - V_{BE_2} = 380.28 \text{ mV}$$

Hence, the difference between the base-emitter voltages of the two BJTs (i.e. $V_{BE_1} - V_{BE_2}$) is **380.28 mV**.

3.24 If the base width in a bipolar junction transistor is doubled, which one of the following statements will be TRUE?

[Set - 03]

- (A) Current gain will increase.
- (B) Unity gain frequency will increase.
- (C) Emitter-base junction capacitance will increase.
- (D) Early voltage will increase.

1. Current gain (β) :

If base width will increase then base recombination of BJT will increase and hence, recombination of emitter carrier entering into base region will increase and therefore, number of carrier going from emitter region to collector region will decrease and hence, collector current I_C will decrease. Therefore, current gain

$$\left(\beta = \frac{I_C}{I_B} \right) \text{ will decrease.}$$

2. Unity gain cutoff frequency (f_T) :

Unity gain cutoff frequency is given by,

$$f_T = h_{fe} f_\beta = \frac{g_m}{2\pi(C_e + C_c)}$$

where, f_β = Cutoff frequency,

C_e = Diffusion capacitance across emitter base junction,

C_c = Transition capacitance across collector base junction.

$$\text{Transconductance, } g_m = \frac{I_C}{V_T}$$

$$f_T = \frac{I_C}{2\pi V_T (C_e + C_c)}$$

When base width will increase then recombination in base region of BJT will increase and hence, I_C will decrease and therefore, f_T will decrease.

3.

Emitter base junction capacitance (C_e):

Emitter base junction capacitance, depends on the depletion region width around emitter base junction.

Depletion region width around emitter base junction will depend on factors given below,

(i) **Doping concentration of emitter and base :** If doping concentration of either emitter or base will increase then width of depletion region around emitter base junction will decrease and hence, emitter base junction capacitance will increase.

(ii) **Voltage applied across emitter base junction :** If voltage across emitter base junction will increase then width of depletion region around emitter base junction will decrease and hence, emitter base junction capacitance will increase.

4.

Early voltage (V_A) :

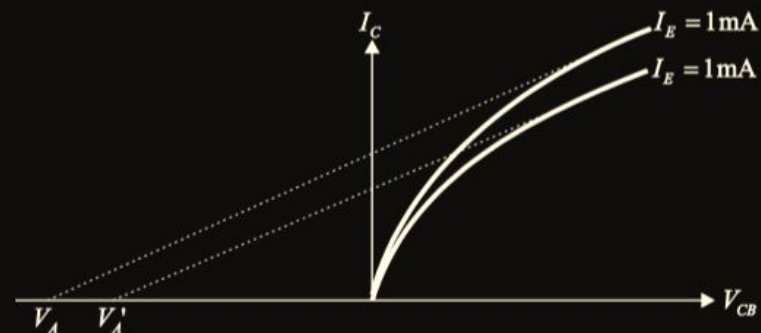


Fig. Output characteristics of CB configuration

If base width will increase then due to recombination in base region I_B will increase and hence, I_C will decrease for same value of I_E . Hence, slope of output characteristic will decrease and therefore intersection point of output characteristic on negative real axis will decrease.

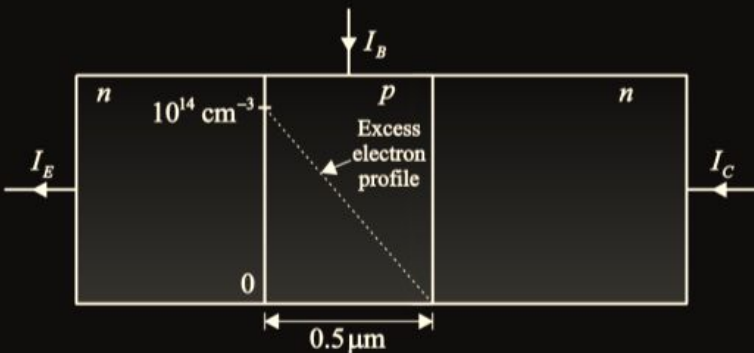
Therefore, early voltage will increase.

Hence, the correct option is (D).

3.27 The injected excess electron concentration profile in the base region of an n - p - n BJT, biased in the active region, is linear, as shown in the figure. If the area of the emitter-base junction is 0.001 cm^2 , $\mu_n = 800 \text{ cm}^2/(\text{V}\cdot\text{s})$ in the base region and depletion layer widths are negligible, then the collector current I_C (in mA) at room temperature is _____.

(Given : thermal voltage $V_T = 26 \text{ mV}$ at room temperature, electronic charge $q = 1.6 \times 10^{-19} \text{ C}$)

[Set - 03]



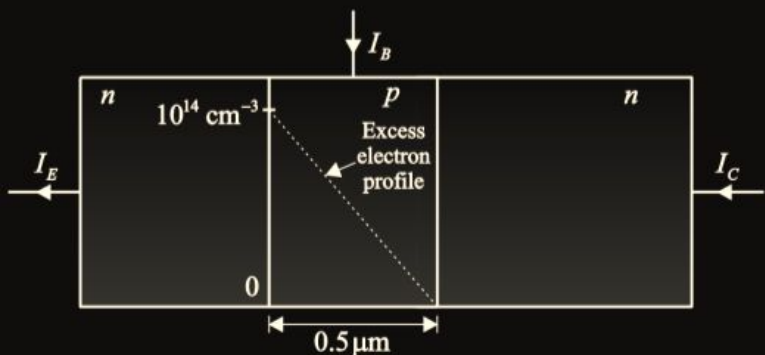
3.27 6.656

Given : The area of the emitter-base junction = 0.001 cm^2

$$\mu_n = 800 \text{ cm}^2/\text{V}\cdot\text{s}$$

$$V_T = 26 \text{ mV}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$



$$\frac{dn}{dx} = \frac{0 - 10^{14}}{0.5 \times 10^{-5} - 0} = \frac{-10^{14}}{0.5 \times 10^{-4}}$$

Electron current density is given by,

$$J_n = qD_n \frac{dn}{dx}$$

$$I = AqD_n \frac{dn}{dx}$$

where, $D_n = \mu_n V_T$

$$I = 0.001 \times 1.6 \times 10^{-19}$$

$$\times 800 \times 26 \times 10^{-3} \times \left[\frac{-10^{14}}{0.5 \times 10^{-4}} \right]$$

$$I = -66.56 \times 10^{-4} \text{ A} = -6.656 \text{ mA}$$

The negative sign indicates that the diffusion of electrons is opposite to direction of current constituted by them.

The electron current density at the base is negative as the minority charge carriers at the base are diffusing from higher concentration to lower concentration but the current constituted by them is opposite to the direction of diffusion. Direction of current due to these minority carrier is as same as conventional collector current. Hence, collector current is given by,

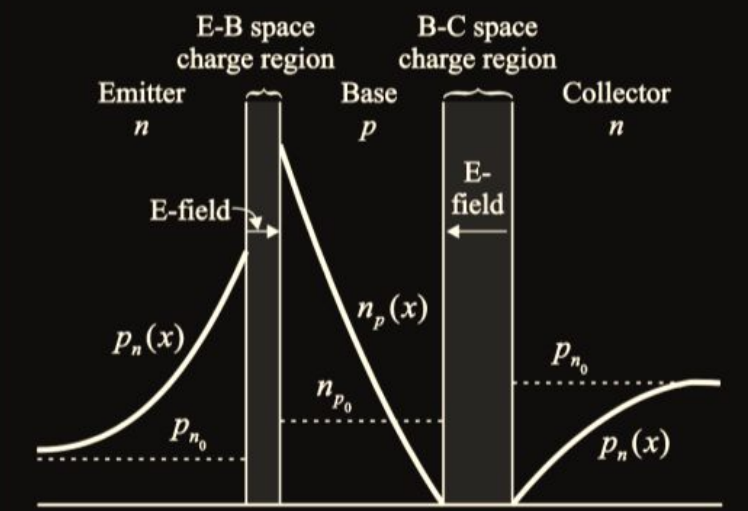
$$I_C = |I| = 6.656 \text{ mA}$$

Hence, the collector current I_C at room temperature is **6.656 mA**.

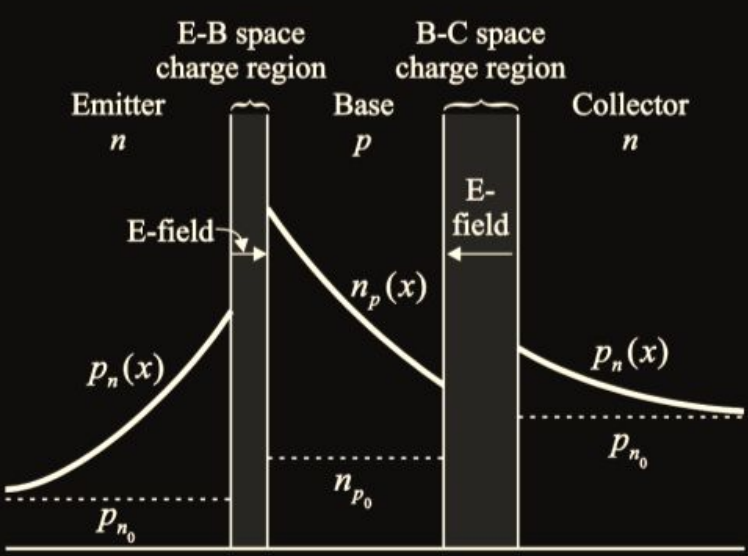
(i) For $n-p-n$ transistor :

Variation of minority carrier concentration (in different operating mode) is shown below,

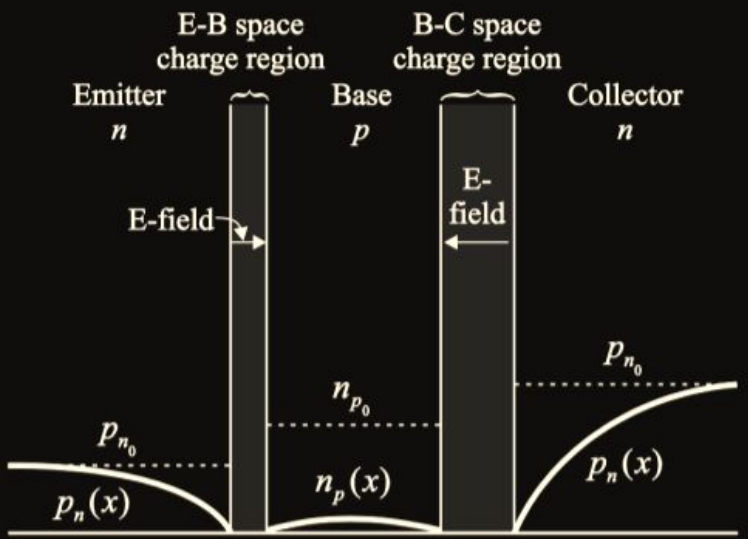
(a) Forward active mode :



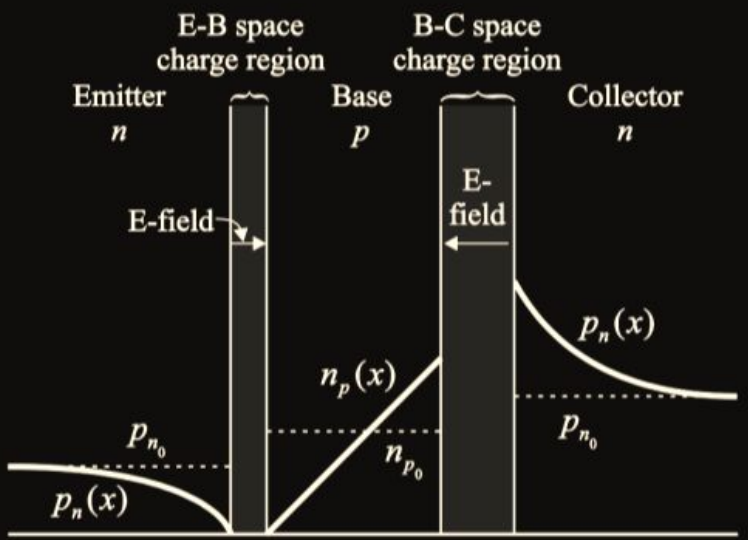
(b) Saturation mode :



(c) Cutoff mode :



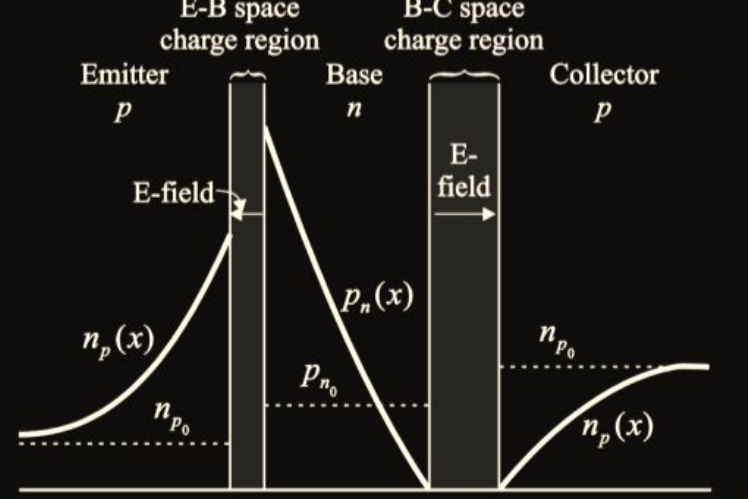
(d) Inverse active mode :



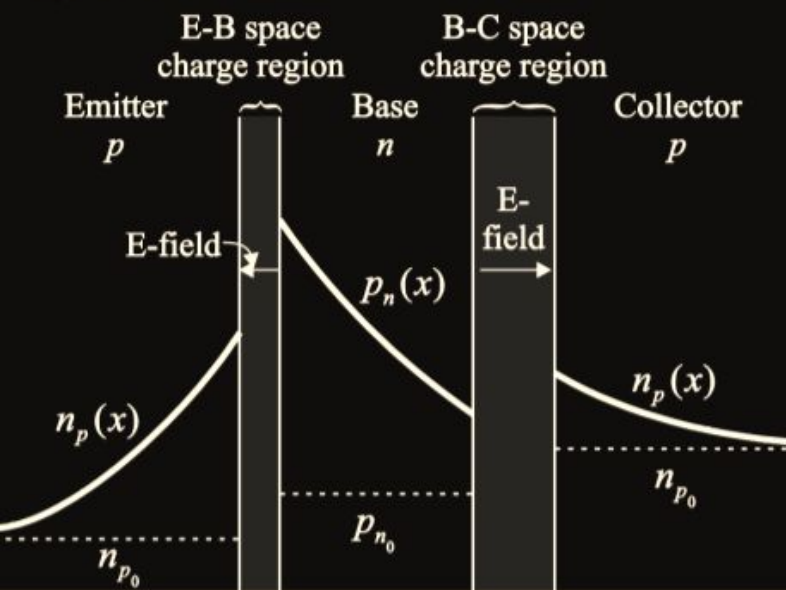
(ii) For $p-n-p$ transistor :

Variation of minority carrier concentration (in different operating mode) is shown below,

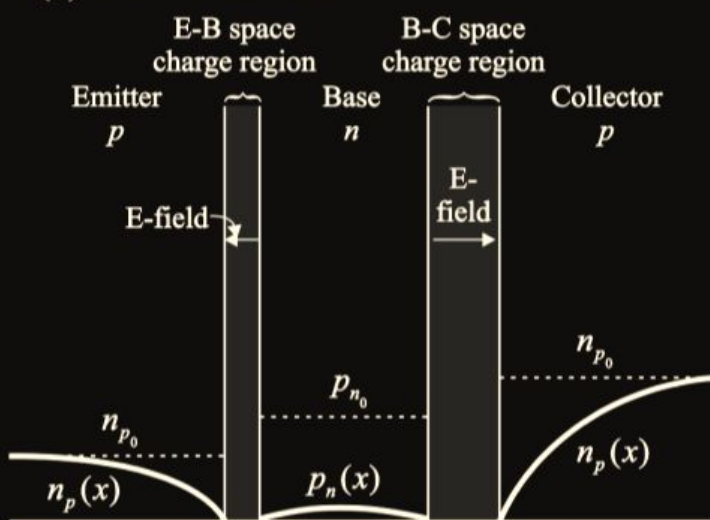
(a) Forward active mode :



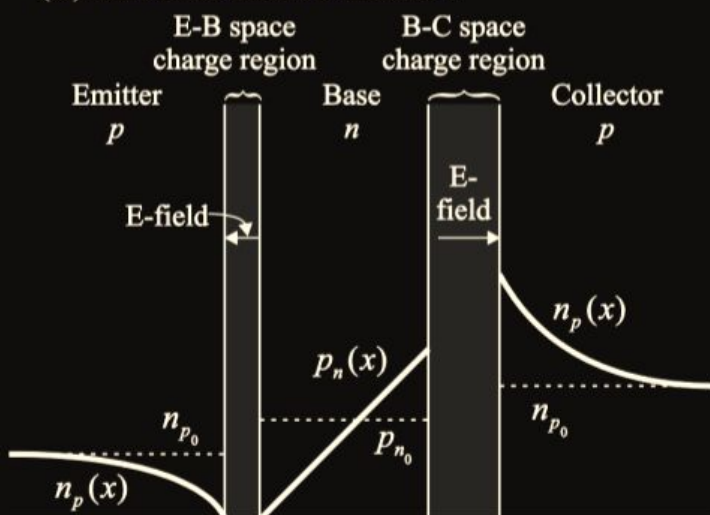
(b) Saturation mode :



(c) Cutoff mode :



(d) Inverse active mode :



3.29

An $n-p-n$ bipolar junction transistor (BJT) is operating in the active region. If the reverse bias across the base collector junction is increased, then [Set - 02]

- (A) the effective base width increases and common-emitter current gain increases.
- (B) the effective base width increases and common-emitter current gain decreases.
- (C) the effective base width decreases and common-emitter current gain increases.
- (D) the effective base width decreases and common-emitter current gain decreases.

3.29 (C)

If reverse bias across base collector junction will increase, then

- (i) Depletion width across base collector junction will increase.
- (ii) **Effective base width will decrease.**
- (iii) Recombination of carrier in base region will decrease.
- (iv) Base current I_B will decrease.
- (v) **Common emitter current gain** $\left(\beta = \frac{I_C}{I_B} \right)$ **will increase.**
- (vi) Collector current $[I_C = \beta I_B + (\beta + 1)I_{CO}]$ will increase, because increase in β dominates on decrease in I_B .

Hence, the correct option is (C).

3.31 Which of the following statements is/are true for a BJT with respect to its *DC* current gain β ?

(A) Under high-level injection condition in forward active mode, β will decrease with increase in the magnitude of collector current.

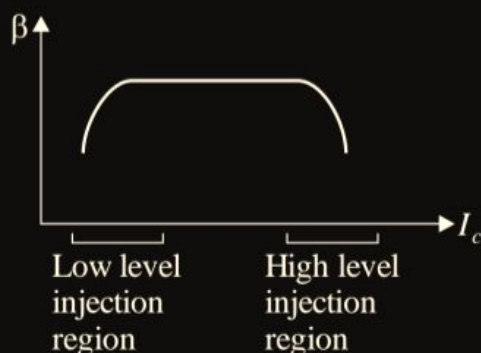
(B) Under low-level injection condition in forward active mode, where the current at the emitter-base junction is dominated by recombination-generation process, β will decrease with increase in the magnitude of collector current.

(C) β will be lower when the BJT is in saturation region compared to when it is in active region.

(D) A higher value of β will lead to a lower value of the collector-to-emitter breakdown voltage.

3.31 (A), (C), (D)

(i) For a BJT the plot of β and magnitude of collector current is



From the above plot we can conclude that.

- Under low level injection β increases with increase in the magnitude of collector current.
- Under high level injection β decreases with increase in the magnitude of collector current.

Hence, option (A) is correct and option (B) is not correct.

(ii) Since, $\beta_{sat} < \beta_{active}$

i.e., β is lower when BJT is in saturation region compared to when BJT is in active region.

Hence, option (C) is correct.

(iii) Breakdown voltage of collector to emitter region is given by

$$BV_{CEO} = BV_{CBO} \left(\frac{1}{\beta} \right)^{1/n}$$

From the above relationship. As β increases breakdown voltage of collector to emitter decreases.

Hence, option (D) is correct.

Hence, correct options are (A), (C) and (D).

MOSFET

The effective channel length decreases with an increase in V_{DS} in saturation region. This is also called **channel length modulation**.

- 5.10** For an n -channel enhancement type MOSFET, if the source is connected at a higher potential than that of the bulk (i.e. $V_{SB} > 0$), the threshold voltage V_{Th} of the MOSFET will
- (A) remain unchanged.
 - (B) decrease.
 - (C) change polarity.
 - (D) increase.

5.10 (D)

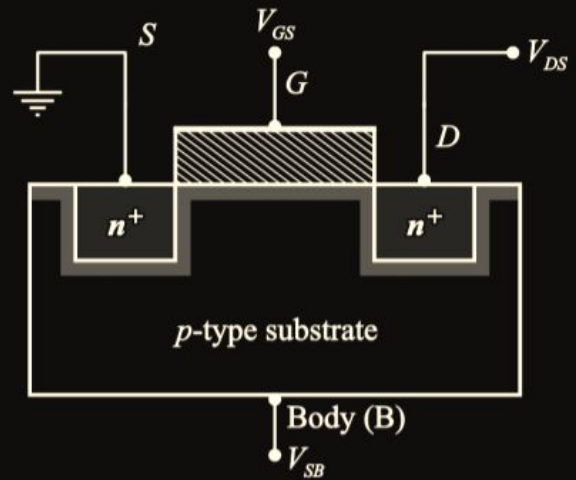
For n -channel enhancement type MOSFET, threshold voltage is given by,

$$V_{Th} = V_{T_0} + \gamma \left[\sqrt{2\phi_{Fp} + |V_{SB}|} - \sqrt{2\phi_{Fp}} \right] \quad \dots (i)$$

where, V_{SB} = Reverse bias voltage between source and body

$$V_{T_0} = \text{Threshold voltage for } V_{SB} = 0$$

ϕ_{Fp} = Physical parameter/Fermi potential of p -type substrate
 γ = Fabrication process parameter or body effect parameter



In the expression of threshold voltage, when source to body (bulk) voltage is zero,

Then threshold voltage, $V_{Th} = V_{T_0}$.

When V_{SB} is greater than zero (Given),

From equation (i),

$$V_{Th} > V_{T_0}$$

Therefore, threshold voltage increases when source to body voltage is applied i.e. when source is connected at a higher potential than bulk.

Hence, the correct option is (D).

5.18 Consider the following two statements about the internal conditions in an n -channel MOSFET operating in the active region.

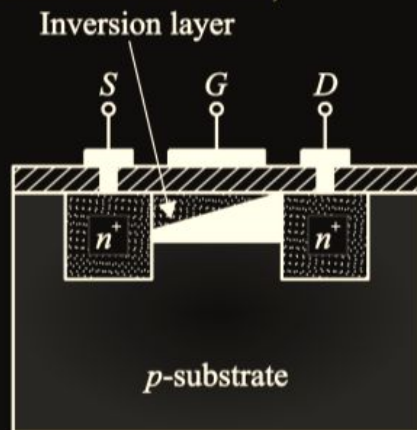
S_1 : The inversion charge decreases from source to drain.

S_2 : The channel potential increases from source to drain.

Which of the following is correct?

5.18 (D)

n -channel MOSFET with channel length modulation is shown below,



V_{DS} appears as a voltage drop across the length of the channel. i.e. as we travel along the channel from source to drain, the voltage (measured relative to the source) increases from 0 to V_{DS} .

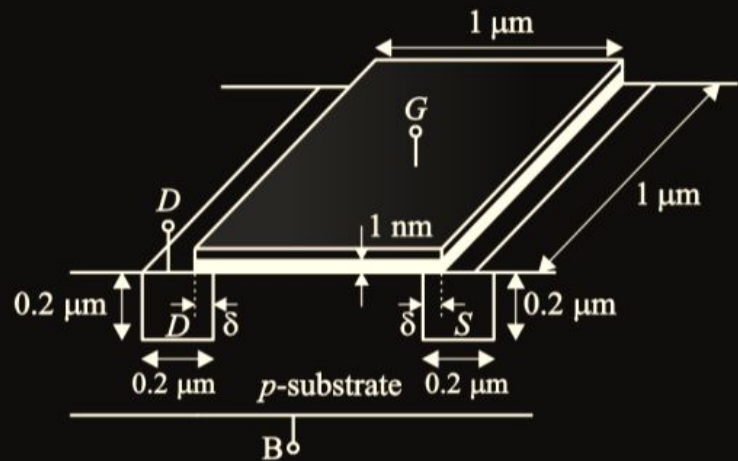
Hence, statement II is true.

Since, as the reverse bias voltage increases, the depletion width across the p - n junction will increase, hence depletion region width will increase from source to drain i.e. depletion region width will be lowest at the source end and it will be highest at drain end. As the depletion width will increase, the width of channel region i.e. inversion region will decrease. Therefore, the width of inversion region will decrease from source to drain.

Therefore, statement 1 and statement 2, both are correct and statement 2 is correct reason of statement 1.

Hence, the correct option is (D).

In the three dimensional view of a silicon n -channel MOS transistor shown below, $\delta = 20$ nm. The transistor is of width $1\ \mu\text{m}$. The depletion width formed at every p - n junction is 10 nm. The relative permittivities of Si and SiO_2 , respectively, are 11.7 and 3.9 , and $\epsilon_0 = 8.9 \times 10^{-12} \text{ F/m}$.



5.22 The source-body junction capacitance is approximately

- (A) 2 fF (B) 7 fF
(C) 2 pF (D) 7 pF

5.23 The gate-source overlap capacitance is approximately

- (A) 0.7 fF (B) 0.7 pF
(C) 0.35 fF (D) 0.24 pF

5.22 (B)

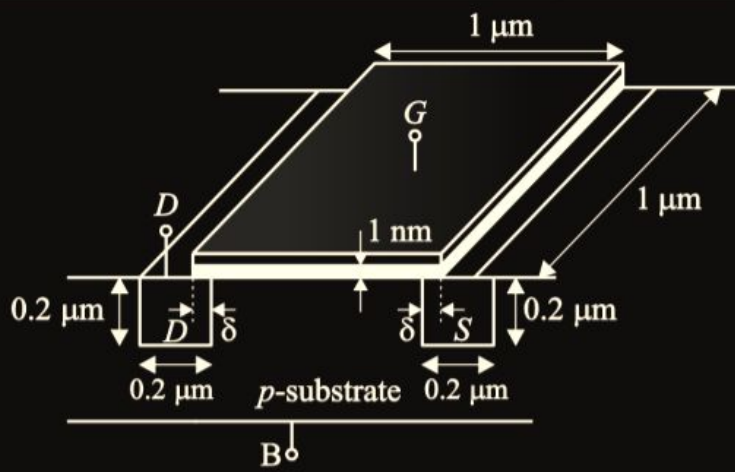
Given : $\delta = 20$ nm

Width of transistor = $1\ \mu\text{m}$

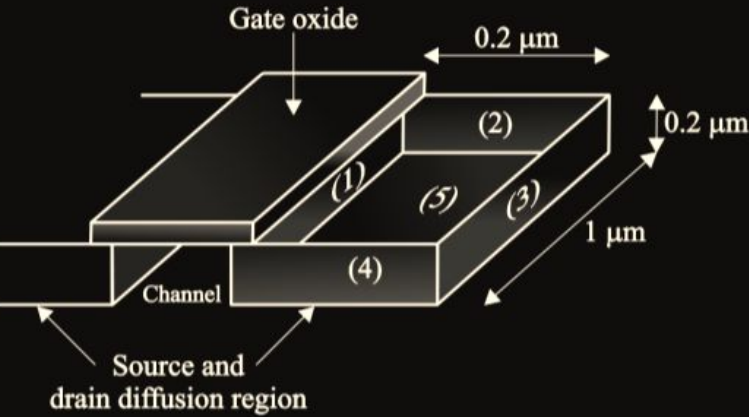
Depletion width at every p - n junction
= 10 nm.

$$\epsilon_{r(\text{Si})} = 11.7, \epsilon_{r(\text{SiO}_2)} = 3.9$$

$$\epsilon_0 = 8.9 \times 10^{-12} \text{ F/m}$$



Source is inserted into *p*-type substrate. 3-dimensional view of source is shown below, which has 6 faces.



All five faces of source is shown in above figure. The sixth face is the top face. Since, it is not possible to show top face, hence we have not shown sixth face.

It has rectangular shape in 3-dimensional view. Except for the top face, all the faces are in connection with the *p*-type substrate and forming the *p-n* junction diode.

Source body junction capacitance is given by,

$$C = \frac{\epsilon A}{d} = \frac{\epsilon_0 \epsilon_{r(Si)} A}{d} \quad \dots (i)$$

where,

d = Width of depletion region = 10 nm

$A = A_1 + A_2 + A_3 + A_4 + A_5$ [Given]

$A = 2A_1 + 2A_2 + A_5$

[Since $A_1 = A_3$, $A_2 = A_4$]

$A = 2[0.2 \times 1] + 2[0.2 \times 0.2] + 1[0.2 \times 1]$

$A = 0.68 \mu m^2$

From equation (i),

$$C = \frac{11.7 \times 8.9 \times 10^{-12} \times 0.68 \times 10^{-12}}{10^{-8}}$$

$$C = 7 \times 10^{-15} \text{ F} = 7 \text{ fF}$$

Hence, the correct option is (B).

5.23 (A)

Gate-source overlap capacitance is given by,

$$C_{ov} = W L_{ov} C_{ox} \quad \dots (i)$$

where, W = Width of transistor

$$W = 1 \mu m = 10^{-6} \text{ m}$$

L_{ov} = overlap length

$$L_{ov} = \delta = 20 \text{ nm} = 20 \times 10^{-9} \text{ m}$$

C_{ox} = Oxide capacitance per unit area

$$C_{ox} = \frac{\epsilon_0 \epsilon_{r(SiO_2)}}{t_{ox}} \text{ F/m}^2 \quad \dots (ii)$$

where, $t_{ox} = 1 \text{ nm} = 10^{-9} \text{ m}$

From equations (i) and (ii),

$$C_{ov} = W L_{ov} \times \frac{\epsilon_0 \epsilon_{r(SiO_2)}}{t_{ox}}$$

$$C_{ov} = 10^{-6} \times 20 \times 10^{-9} \times \frac{8.9 \times 10^{-12} \times 3.9}{10^{-9}}$$

$$C_{ov} = 694.2 \times 10^{-18} \text{ F}$$

$$C_{ov} = 0.6942 \times 10^{-15} \text{ F}$$

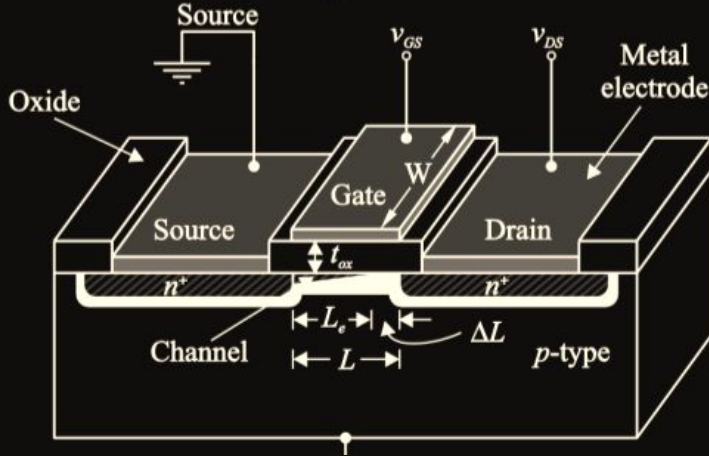
$$C_{ov} \approx 0.7 \text{ fF}$$

Hence, the correct option is (A).

- 5.24** In a MOSFET operating in the saturation region, the channel length modulation effect causes
- an increase in the gate-source capacitance.
 - a decrease in the transconductance.
 - a decrease in the unity-gain cutoff frequency.
 - a decrease in the output resistance.

5.24 (D)

Effect of channel length modulation :



From above figure,

$$L_e = L - \Delta L$$

Because of channel length modulation, effective channel length will reduce.

- In saturation region drain current is given by,

$$I_D = \frac{1}{2} k_n (V_{GS} - V_{Th})^2$$

where, $k_n = \mu_n C_{ox} \frac{W}{L}$

$$I_D \propto \frac{1}{L}$$

Hence, drain current I_D increases.

- Output resistance, $r_0 = \frac{1}{\lambda I_D}$

Hence, $r_0 \propto \frac{1}{I_D}$

Therefore, output resistance r_0 will decrease.

- Slope of output characteristic $= \frac{1}{r_0}$

Hence, slope of output characteristic will increase.

- Transconductance is given by,

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = K'_n \frac{W}{L} (V_{GS} - V_{Th})$$

Therefore, $g_m \propto \frac{1}{L}$

Hence, transconductance g_m will increase.

- Unity gain cutoff frequency is given by,

$$f_T = \frac{g_m}{2\pi(C_{GS} + C_{GD})} = \frac{g_m}{2\pi C_G} \quad \dots (i)$$

where,

$$C_G = C_{GS} + C_{GD}$$

= Total gate to channel capacitance $\dots (ii)$

$$C_G = C_{ox} \times \text{Gate area}$$

$$C_G = C_{ox} \times WL \quad \dots (iii)$$

where, C_{ox} = Gate oxide capacitance

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} \quad \dots (iv)$$

where, ϵ_{ox} = Oxide permittivity

t_{ox} = Oxide thickness

Transconductance,

$$g_m = k_n (V_{GS} - V_{Th})$$

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{Th}) \quad \dots (v)$$

From equation (i), (iii) and (v),

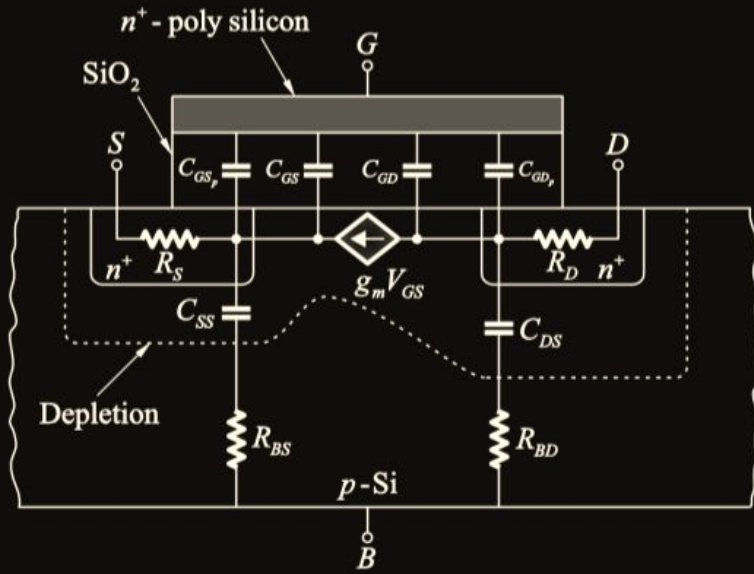
$$f_T = \frac{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{Th})}{2\pi C_{ox} \times WL}$$

$$f_T = \frac{\mu_n}{2\pi L^2} (V_{GS} - V_{Th})$$

Hence, $f_T \propto \frac{1}{L^2}$

Therefore, f_T will increase.

6. Dependence of capacitance on channel length modulation :



(i) Gate to source capacitance, C_{GS} and gate to drain capacitance, C_{GD} ;

(a) If device is in non-saturation region and V_{DS} is small, then the channel inversion charge is approximately uniform.

In this region, $C_{GS} = C_{GD} = \frac{1}{2} WLC_{ox}$

Hence, $C_{GD} \propto L$ and $C_{GS} \propto L$

Therefore, C_{GS} and C_{GD} will decrease.

(b) If device is in saturation region, then the channel is pinched-off at the drain and the inversion charge is no longer uniform.

Hence, in this region,

$$C_{GD} = 0$$

$$C_{GS} = \frac{2}{3} WLC_{ox}$$

Therefore, C_{GS} will decrease and C_{GD} will not change.

(ii) Gate to channel capacitance, C_G ;

$$C_G = C_{GS} + C_{GD}$$

$$C_G = C_{ox} \times \text{Gate area}$$

$$C_G = \frac{\epsilon_{ox}}{t_{ox}} \times WL$$

Hence, $C_G \propto L$

Therefore, C_G will decrease

(iii) Drain to substrate capacitance, C_{DS} ;

It is the drain to substrate pn junction capacitance and it does not depend on channel length. Hence, due to channel length modulation C_{DS} will not change.

(iv) Gate to source parasitic or overlap capacitance, C_{GS_p} ;

It exist because of gate oxide overlap with source, because of tolerance or other fabrication factor. It does not depend on channel length. Hence due to channel length modulation, it will not change.

(v) Gate to drain parasitic or overlap capacitance, C_{GD_p} ;

It exist because of gate oxide overlap with drain, because of tolerance or other fabrication factor. It does not depend on channel length. Hence due to channel length modulation, it will not change.

Hence, the correct option is (D).

Key Point

- (i) When there is no channel length modulation, $\lambda = 0$ and hence $r_0 = \infty$.
- (ii) Channel length modulation occurs only in E-MOSFET.
- (iii) Channel length modulation does not exist in depletion MOSFET.

5.33 A MOSFET in saturation has a drain current of 1 mA for $V_{DS} = 0.5$ V. If the channel length modulation coefficient is 0.05 V^{-1} , the output resistance (in $\text{k}\Omega$) of the MOSFET is _____. [Set - 01]

5.33 20

Given : MOSFET is in saturation region has,

- (i) $I_D = 1 \text{ mA}$ for $V_{DS} = 0.5 \text{ V}$
- (ii) Channel length modulation coefficient, $\lambda = 0.05 / \text{V}$

Drain current (with channel length modulation effect) is given by,

$$I_D' = I_D(1 + \lambda V_{DS}) \quad \dots (i)$$

where, I_D = Drain current without channel length modulation effect

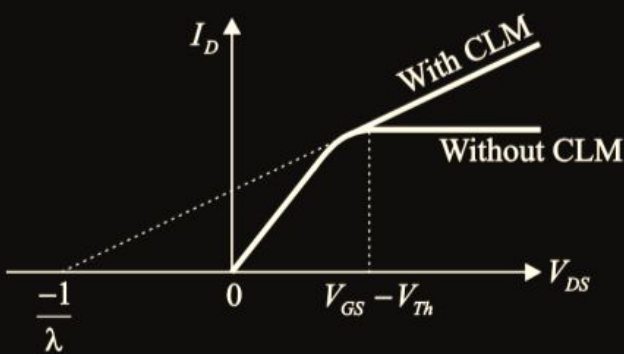
Drain saturation current (without channel length modulation effect) is given by,

$$I_D = \frac{k_n}{2} (V_{GS} - V_{Th})^2 \quad \dots (ii)$$

From equations (i) and (ii),

$$I_D' = \frac{k_n}{2} (V_{GS} - V_{Th})^2 (1 + \lambda V_{DS}) \quad \dots (iii)$$

Output characteristics of MOSFET is shown below,



Output resistance is given by,

$$r_0 = r_{ds} = r_{d(\text{ON})} = \frac{\partial V_{DS}}{\partial I_D'} \quad \dots (iv)$$

From equation (iii),

$$\frac{\partial I_D'}{\partial V_{DS}} = \frac{k_n}{2} (V_{GS} - V_{Th})^2 \times \lambda \quad \dots (v)$$

From equation (iv) and (v),

$$r_0 = \frac{1}{\frac{k_n}{2} (V_{GS} - V_{Th})^2 \times \lambda} = \frac{1}{10^{-3} \times 0.05}$$

[From equation (ii), $I_D = 1 \text{ mA}$]

$$r_0 = 20 \text{ k}\Omega$$

Hence, the output resistance of the MOSFET is **20 k Ω** .

5.37 Consider the following statements for a metal oxide semiconductor field effect transistor (MOSFET) : [Set - 01]

P : As channel length reduces, OFF-state current increases.

Q : As channel length reduces, output resistance increases.

R : As channel length reduces, threshold voltage remains constant.

S : As channel length reduces, ON current increases.

Which of the above statements are **INCORRECT**?

(A) P and Q

(B) P and S

(C) Q and R

(D) R and S

P : As channel length reduces, OFF-state current increases.

R : As channel length reduces, threshold voltage remains constant.

OFF-state current is defined as a small amount of current which flows from drain to source (NMOS) when channel is not completely formed i.e. OFF-state current flows for $0 < V_{GS} < V_{Th}$.

From figure (a) and (b),

The charge carrier density below the oxide layer is greater in MOSFET with channel length L_2 ($L_2 \ll L_1$).

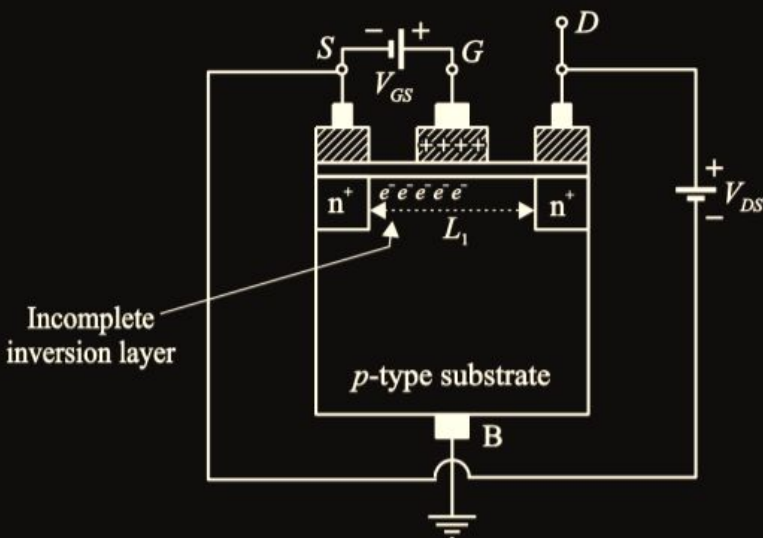


Fig. (a) NMOS transistor

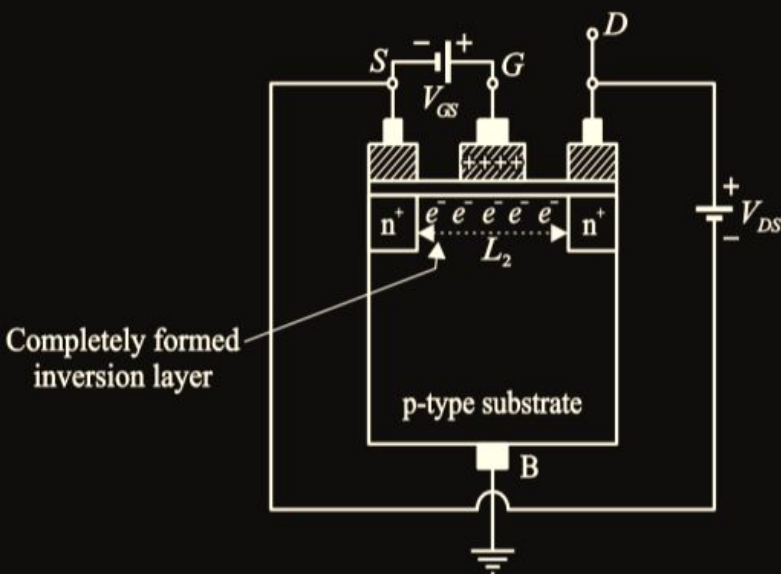


Fig. (b) NMOS transistor

Hence,

(i) For $0 < V_{GS} < V_{Th}$, due to increased charge density in figure (b), the OFF-state current will increase.

(ii) Channel will induce faster in figure (b) and current will start to flow for smaller value of V_{GS} i.e. threshold voltage will be reduced.

Hence, statement **P** is correct and statement **R** is incorrect.

Q : As channel length reduces, output resistance increases.

S : As channel length reduces, ON current increases.

In saturation region, drain current is given by,

$$I_D = \frac{1}{2} k_n (V_{GS} - V_{Th})^2$$

$$\text{where, } k_n = \mu_n C_{ox} \frac{W}{L}$$

$$\text{ON current, } I_D \propto \frac{1}{L} \quad \dots (i)$$

Hence, as channel length reduces, drain current I_D (ON current) increases.

Output resistance is given by,

$$r_o = \frac{1}{\lambda I_D} \quad \dots (ii)$$

From equation (i) and (ii),

$$r_o \propto L$$

From above expression, output resistance decreases with decrease in channel length.

Hence, statement **Q** is incorrect and statement **S** is correct.

Hence, the correct option is (C).

$$\text{Transconductance in MOSFET} = g_m = \frac{\partial I_D}{\partial V_{GS}}$$

$$\text{Drain conductance in MOSFET} = g_d = \frac{\partial I_D}{\partial V_{DS}}$$

5.39 A long-channel NMOS transistor is biased in the linear region with $V_{DS} = 50 \text{ mV}$ and is used as a resistance. Which one of the following statements is NOT correct? **[Set - 01]**

- (A) If the device width W is increased, the resistance decreases.
- (B) If the threshold voltage is reduced, the resistance decreases.
- (C) If the device length L is increased, the resistance increases.
- (D) If V_{GS} is increased, the resistance increases.

For NMOS transistor, drain current in linear region is given by,

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} [2(V_{GS} - V_{Th})V_{DS} - V_{DS}^2]$$

For linear region, $V_{DS} < V_{GS} - V_{Th}$

$$\text{Hence, } I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} [2(V_{GS} - V_{Th})V_{DS}]$$

Differentiating above expression with respect to V_{DS} ,

$$\frac{\partial I_D}{\partial V_{DS}} = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{Th})$$

Hence, for NMOS transistor r_{ds} in linear region is given by,

$$\frac{\partial V_{DS}}{\partial I_D} = r_{ds} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{Th})} \dots (i)$$

where, r_{ds} is Channel Resistance.

From the above equation (i), if

- (i) W increases then r_{ds} decreases.
- (ii) V_{Th} decreases then r_{ds} decreases.
- (iii) L increases then r_{ds} increases.
- (iv) V_{GS} increases then r_{ds} decreases.

Hence, the correct option is (D).

5.42 Consider an n -channel MOSFET having width W , length L , electron mobility in the channel μ_n and oxide capacitance per unit area C_{ox} . If gate-to-source voltage $V_{GS} = 0.7 \text{ V}$, drain to source voltage $V_{DS} = 0.1 \text{ V}$, $\mu_n C_{ox} = 100 \mu\text{A/V}^2$, threshold voltage $V_{Th} = 0.3 \text{ V}$ and

$\frac{W}{L} = 50$, then the transconductance g_m (in mA/V) is _____. [Set - 01]

5.42 0.5

Given : For n -channel MOSFET,

- (i) $V_{DS} = 0.1 \text{ V}$, $V_{GS} = 0.7 \text{ V}$
- (ii) $V_{Th} = 0.3 \text{ V}$
- (iii) $\mu_n C_{ox} = 100 \mu\text{A/V}^2 = 0.1 \text{ mA/V}^2$,
- (iv) $\frac{W}{L} = 50$

From given data,

Over-drive voltage, $V_{OV} = V_{GS} - V_{Th} = 0.4 \text{ V}$

Hence, $V_{GS} > V_{Th}$ and $V_{DS} < V_{GS} - V_{Th}$

So, n -channel MOSFET is in linear region.

In linear region, drain current is given by,

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{Th}) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

Transconductance is given by,

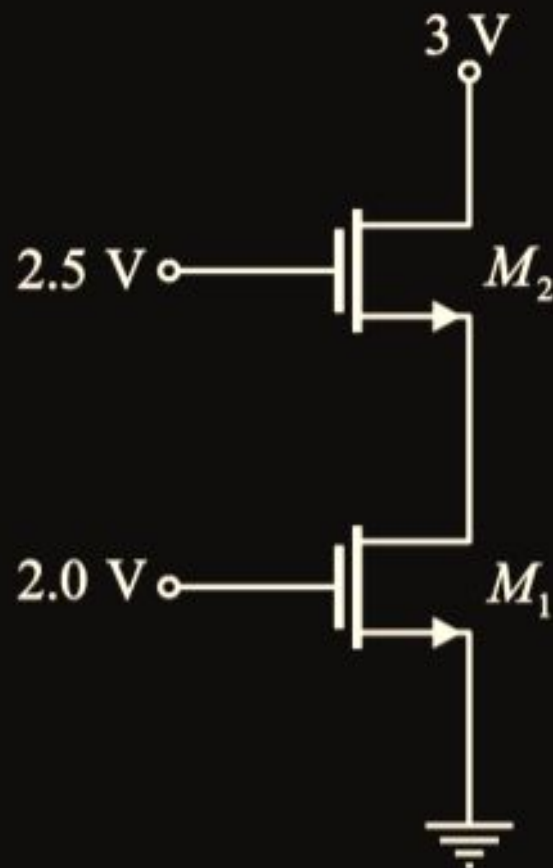
$$g_m = \frac{\partial I_D}{\partial V_{GS}}$$

Hence, $g_m = \mu_n C_{ox} \frac{W}{L} V_{DS}$

$$g_m = 0.1 \times 50 \times 0.1 = 0.5 \text{ mA/V}$$

Hence, the transconductance g_m is **0.5 mA/V**.

5.45 Assuming that transistors M_1 and M_2 are identical and have a threshold voltage of 1 V , the state of transistors M_1 and M_2 are respectively [Set - 02]



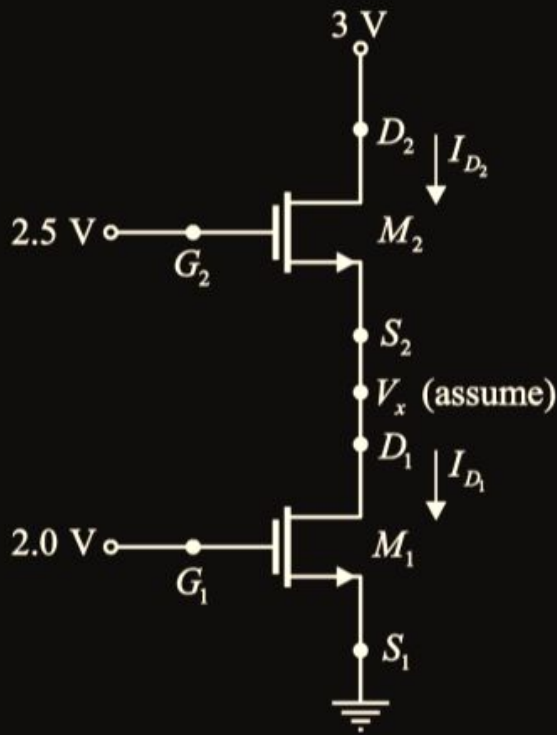
(A) saturation, saturation

(B) linear, linear

(C) linear, saturation

(D) saturation, linear

Given circuit is shown below,



(i) $M_1 = \text{NMOS}$

(ii) $M_2 = \text{NMOS}$

(iii) $V_{Th} = 1 \text{ V}$

For transistor M_2 :

$$V_{DS_2} = V_{D_2} - V_{S_2} = 3 - V_x$$

$$V_{GS_2} = V_{G_2} - V_{S_2} = 2.5 - V_x$$

Over-drive voltage,

$$V_{OV_2} = V_{GS_2} - V_{Th} = 2.5 - V_x - 1$$

$$V_{OV_2} = 1.5 - V_x$$

Since, $V_{GS_2} > V_{Th}$ and $V_{DS_2} \geq V_{OV_2}$

So, transistor M_2 is in saturation.

For transistor M_1 :

Assume M_1 is also in saturation.

$$V_{DS_1} = V_{D_1} - V_{S_1} = V_x - 0 = V_x$$

$$V_{GS_1} = V_{G_1} - V_{S_1} = 2 - 0 = 2 \text{ V} \quad \dots (i)$$

$$V_{OV_1} = V_{GS_1} - V_{Th} = 1 \quad \dots (ii)$$

Since, transistors are connected in series,

$$\text{Hence, } I_{D_2}|_{\text{saturation}} = I_{D_1}|_{\text{saturation}}$$

$$K_n (V_{GS_2} - V_{Th})^2 = K_n (V_{GS_1} - V_{Th})^2$$

$$K_n (2.5 - V_x - 1)^2 = K_n (2 - 1)^2$$

$$1.5 - V_x = \pm 1$$

$$\text{Hence, } V_x = 0.5 \text{ V, } 2.5 \text{ V}$$

Case 1 : For $V_x = 2.5 \text{ V}$;

For transistor M_2 ,

$$V_{GS_2} = V_{G_2} - V_{S_2} = V_{G_2} - V_x$$

$$V_{GS_2} = 2.5 - 2.5 = 0 \text{ V}$$

$$\text{Hence, } V_{GS_2} < V_{Th}$$

So, M_2 is in cutoff region.

$$\text{So, } V_x \neq 2.5 \text{ V}$$

Case 2 : For $V_x = 0.5 \text{ V}$;

(i) For transistor M_2 ,

$$V_{GS_2} = V_{G_2} - V_{S_2} = V_{G_2} - V_x$$

$$V_{GS_2} = 2.5 - 0.5 = 2 \text{ V}$$

$$V_{DS_2} = V_{D_2} - V_{S_2}$$

$$V_{DS_2} = 3 - 0.5 = 2.5 \text{ V}$$

$$V_{OV_2} = V_{GS_2} - V_{Th}$$

$$V_{OV_2} = 2 - 1 = 1 \text{ V}$$

$$\text{Hence, } V_{GS_2} > V_{Th} \text{ and } V_{DS_2} > V_{OV_2}$$

So, M_2 is in saturation region.

(ii) For transistor M_1 ,

$$V_{GS_1} = V_{G_1} - V_{S_1}$$

$$V_{GS_1} = 2 - 0 = 2 \text{ V}$$

$$V_{DS_1} = V_{D_1} - V_{S_1} = V_x - 0$$

$$V_{DS_1} = 0.5 \text{ V} \quad \dots \text{ (iii)}$$

From equation (i), (ii) and (iii),

Hence, $V_{GS_1} > V_{Th}$ and $V_{DS_1} < V_{OV_1}$

So, our assumption is wrong and hence, M_1 is in linear region.

Hence, the correct option is (C).

5.48 For an n channel silicon MOSFET with 10 nm gate oxide thickness, the substrate sensitivity $(\partial V_T / \partial |V_{BS}|)$ is found to be 50 mV/V at a substrate voltage $|V_{BS}| = 2 \text{ V}$, where V_T is the threshold voltage of the MOSFET. Assume that $|V_{BS}| \gg 2\phi_B$, where $q\phi_B$ is the separation between the fermi energy level E_F and the intrinsic level E_i in the bulk. Parameters given are :

Electron charge $(q) = 1.6 \times 10^{-19} \text{ C}$

Vacuum permittivity,

$$(\epsilon_0) = 8.85 \times 10^{-12} \text{ F/m}$$

Relative permittivity of silicon $(\epsilon_{si}) = 12$

Relative permittivity of oxide $(\epsilon_{ox}) = 4$

The doping concentration of the substrate is

(A) $7.37 \times 10^{15} \text{ cm}^{-3}$

(B) $4.37 \times 10^{15} \text{ cm}^{-3}$

(C) $2.37 \times 10^{15} \text{ cm}^{-3}$

(D) $9.37 \times 10^{15} \text{ cm}^{-3}$

5.48 (A)

Given :

Gate-oxide thickness, $t_{ox} = 10 \text{ nm}$

Substrate sensitivity, $\frac{\partial V_T}{\partial |V_{BS}|} = 50 \text{ mV/V}$

Substrate voltage, $|V_{BS}| = 2 \text{ V}$

Electron charge $(q) = 1.6 \times 10^{-19} \text{ C}$

Vacuum permittivity $(\epsilon_0) = 8.85 \times 10^{-12} \text{ F/m}$

Relative permittivity of silicon $(\epsilon_{si}) = 12$

Relative permittivity of oxide $(\epsilon_{ox}) = 4$

Threshold voltage when body effect is consider,

$$V_T = V_{To} + \gamma \left[\sqrt{|2\phi_F| + |V_{SB}|} - \sqrt{|2\phi_F|} \right] \quad \dots \text{(i)}$$

$$\text{Where, } \gamma = \frac{\sqrt{2qN_A \epsilon_{si}}}{C_{ox}} = \frac{\sqrt{2qN_A \epsilon_{si}}}{\epsilon_{ox}} t_{ox}$$

$$\left[\because C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} \right]$$

As per the question, $\phi_F = \phi_B$

(Both are same)

Differentiate equation (i), with respect to V_{BS} ,

$$\frac{\partial V_T}{\partial V_{BS}} = 0 + \gamma \frac{1}{2} (|2\phi_F| + |V_{SB}|)^{-\frac{1}{2}} \quad \dots(ii)$$

Since $|V_{SB}| \gg |2\phi_F|$, so equation (ii) becomes as,

$$\frac{\partial V_T}{\partial V_{BS}} = \gamma \left(\frac{1}{2} \right) (|V_{SB}|)^{-\frac{1}{2}}$$

$$50 \times 10^{-3} = \gamma \left(\frac{1}{2} \right) (2)^{-\frac{1}{2}}$$

$$50 \times 10^{-3} = \gamma \frac{1}{2\sqrt{2}}$$

$$\gamma = 2\sqrt{2} \times 50 \times 10^{-3} = 0.14142$$

$$\text{As } \gamma = \sqrt{2qN_A \epsilon_{si}} \left[\frac{t_{ox}}{\epsilon_{ox}} \right]$$

$$\sqrt{2qN_A \epsilon_{si}} \times \frac{t_{ox}}{\epsilon_{ox}} = 0.14142$$

$$2qN_A \epsilon_{si} = \left[0.14142 \times \left(\frac{\epsilon_{ox}}{t_{ox}} \right) \right]^2$$

$$N_A = \frac{1}{2q(12\epsilon_0)} \times \left[0.14142 \times \left(\frac{4\epsilon_0}{t_{ox}} \right) \right]^2$$

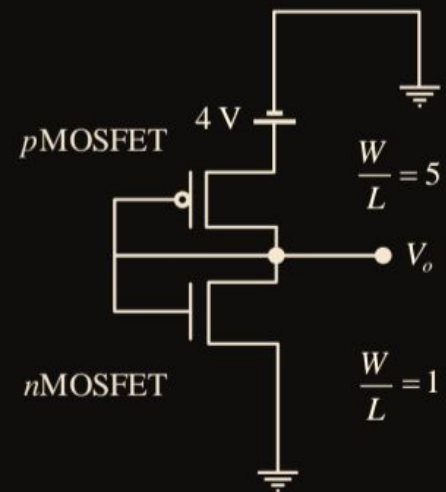
$(\because \epsilon_{si} = 12\epsilon_0, \epsilon_{ox} = 4\epsilon_0)$

$$N_A = \frac{(0.14142)^2 \times (4 \times 8.85 \times 10^{-14})^2}{2 \times 1.6 \times 10^{-19} \times 12 \times 8.55 \times 10^{-14} \times (10 \times 10^{-9} \times 100)^2}$$

$$N_A = 7.37 \times 10^{15} \text{ cm}^{-3}$$

Hence, the correct option is (A).

5.49 Consider the CMOS circuit shown in the figure (substrates are connected to their respective sources). The gate width (W) to gate length (L) ratios $\left(\frac{W}{L} \right)$ of the transistors are as shown. Both the transistors have the same gate oxide capacitance per unit area. For the p MOSFET, the threshold voltage is -1 V and the mobility of holes is $40 \frac{\text{cm}^2}{\text{V-s}}$. For the n MOSFET, the threshold voltage is 1 V the mobility of electrons is $300 \frac{\text{cm}^2}{\text{V-s}}$. The steady state output voltage V_o is



- (A) Equal to 0 V (B) Equal to 2 V
(C) More than 2 V (D) Less than 2 V

5.49 (D)

Given :

- (i) Both transistors have same gate oxide capacitance
(ii) $V_{Th_p} = -1$ V
(iii) $V_{Th_N} = 1$ V
(iv) $\mu_p = 40 \frac{\text{cm}^2}{\text{V-s}}$

$$(v) \quad \mu_n = 300 \frac{\text{cm}^2}{\text{V}\cdot\text{s}}$$

$$(vi) \quad \left(\frac{W}{L}\right)_p = 5$$

$$(vii) \quad \left(\frac{W}{L}\right)_n = 1$$

For n MOSFET :

Condition for saturation is given by,

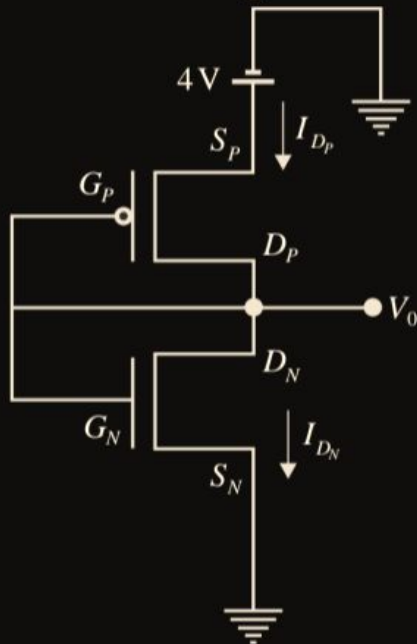
$$V_{DS_N} \geq V_{GS_N} - V_{Th_N}$$

$$V_{D_N} - V_{S_N} \geq V_{G_N} - V_{S_N} - V_{Th_N}$$

$$V_{D_N} \geq V_{G_N} - V_{Th_N}$$

$$V_{D_N} - V_{G_N} \geq -V_{Th_N}$$

If this inequality is satisfied then n MOSFET will be operate in saturation region.



From given circuit, $V_{G_N} = V_{D_N}$

$$\text{So, } V_{D_N} - V_{G_N} \geq -V_{Th_N}$$

$$0 \geq -1$$

Therefore n MOSFET is operated in saturation region.

For p MOSFET :

Condition for saturation is given by

$$V_{SD_p} \geq V_{SG_p} - |V_{Th_p}|$$

$$V_{S_p} - V_{D_p} \geq V_{S_p} - V_{G_p} - |V_{Th_p}|$$

$$V_{G_p} - V_{D_p} \geq -|V_{Th_p}|$$

If this inequality is satisfied then p MOSFET will be operate in saturation region.

From given circuit, $V_{G_p} = V_{D_p}$

$$\text{Hence, } V_0 \geq -|V_{Th_p}|$$

$$0 \geq -1$$

Therefore p MOSFET is operated in saturation region.

We can say that, whenever gate and drain terminal of n MOSFET and p MOSFET connected together, both MOSFET will be operated in saturation region.

Since, current flow through gate terminal of MOSFETs is zero, hence both MOSFET will have same drain current.

$$I_{D_N} = I_{D_p}$$

$$\begin{aligned} \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_n (V_{GS} - V_{Th_N})^2 \\ = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L}\right)_p (V_{SG} - |V_{Th_p}|)^2 \end{aligned}$$

$$\begin{aligned} \frac{1}{2} \times 300 \times C_{ox} \times 1 (V_0 - 1)^2 \\ = \frac{1}{2} \times 40 \times C_{ox} \times 5 (4 - V_0 - 1)^2 \end{aligned}$$

$$300(V_0 - 1)^2 = 40 \times 5 (4 - V_0 - 1)^2$$

$$3(V_0 - 1)^2 = 2(3 - V_0)^2$$

$$3V_0^2 + 3 - 6V_0 = 18 + 2V_0^2 - 12V_0$$

$$V_0^2 + 6V_0 - 15 = 0$$

$$V_0 = \frac{-6 \pm \sqrt{36 + 60}}{2}$$

$$V_0 = 1.89\text{V}, -7.89\text{V}$$

Taking negative value of V_0 .

$$V_0 = -7.89\text{V}$$

For n MOSFET :

$$V_0 = V_{G_N} = -7.89\text{V}$$

$$V_{S_N} = 0\text{V}$$

$$V_{GS_N} = V_{G_N} - V_{S_N}$$

$$V_{GS_N} = -7.89 \text{ V} - 0 \text{ V} = -7.89 \text{ V}$$

$$V_{Th_N} = 1 \text{ V}$$

$$V_{GS_N} < V_{Th_N}$$

Hence, n MOSFET will be OFF and drain current $I_{D_N} = 0$.

Since, drain current through n MOSFET is zero so we do not need to check whether p MOSFET is ON or OFF.

Taking positive value of V_0 .

$$V_0 = 1.89 \text{ V}$$

For n MOSFET :

$$V_0 = V_{G_N} = 1.89 \text{ V}$$

$$V_{D_N} = 1.89 \text{ V}$$

$$V_{GS_N} = V_{G_N} - V_{S_N} = 1.89 \text{ V} - 0 \text{ V}$$

$$V_{GS_N} = 1.89 \text{ V}$$

$$V_{Th_N} = 1 \text{ V}$$

$$V_{GS_N} > V_{Th_N}$$

Hence, n MOSFET will be ON for $V_0 = 1.89 \text{ V}$

Now we will check n MOSFET is operate either in saturation or linear region

$$V_{G_N} = 1.89 \text{ V}$$

$$V_{D_N} = 1.89 \text{ V}$$

$$V_{S_N} = 0 \text{ V}$$

$$V_{DS_N} = V_{D_N} - V_{S_N} = 1.89 \text{ V} - 0 \text{ V} = 1.89 \text{ V}$$

$$V_{GS_N} = V_{G_N} - V_{S_N} = 1.89 \text{ V} - 0 \text{ V} = 1.89 \text{ V}$$

The condition for saturation is given by,

$$V_{DS_N} \geq V_{GS_N} - V_{Th_N}$$

$$V_{DS_N} = 1.89 \text{ V}$$

$$V_{GS_N} = 1.89 \text{ V}$$

$$1.89 \text{ V} \geq 1.89 - 1 \text{ V}$$

$$1.89 \geq 0.89$$

Therefore, n MOSFET is operates in saturation region for $V_0 = 1.89 \text{ V}$

For p MOSFET :

$$V_0 = V_{G_p} = 1.89 \text{ V}$$

$$V_{S_p} = 4 \text{ V}$$

$$V_{SG_p} = V_{S_p} - V_{G_p} = 4 \text{ V} - 1.89 \text{ V} = 2.11 \text{ V}$$

$$V_{SG_p} = 2.11 \text{ V}$$

$$V_{Th_p} = -1 \text{ V}$$

$$|V_{Th_p}| = 1 \text{ V}$$

$$V_{SG_p} > |V_{Th_p}|$$

Hence, p MOSFET will be ON for $V_0 = 1.89 \text{ V}$

Now we will check p MOSFET is operate either in saturation or in linear region. $V_0 = 1.89 \text{ V}$

$$V_{G_p} = 1.89 \text{ V}$$

$$V_{D_p} = 1.89 \text{ V}$$

$$V_{S_p} = 4 \text{ V}$$

$$|V_{Th_p}| = 1 \text{ V}$$

$$V_{SG_p} = V_{S_p} - V_{G_p} = 4 \text{ V} - 1.89 \text{ V} = 2.11 \text{ V}$$

$$V_{SD_p} = V_{S_p} - V_{D_p} = 4 \text{ V} - 1.89 \text{ V} = 2.11 \text{ V}$$

The conditions for saturation is given by,

$$V_{SD_p} \geq V_{SG_p} - |V_{Th_p}|$$

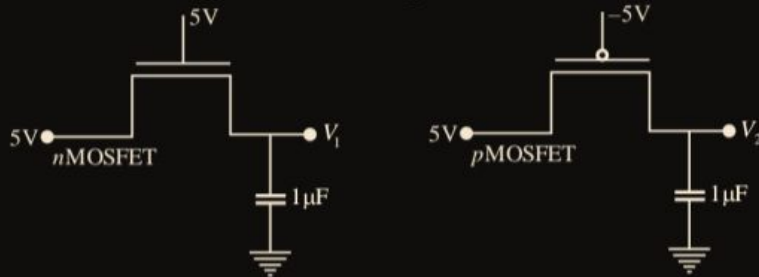
$$2.11 \text{ V} \geq 2.11 \text{ V} - 1 \text{ V}$$

$$2.11 \geq 1.11$$

Therefore, p MOSFET is operates in saturation region for $V_0 = 1.89 \text{ V}$.

Hence, the correct option is (D).

5.50 The ideal long channel n MOSFET and p MOSFET devices shown in the circuit have threshold voltages of 1 V and -1 V respectively. The MOSFET substrate are connected to their respective sources. Ignore leakage currents and assume that the capacitors are initially discharged. For the applied voltages as shown, the steady state voltages are



- (A) $V_1 = 5\text{ V}$, $V_2 = 4\text{ V}$
 (B) $V_1 = 4\text{ V}$, $V_2 = 5\text{ V}$
 (C) $V_1 = 4\text{ V}$, $V_2 = -5\text{ V}$
 (D) $V_1 = 5\text{ V}$, $V_2 = 5\text{ V}$

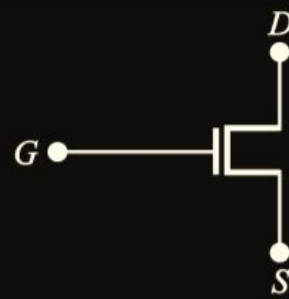
5.50

(B)

Given :

(i) $V_{Tn} = 1\text{ V}$

(ii) $V_{Tp} = -1\text{ V}$



In NMOS transistor, gate voltage work as a controlled input. When NMOS work as a switch then,

If $V_G = 0\text{ V}$, then NMOS will be OFF and

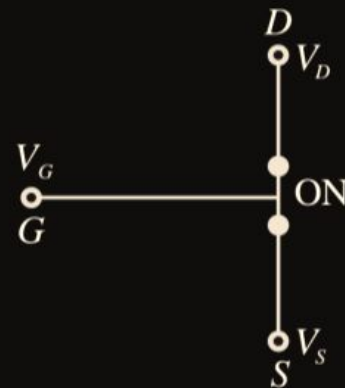
$V_G = +5\text{ V}$, then NMOS will be ON.

(i)

For $V_{DS} < V_{GS} - V_{Tn}$

$$V_D - V_S < V_G - V_S - V_{Tn}$$

$$V_D < V_G - V_{Tn}$$



When NMOS Will satisfy above equation, then NMOS will be in linear region and hence it will be ON.

Therefore, $V_D = V_S$

(ii)

For $V_{DS} \geq V_{GS} - V_{Tn}$

$$V_D - V_S \geq V_G - V_S - V_{Tn}$$

$$V_D \geq V_G - V_{Tn}$$

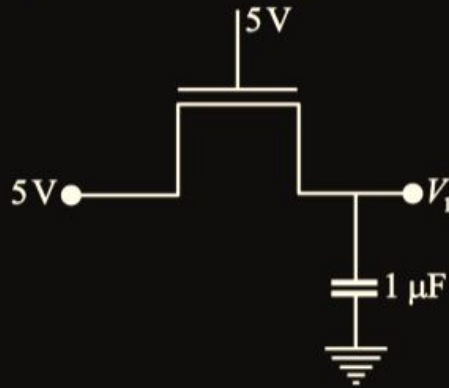


When NMOS will satisfy above equation, then NMOS will be in saturation region and hence it will be ON.

Therefore, $V_D = V_S = V_G - V_{Tn}$

$$V_s = V_G - V_{Tn}$$

For NMOS pass transistor logic,



If $V_D - V_G \geq -V_{Tn}$

Then, $V_G - V_S = V_{Tn}$

If $V_D - V_G < -V_{Tn}$

Then, $V_S = V_D$

$$V_D = 5 \text{ V}$$

$$V_G = 5 \text{ V}$$

$$V_S = V_1$$

$$V_D - V_G = 5 - 5 = 0$$

$$V_D - V_G \geq -V_{Tn}$$

$$0 \geq -1 \quad [\text{True}]$$

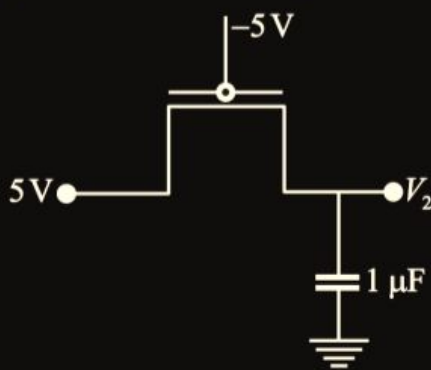
Hence, $V_G - V_S = V_{Tn}$

$$5 - V_1 = 1$$

$$5 - 1 = V_1$$

$$V_1 = 4 \text{ V}$$

For PMOS pass transistor logic,



If $V_G - V_D \geq -|V_{Tp}|$

Then, $V_S - V_G = -|V_{Tp}|$

If $V_G - V_D < -|V_{Tp}|$

Then, $V_S = V_D$

$$V_G = -5 \text{ V}$$

$$V_D = 5 \text{ V}$$

$$V_S = V_2$$

$$V_G - V_D = -5 - 5 = -10$$

$$V_G - V_D < -|V_{Tp}|$$

$$-10 < -1 \quad [\text{True}]$$

Hence, $V_S = V_D$

$$V_2 = 5 \text{ V}$$

Hence, the correct option is (B).

